

**MEGA FILE**

**FOR MID TERM (FALL 2021,22)**

**Course Code: MTH601**

**EDIT BY:**

**SYED HUSSNAIN ALI NAQVI**

**MSC MATHEMATICS**



**Note:**

This file is for preparation of midterm only. This file is made up of different small files. So just read the things that are needed in the file. If irrelevant materials are found then avoid it.



# Grand Quiz Spring 2021

Subject Code MTH601 lecture 1 to 22

Solved By Riz Mughal



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<https://www.facebook.com/groups/923887914750307>



<https://www.youtube.com/channel/UCINsFwDiB62SVaCCPDZbRQ/playlists>

## Dear Viewers:

I'm providing 100% correct quiz solution. You can visit my YouTube channel and get more information about all other subjects' quizzes and final year project (CS619).

RIZ MUGHAL (SQA ENGINEER)

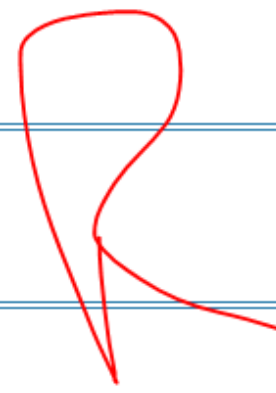
Question # 1 of 30 ( Start time: 09:00:15 AM, 29 June 2021 )

Tot

Which of the following correspond to the practical limitations of available resources while modeling a real life situation into a linear program?

Select the correct option

|                                  |                    |
|----------------------------------|--------------------|
| <input checked="" type="radio"/> | Constraints        |
| <input type="radio"/>            | Objective function |
| <input type="radio"/>            | Linear function    |
| <input type="radio"/>            | Decision variables |



MTH601:Grand Quiz

Question # 2 of 30 ( **Start time: 09:00:43 AM, 29 June 2021** )

Which of the following method follows iterative procedure to solve a Linear Programming problem?

Select the correct option

|                                  |                     |
|----------------------------------|---------------------|
| <input type="radio"/>            | Graphical           |
| <input checked="" type="radio"/> | Algebraic (Simplex) |



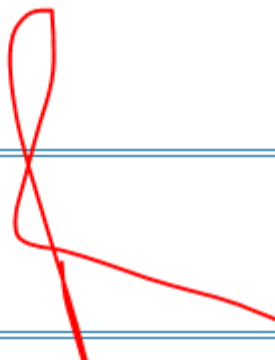


## MTH601:Grand Quiz

Question # 3 of 30 ( Start time: 09:00:57 AM, 29 June 2021 )

Solution region of the constraints: ' $2x+3y>12$ ' and ' $2x+3y<12$ ' will be the-----.

Select the correct option

- |                                  |                           |
|----------------------------------|---------------------------|
| <input type="radio"/>            | point(6,4)                |
| <input type="radio"/>            | straight line: $2x+3y=12$ |
| <input checked="" type="radio"/> | empty set                 |
| <input type="radio"/>            | all first quadrant        |
- 

## MTH601:Grand Quiz

Question # 4 of 30 ( Start time: 09:01:13 AM, 29 June 2021 )

In the graph of the Purchasing Model with shortages, the area under the horizontal axis represents -----

Select the correct option

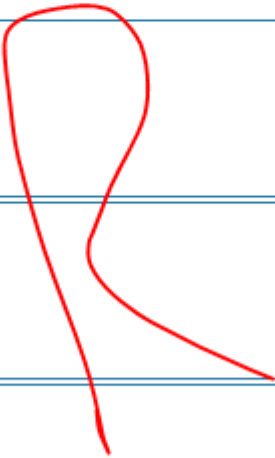
- |                                  |                       |
|----------------------------------|-----------------------|
| <input type="radio"/>            | set up cost of item   |
| <input type="radio"/>            | purchase cost of item |
| <input type="radio"/>            | carrying cost of item |
| <input checked="" type="radio"/> | shortage cost of item |
- 

Question # 5 of 30 ( Start time: 09:01:27 AM, 29 June 2021 )

The straight line associated with the constraint;  $2x+3y<12$ , will meet x-axis at a distance of----- units from origin.

Select the correct option

|                                  |   |
|----------------------------------|---|
| <input type="radio"/>            | 2 |
| <input type="radio"/>            | 4 |
| <input checked="" type="radio"/> | 6 |
| <input type="radio"/>            | 3 |

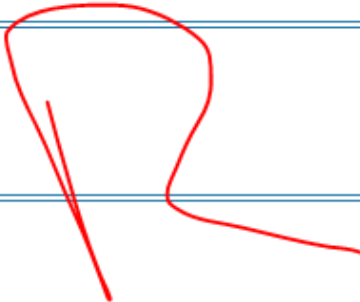


## Question # 6 of 30 ( Start time: 09:01:51 AM, 29 June 2021 )

Total |

Which of the following value is correct for the expected time of an activity having optimistic, pessimistic and most likely times as 4, 8 and 6 days respectively?

Select the correct option

- |                                  |           |
|----------------------------------|-----------|
| <input type="radio"/>            | 6.66 days |
| <input type="radio"/>            | 4.66 days |
| <input checked="" type="radio"/> | 6 days    |
| <input type="radio"/>            | 9.33 days |
- 

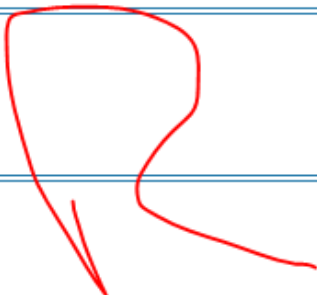
Question # 7 of 30 ( Start time: 09:02:10 AM, 29 June 2021 )

Total Mark

In a network flow diagram, for an activity (i,j) of duration of three days, if its earliest start time is of two days then which of the following will be its early finish time?

Select the correct option

|                                  |                  |
|----------------------------------|------------------|
| <input type="radio"/>            | One and half day |
| <input type="radio"/>            | One day          |
| <input type="radio"/>            | Six days         |
| <input checked="" type="radio"/> | Five days        |

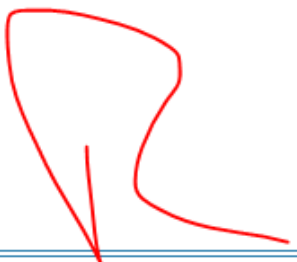


Question # 8 of 30 ( Start time: 09:02:26 AM, 29 June 2021 )

Total Marks: 1

Which of the following method would remain impractical to solve a Linear Programming problem when there are more than two decision variable?

Select the correct option

|                                  |                     |                                                                                    |    |
|----------------------------------|---------------------|------------------------------------------------------------------------------------|----|
| <input type="radio"/>            | Algebraic (Simplex) |  | // |
| <input checked="" type="radio"/> | Graphical           |                                                                                    | // |



## Question # 9 of 30 ( Start time: 09:02:44 AM, 29 June 2021 )

Total

In a development project, if an activity (m,n) of seven days duration, starts late on 5rd day then which of the following will be its latest finish time?

Select the correct option

|                                  |                |
|----------------------------------|----------------|
| <input type="radio"/>            | 7th day        |
| <input type="radio"/>            | 5th or 7th day |
| <input checked="" type="radio"/> | 12th day       |
| <input type="radio"/>            | 5th day        |



Question # 10 of 30 ( Start time: 09:03:00 AM, 29 June 2021 )

Which of the following is an intermediate step to model a linear programming problem?

Select the correct option

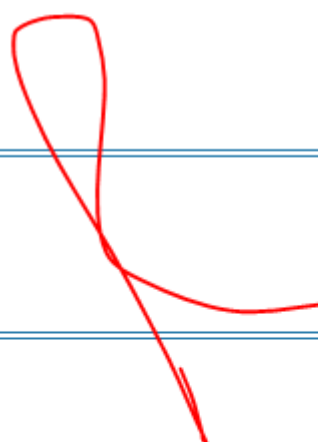
|                                  |                                                        |
|----------------------------------|--------------------------------------------------------|
| <input type="radio"/>            | Identify the objective function                        |
| <input checked="" type="radio"/> | Identify the non-negative restrictions and constraints |
| <input type="radio"/>            | Identify the unknown decision variables                |
| <input type="radio"/>            | None of these                                          |

Question # 11 of 30 ( Start time: 09:03:16 AM, 29 June 2021 )

A/An ----- is the collection of inter related activities to be performed in a particular sequence to completion.

Select the correct option

|                                  |         |
|----------------------------------|---------|
| <input checked="" type="radio"/> | project |
| <input type="radio"/>            | branch  |
| <input type="radio"/>            | event   |
| <input type="radio"/>            | node    |

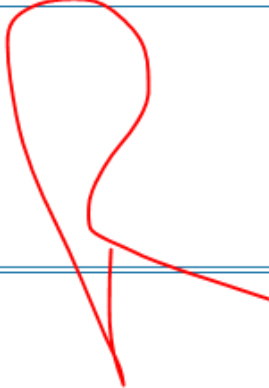


Question # 12 of 30 ( Start time: 09:03:31 AM, 29 June 2021 )

Total Marks: 1

Which of the following times are directly related to the activity cost that is by reducing the activity duration, the direct cost of the corresponding activity is increased?

Select the correct option

|                                  |            |                                                                                     |
|----------------------------------|------------|-------------------------------------------------------------------------------------|
| <input type="radio"/>            | PERT Times |  |
| <input checked="" type="radio"/> | CPM Times  |                                                                                     |

[Click to Save Answer & Move to Next Question](#)

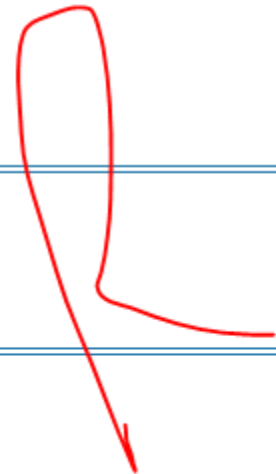
## MTH601:Grand Quiz

Question # 13 of 30 ( Start time: 09:03:49 AM, 29 June 2021 )

Which of the following is an operations research (OR) process?

Select the correct option

|                                  |                    |
|----------------------------------|--------------------|
| <input type="radio"/>            | Simplex method     |
| <input type="radio"/>            | Linear programming |
| <input checked="" type="radio"/> | Observation        |
| <input type="radio"/>            | Networking         |



Question # 14 of 30 ( Start time: 09:04:04 AM, 29 June 2021 )

In the Purchasing Model with Shortages the Cost Function ' $C(S,Q)$ ' can also be expressed as function of -----.

Select the correct option

- |                                  |       |
|----------------------------------|-------|
| <input type="radio"/>            | (S,D) |
| <input type="radio"/>            | (Q,D) |
| <input checked="" type="radio"/> | (S,t) |
| <input type="radio"/>            | (t,D) |

Question # 15 of 30 ( Start time: 09:04:21 AM, 29 June 2021 )

Total Marks

Which of the following would be the objective of the yield per minute in a chemical process which depends upon the temperature 'x' and pressure 'y'?

Select the correct option

|                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | Minimization |
| <input type="radio"/>            | Inflection   |
| <input checked="" type="radio"/> | Maximization |
| <input type="radio"/>            | Average      |

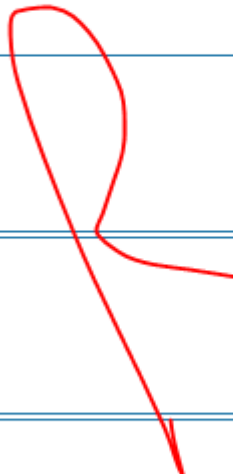
## MTH601:Grand Quiz

Question # 16 of 30 ( Start time: 09:04:38 AM, 29 June 2021 )

Which of the following is an operations research (OR) process?

Select the correct option

|                                  |                    |
|----------------------------------|--------------------|
| <input type="radio"/>            | Linear programming |
| <input type="radio"/>            | Simplex method     |
| <input type="radio"/>            | Networking         |
| <input checked="" type="radio"/> | Solution           |



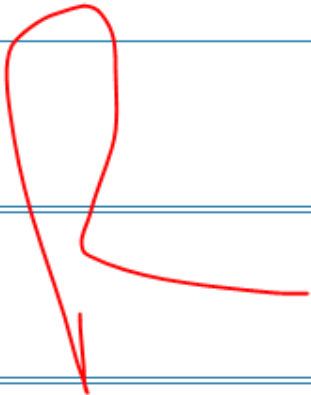


Question # 17 of 30 ( Start time: 09:04:54 AM, 29 June 2021 )

Total M

In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 130 respectively then which of the following is true about the 2nd month's requirement?

Select the correct option

- |                                  |                                                           |
|----------------------------------|-----------------------------------------------------------|
| <input checked="" type="radio"/> | 2nd month demand can be included in 1st month             |
| <input type="radio"/>            | 2nd month demand can be included in 3rd month             |
| <input type="radio"/>            | 2nd month demand can be included in any month of the year |
| <input type="radio"/>            | 2nd month demand will have to fulfill in 2nd month        |
- 

Question # 18 of 30 ( Start time: 09:05:09 AM, 29 June 2021 )

Total Marks:

The mathematical technique which is used to solve a wide class of problems such as allocating scarce resources among competitive activities is known as -----.

Select the correct option

- |                                  |                          |
|----------------------------------|--------------------------|
| <input type="radio"/>            | Classical model          |
| <input type="radio"/>            | Descriptive model        |
| <input checked="" type="radio"/> | Linear Programming model |
| <input type="radio"/>            | Mathematical Model       |

## MTH601:Grand Quiz

Question # 19 of 30 ( Start time: 09:05:28 AM, 29 June 2021 )

Which of the following would be the objective of the cost per unit of producing certain cameras?

Select the correct option

|                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | Average      |
| <input checked="" type="radio"/> | Minimization |
| <input type="radio"/>            | Inflection   |
| <input type="radio"/>            | Maximization |

Question # 20 of 30 ( Start time: 09:05:50 AM, 29 June 2021 )

In CPM each activity has one deterministic time while in PERT each activity has -----probabilistic time/times.

Select the correct option

|                                  |          |
|----------------------------------|----------|
| <input checked="" type="radio"/> | three    |
| <input type="radio"/>            | also one |
| <input type="radio"/>            | two      |
| <input type="radio"/>            | no       |

Question # 21 of 30 ( Start time: 09:06:03 AM, 29 June 2021 )

Total Marks: 1

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, the computations are proceeded from ----- to the final event.

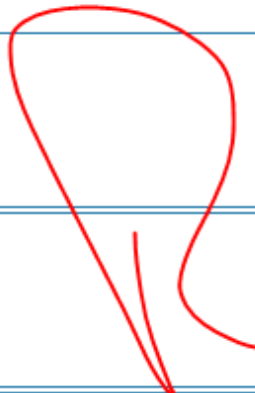
Select the correct option

|                                  |               |                                                                                     |
|----------------------------------|---------------|-------------------------------------------------------------------------------------|
| <input checked="" type="radio"/> | left to right |  |
| <input type="radio"/>            | bottom to top |                                                                                     |
| <input type="radio"/>            | top to bottom |                                                                                     |
| <input type="radio"/>            | right to left |                                                                                     |

## Question # 22 of 30 ( Start time: 09:06:19 AM, 29 June 2021 )

In a development project, if an activity (i,j) of six days duration, starts late on 3rd day then which of the following will be its latest finish time?

Select the correct option

- |                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | 18th day |
| <input checked="" type="radio"/> | 9th day  |
| <input type="radio"/>            | 3rd day  |
| <input type="radio"/>            | 2nd day  |
- 

Question # 23 of 30 ( Start time: 09:06:33 AM, 29 June 2021 )

Total Mark

If the annual demand of a product is 10000 items, and its set up and inventory costs are 200 and 100 respectively then its economic order quantity is ----  
-----provided that shortage is fulfilled INSTANTANEOUSLY.

Select the correct option

- |                                  |             |
|----------------------------------|-------------|
| <input checked="" type="radio"/> | 200 units   |
| <input type="radio"/>            | 20000 units |
| <input type="radio"/>            | 40000 units |
| <input type="radio"/>            | 100 units   |

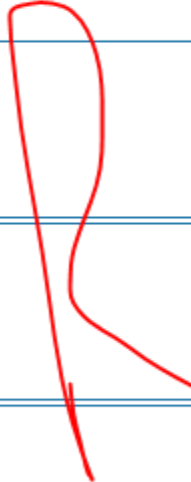
MTH601:Grand Quiz

Question # 24 of 30 ( Start time: 09:06:49 AM, 29 June 2021 )

For a LP problem say,  $\text{Max: } z = x + y$ , under the constraints  $x, y \geq 0$ , the feasible region would be-----.

Select the correct option

|                                  |                        |
|----------------------------------|------------------------|
| <input checked="" type="radio"/> | all the first quadrant |
| <input type="radio"/>            | all xy-plane           |
| <input type="radio"/>            | empty                  |
| <input type="radio"/>            | point(0,0)             |





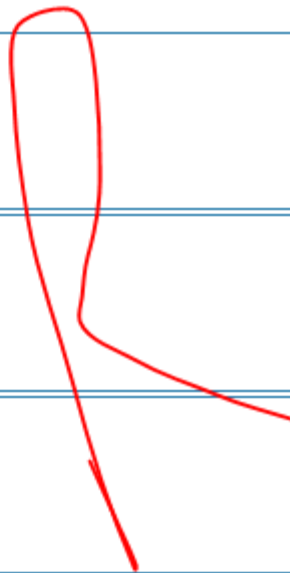
## MTH601:Grand Quiz

Question # 25 of 30 ( Start time: 09:07:06 AM, 29 June 2021 )

By which of the following method, any complex linear programming problem can be handled?

Select the correct option

|                                  |                   |
|----------------------------------|-------------------|
| <input type="radio"/>            | Degenerate method |
| <input type="radio"/>            | Graphical method  |
| <input checked="" type="radio"/> | Simplex method    |
| <input type="radio"/>            | Dual method       |

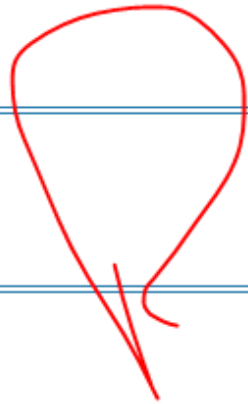


## Question # 26 of 30 ( Start time: 09:07:22 AM, 29 June 2021 )

To

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, for the first node(event), we start with time -----

Select the correct option

- |                                  |                            |
|----------------------------------|----------------------------|
| <input type="radio"/>            | with strict positive value |
| <input checked="" type="radio"/> | $t = 0$                    |
| <input type="radio"/>            | $t = a$ (arbitrary)        |
| <input type="radio"/>            | $t = \text{infinity}$      |
- 

Question # 27 of 30 ( Start time: 09:07:36 AM, 29 June 2021 )

To

Solution region of the constraints:  $2x+3y>12$  or  $6x+9y=36$ , will be the half plane bisected by ' $2x+3y=12$ ', ----- all the points on ' $2x+3y=12$ '.

Select the correct option

|                                  |           |
|----------------------------------|-----------|
| <input type="radio"/>            | excluding |
| <input checked="" type="radio"/> | including |



## MTH601:Grand Quiz

Question # 28 of 30 ( Start time: 09:07:50 AM, 29 June 2021 )

Which of the following event has no immediate predecessor in any network flow diagram?

Select the correct option

|                                  |                    |
|----------------------------------|--------------------|
| <input type="radio"/>            | Critical Event     |
| <input checked="" type="radio"/> | Head Event         |
| <input type="radio"/>            | Tail Event         |
| <input type="radio"/>            | Non-Critical Event |

## MTH601:Grand Quiz

Question # 29 of 30 ( Start time: 09:08:06 AM, 29 June 2021 )

Which one of the following is not an operations research problem solving steps?

Select the correct option

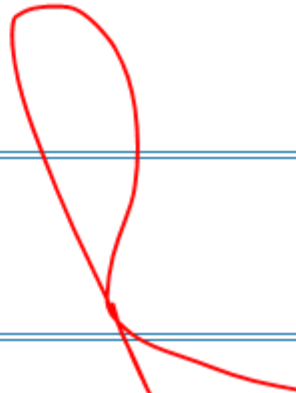
|                                  |                            |
|----------------------------------|----------------------------|
| <input checked="" type="radio"/> | variable selections        |
| <input type="radio"/>            | definition of the problems |
| <input type="radio"/>            | observation                |
| <input type="radio"/>            | model construction         |

## MTH601:Grand Quiz

Question # 30 of 30 ( Start time: 09:08:23 AM, 29 June 2021 )

For critical path of a network, which of the followings is the best suitable answer.

Select the correct option

- |                                  |                               |
|----------------------------------|-------------------------------|
| <input type="radio"/>            | Its total float is zero       |
| <input type="radio"/>            | Its free float is zero        |
| <input checked="" type="radio"/> | All of the above              |
| <input type="radio"/>            | Its independent float is zero |
- 



# Thank you for watching

Please share it with your friends 😊

RIZ MUGHAL (SQA ENGINEER)



**RIZ MUGHAL**

# QUIZ MASTER

## Grand Quiz(MTH601)

100% correct solution.

For more information you can visit my channel and for any type of help related to CS619 you can contact me.



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<https://www.youtube.com/channel/UCINsFwDiB62SValCcPDZbRQ/playlists>

### **FACEBOOK GROUP:**

<https://www.facebook.com/groups/923887914750307>



Question # 1 of 30 ( Start time: 01:06:58 AM, 31 December 2020 )

Total Marks:

By putting the Replacement Rate 'R?Infinity' and Shortage Cost 'C4?Infinity' in the Economic Order Quantity(EOQ) of Manufacturing Model with allowed Shortage, then the result is -----

Select the correct option

- |                                  |                                               |
|----------------------------------|-----------------------------------------------|
| <input checked="" type="radio"/> | EOQ of Wilson's Model                         |
| <input type="radio"/>            | EOQ of Manufacturing Model without Shortage   |
| <input type="radio"/>            | EOQ of Purchasing Model with allowed Shortage |
| <input type="radio"/>            | Invariant                                     |

As, EOQ of Wilson's Model =  $Q^* = \sqrt{\frac{2C_2D}{C_3}}$

Now,

Manufacturing Model with shortage

$$Q^* = \sqrt{\frac{2C_2D}{C_3(1-\frac{D}{R})}} \times \sqrt{\frac{C_3+C_4}{C_4}}$$

According to question

$$R = \infty$$
$$C_4 = \infty$$

then

$$Q^* = \sqrt{\frac{2C_2D}{C_3(1-\frac{D}{\infty})}} \times \sqrt{\frac{C_3+\infty}{\infty}}$$
$$Q^* = \sqrt{\frac{2C_2D}{C_3(1-0)}} \times 1$$
$$Q^* = \sqrt{\frac{2C_2D}{C_3}}$$

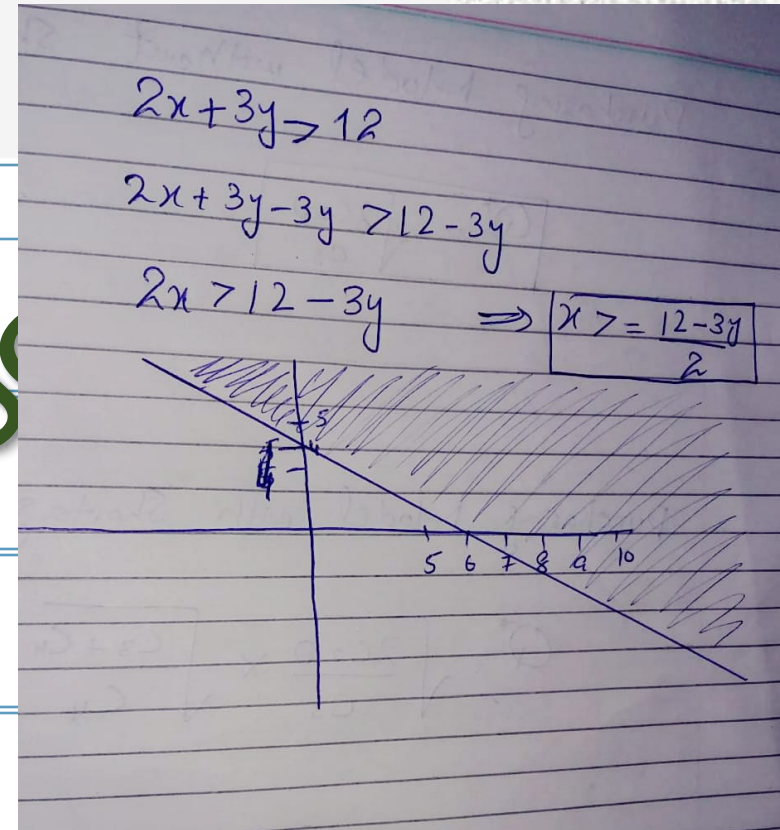
✓

Question # 2 of 30 ( Start time: 01:07:22 AM, 31 December 2020 )

The solution set of the constraint:  $2x+3y>12$ , will lie in ----- bisected by the line:  $2x+3y=12$ .

Select the correct option

- |                                  |                                           |
|----------------------------------|-------------------------------------------|
| <input type="radio"/>            | origin side of half plane                 |
| <input checked="" type="radio"/> | opposite to the origin side in half plane |
| <input type="radio"/>            | both half planes                          |
| <input type="radio"/>            | first quadrant only                       |





## Question # 3 of 30 ( Start time: 01:07:39 AM, 31 December 2020 )

Total Marks:

If the variance of the project length = 25, expected and scheduled times are 40 and 36 respectively, then the Standard Normal Variable of the given project is-----.

Select the correct option

☐ 0.80☒ -0.80☐ 0.16☐ -0.16

Handwritten calculation of the Standard Normal Variable  $Z$ :

$$Z = \frac{T - t_e}{S.D}$$
$$Z = \frac{36 - 40}{\sqrt{25}}$$
$$= \frac{-4}{5}$$
$$= -0.8$$

Side notes:

- $\sqrt{(S.D)^2} = \sqrt{\text{Variance}}$
- $\therefore S.D = \sqrt{\text{Variance}}$

Question # 4 of 30 ( Start time: 01:08:01 AM, 31 December 2020 )

Under which of the following condition, a mathematical Program would be non-linear?

Select the correct option

|                                  |                                                           |
|----------------------------------|-----------------------------------------------------------|
| <input checked="" type="radio"/> | If least Objective function is non-linear                 |
| <input type="radio"/>            | If both Objective function and constraints are non-linear |

5) Other techniques consist of all the remaining techniques, which do not come under the four heads mentioned above. For example, Dynamic programming employs a different modeling and solution logic than linear programming. In non-linear programming either the objective function or the constraints or both can be non-linear functions, which would require altogether different solution technique.

#### 1.7 Uses of OR

In its recent years of organized development, OR has entered successfully many different areas of research for

Question # 5 of 30 ( Start time: 01:08:13 AM, 31 December 2020 )

Which of the following would be the objective of the cost per unit of producing certain cameras?

Select the correct option

|                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | Maximization |
| <input checked="" type="radio"/> | Minimization |
| <input type="radio"/>            | Inflection   |
| <input type="radio"/>            | Average      |



Question # 6 of 30 ( Start time: 01:08:26 AM, 31 December 2020 )

Total Mark

In a development project, if an activity (i,j) of six days duration, starts late on 3rd day then which of the following will be its latest finish time?

Select the correct option

☒

9th day

☐

2nd day

☐

3rd day

☐

18th day

RIZ MUGHAL

## MTH601:Grand Quiz

Question # 7 of 30 ( Start time: 01:08:44 AM, 31 December 2020 )

Solution of a Linear Programming problem is found by ----- methods.

The second method is the algebraic procedure or simplex method, which involves an iterative procedure. This can be applied to linear programming problems involving two or more variables. This is the most versatile method of solving linear programming models. In this method, an initial solution is assumed and this solution is modified progressively through a well defined iterative process until we get the optimal solution. The procedure is

Select the correct option

|                                  |                |
|----------------------------------|----------------|
| <input type="radio"/>            | Analytical     |
| <input type="radio"/>            | Probabilistic  |
| <input checked="" type="radio"/> | Iterative      |
| <input type="radio"/>            | Pure algebraic |

Question # 8 of 30 ( Start time: 01:08:57 AM, 31 December 2020 )

Total Marks:

In a development project, if an activity (q,r) of five days duration, starts late on 2nd day then which of the following will be its latest finish time?

Select the correct option

- |                                  |          |    |
|----------------------------------|----------|----|
| <input type="radio"/>            | 3rd day  | // |
| <input type="radio"/>            | 4th day  | // |
| <input checked="" type="radio"/> | 7th day  | // |
| <input type="radio"/>            | 10th day | // |



Question # 9 of 30 ( Start time: 01:09:13 AM, 31 December 2020 )

Total Marks: 1

Objective of inventory control is to balance between the advantage of having inventories and the ----- of carrying them to arrive at an optimal level of inventories to minimize the total inventory cost.

Select the correct option

- |                                  |                |    |
|----------------------------------|----------------|----|
| <input type="radio"/>            | demand         | // |
| <input type="radio"/>            | profit         | // |
| <input checked="" type="radio"/> | cost           | // |
| <input type="radio"/>            | transportation | // |

Question # 10 of 30 ( Start time: 01:09:26 AM, 31 December 2020 )

Total Marks

If the annual demand of a product is 10000 items, and its set up and inventory costs are 200 and 100 respectively then its economic order quantity is -----provided that shortage is fulfilled INSTANTANEOUSLY.

Select the correct option

- ☒ 200 units
- ☐ 100 units
- ☐ 20000 units
- ☐ 40000 units

$$Q = \sqrt{\frac{2 C_2 D}{C_3}}$$

$$Q = \sqrt{\frac{2(200)(10000)}{100}}$$

$$Q = \sqrt{40000}$$

$$Q = 200$$

expression

$$TC = (D/Q) C_0 + (Q/2) C_h$$

where  $TC$  = total variable inventory cost; $D$  = annual demand; $C_0$  = ordering cost per order; $C_h$  = carrying cost per time period; $Q$  = size of an order

be taken and equated to zero as the first step towards calculating,  $EOQ(Q) = \sqrt{2DC_0 / C_h}$ . This is because of the concept of maxima and minima used for optimality.

(ii)  
(iii)

## Question # 11 of 30 ( Start time: 01:09:40 AM, 31 December 2020 )

The task which is executed by the usage of resources and time is called \_\_\_\_\_. Select correct option.

. Activity consumes resources like time, money and materials. Event simply represents either starting or ending of an activity. Event

Select the correct option

|                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | node     |
| <input type="radio"/>            | event    |
| <input type="radio"/>            | project  |
| <input checked="" type="radio"/> | activity |



## MTH601:Grand Quiz

Question # 12 of 30 ( Start time: 01:09:53 AM, 31 December 2020 )

Which of the following is first step to model a linear programming problem?

Select the correct option

- |                                  |                                            |
|----------------------------------|--------------------------------------------|
| <input type="radio"/>            | Identifying the objective function         |
| <input type="radio"/>            | Identifying the non-negative constraints   |
| <input checked="" type="radio"/> | Identifying the unknown decision variables |
| <input type="radio"/>            | Identifying all the restrictions           |

### 2.7 GUIDELINES ON LINEAR PROGRAMMING MODEL FORMULATION

The effective use and application requires, as a **first step**, the mathematical formulation of an LP model. Steps of LP model formulation are summarized as follows:

#### Step 1: Identify the decision variables

- (a) Express each constraint in words. For this you should first see whether the constraint is of the form  $\geq$  (at least as large as), of the form  $\leq$  (no larger than) or of the form  $=$  (exactly equal to).
- (b) Express verbally the objective function.
- (c) Verbally identify the decision variables with the help of Step (a) and (b). For this you need to ask yourself the question – *What decisions must be made in order to optimize the objective function?*

Having followed Step 1(a) to (c) decide the symbolic notation for the decision variables and specify their units of measurement. Such specification of units of measurement would help in interpreting the final solution of the LP problem.

#### Step 2: Identify the problem data

To formulate an LP model, identify the problem data in terms of constants, and parameters associated with decision variables. It may be noted that the decision-maker can control values of the variables but cannot control values in the data set.

#### Step 3: Formulate the constraints

Convert the verbal expression of the constraints in terms of resource requirement and availability of each resource. Then express each of them as linear equality or inequality, in terms of the decision variables defined in Step 1.

Values of these decision variables in the optimal LP problem solution must satisfy these constraints in order to constitute an acceptable (feasible) solution. Wrong formulation can either lead to a solution that is not feasible or to the exclusion of a solution that is actually feasible and possibly optimal.

#### Step 4: Formulate the objective function

Identify whether the objective function is to be maximized or minimized. Then express it in the form of linear mathematical expression in terms of decision variables along with profit (cost) contributions associated with them.

After gaining enough experience in model building, readers may skip verbal description. The following are certain examples of LP model formulation that may be used to strengthen the ability to translate a real

Question # 13 of 30 ( Start time: 01:10:08 AM, 31 December 2020 )

For a LP problem say:  $\text{Max: } z = x + y$ , under the constraints  $x, y \geq 0$ , the feasible region would be-----.

Select the correct option

- |                                  |                        |
|----------------------------------|------------------------|
| <input type="radio"/>            | empty                  |
| <input type="radio"/>            | all xy-plane           |
| <input checked="" type="radio"/> | all the first quadrant |
| <input type="radio"/>            | point(0,0)             |

Both positive and greater than zero first quadrant

Question # 14 of 30 ( Start time: 01:10:26 AM, 31 December 2020 )

Total Marks:

While applying Simplex method to a LP of minimization type, we proceed stepwise from one -----solution to another in such a way that the objective function always increases its value.

Select the correct option

- |                                  |                    |
|----------------------------------|--------------------|
| <input type="radio"/>            | optimal            |
| <input type="radio"/>            | non-basic feasible |
| <input checked="" type="radio"/> | basic feasible     |
| <input type="radio"/>            | degenerate         |

### 3.1. INTRODUCTION

As discussed earlier, there are many methods to solve the Linear Programming Problem, such as Graphical Method, Trial and Error method, Vector method and Simplex Method. Though we use graphical method for solution when we have two problem variables, the other method can be used when there are more than two decision variables in the problem. Among all the methods, **SIMPLEX METHOD** is most powerful method. It deals with iterative process, which consists of **first designing a Basic Feasible Solution** or a **Programme** and proceed towards the **OPTIMAL SOLUTION** and testing each feasible solution for **Optimality** to know whether the solution on hand is optimal or not. If not an optimal solution, redesign the programme, and test for optimality until the test confirms **OPTIMALITY**. Hence we can say that the Simplex Method depends on two concepts known as **Feasibility** and **optimality**.

*The simplex method is based on the property that the optimal solution to a linear programming problem, if it exists, can always be found in one of the basic feasible solution.* The simplex method is quite simple and mechanical in nature. The iterative steps of the simplex method are repeated until a finite optimal solution, if exists, is found. If no optimal solution, the method indicates that no finite



Question # 15 of 30 ( Start time: 01:10:42 AM, 31 December 2020 )

Total Marks:

In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 120 respectively then which of the following is true about the 2nd month's requirement?

Select the correct option

- |                                  |                                                           |    |
|----------------------------------|-----------------------------------------------------------|----|
| <input type="radio"/>            | 2nd month demand can be included in 1st month             | // |
| <input checked="" type="radio"/> | 2nd month demand will have to fulfill in 2nd month        | // |
| <input type="radio"/>            | 2nd month demand can be included in 3rd month             | // |
| <input type="radio"/>            | 2nd month demand can be included in any month of the year | // |

Question # 16 of 30 ( Start time: 01:10:57 AM, 31 December 2020 )

Total Marks

A linear programming problem can be converted into a Transportation problem by restricting the coefficients 'aij' to have the values-----

Select the correct option



Zero only



One only



either '0' or '1'



arbitrarily

### 7.7.3 Plant Location Problem

Suppose there are  $m$  possible sites (locations) at which the plants could be located. Each of these plants produces a single commodity for  $n$  customers (markets or demand points), each with a minimum demand for  $b_j$  units ( $j = 1, 2, \dots, n$ ). Suppose at  $i$ th location, the fixed setup cost (expenses associated with constructing and operating a plant) of a plant is  $f_i$  ( $i = 1, 2, \dots, m$ ). The production capacity of each plant is limited to  $a_i$  units. The unit transportation cost from plant  $i$  to customer  $j$  is  $c_{ij}$ . The problem now is to decide the location of plants in such a way that the sum of the fixed setup costs and transportation cost is lowest (minimum).

Let  $x_{ij}$  be the amount shipped from plant  $i$  to customer  $j$ , and  $y_i$  be the new variable associated with each of the plant locations, such that

$$y_i = \begin{cases} 1, & \text{if plant is located at the } i\text{th location} \\ 0, & \text{otherwise} \end{cases}$$

The value of  $f_i$  is assumed to be fixed and independent of the amount of  $x_{ij}$  shipped so long as  $x_{ij} > 0$ , i.e. for  $x_{ij} = 0$ , the value  $f_i = 0$ . Thus, the objective is to minimize the total cost (variable + fixed) of settings up and operating the network of transportation routes.

The general mathematical 0–1 integer programming model of plant location problem can be stated as follows:



Question # 17 of 30 ( Start time: 01:11:12 AM, 31 December 2020 )

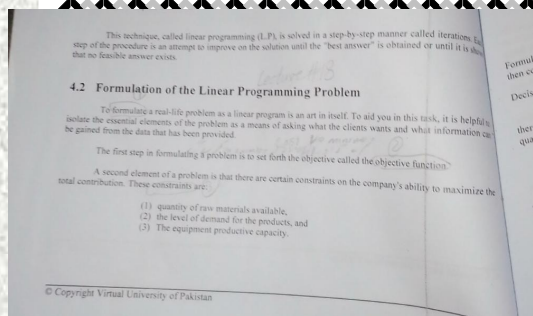
By which of the following method, any complex linear programming problem can be handled?

resources – you see what I mean!

Linear programming (LP) is one of the simplest ways to perform optimization. It helps you solve some very complex optimization problems by making a few simplifying assumptions. As an analyst, you are bound to come across applications and problems to be solved by Linear Programming.

Select the correct option

|                                  |                   |
|----------------------------------|-------------------|
| <input type="radio"/>            | Graphical method  |
| <input checked="" type="radio"/> | Simplex method    |
| <input type="radio"/>            | Dual method       |
| <input type="radio"/>            | Degenerate method |



## MTH601:Grand Quiz

Question # 18 of 30 ( Start time: 01:11:37 AM, 31 December 2020 )

Linear Programming is the backbone of Operations Research models due to-----

Objective of linear operation and constrainLinear programming ki solve karny k lye must agr ye na ho linear programming solve hi nhi hoti

Select the correct option

|                                  |                                                 |
|----------------------------------|-------------------------------------------------|
| <input type="radio"/>            | linearity of objective function and constraints |
| <input checked="" type="radio"/> | its fantastic computational efficiency          |
| <input type="radio"/>            | non-negativity of decision variables            |
| <input type="radio"/>            | unrestricted signs of decision variables        |

Question # 19 of 30 ( Start time: 01:11:48 AM, 31 December 2020 )

Which of the following technique to solve the network flow diagrams, is event oriented?

Select the correct option



Programme Evaluation and Review Technique



Critical Path Method

RIZ MUGHAL



## Question # 20 of 30 ( Start time: 01:12:18 AM, 31 December 2020 )

The parameters like demand and ordering quantity cost are quite ----- in the Probabilistic inventory models.

Select the correct option

☐ certain

☒ uncertain

Certain deterministic  
Uncertainty probability

Question # 21 of 30 ( Start time: 01:12:34 AM, 31 December 2020 )

Total Marks

Which of the following correspond to the economic or physical quantities which are under the control of operations researcher while modeling a real life situation into a linear program?

Select the correct option

|                                  |                    |
|----------------------------------|--------------------|
| <input type="radio"/>            | Objective function |
| <input type="radio"/>            | Constraints        |
| <input checked="" type="radio"/> | Decision variables |
| <input type="radio"/>            | Linear function    |

Model jb bnaty tb decision. Kar k bnaty ... decision laty

Question # 22 of 30 ( Start time: 01:12:53 AM, 31 December 2020 )

The word 'Linear' in Linear programming problem describes ----- between two or more variables.

Select the correct option



that a Correlation exists



a recurrence relation



a direct proportionate relation



an inverse proportionate relation

#### 4 Linear Programming

##### 4.1 Introduction

Linear programming is a mathematical technique designed to aid managers in allocating scarce resources (such as labor, capital, or energy) among competing activities. It reflects, in the form of a model, the organization's attempt to achieve some objective (frequently, maximizing profit contribution, maximizing rate of return, or minimizing costs) in view of limited or constrained resources (available capital or labor, service levels, machine time, budgets).

The linear programming technique can be said to have a linear objective function that is to be maximized (either maximized or minimized) subject to linear equality or inequality constraints and sign restrictions on the variables. The term linear describes the proportionate relationship of two or more variables. Thus, a given change in one variable will always cause a resulting proportional change in another variable.

Some areas in which linear programming have been applied will be helpful in setting the climate for learning about this important technique.

- Application*
- (i) A company produces agricultural fertilizers. It is interested in minimizing costs while meeting specified levels of nitrogen, phosphate, and potash by blending together a number of raw materials.
  - (ii) An investor wants to maximize return on investment by allocating funds among various securities.



Question # 23 of 30 ( Start time: 01:13:07 AM, 31 December 2020 )

Which of the following method follows iterative procedure to solve a Linear Programming problem?

### 5. Simplex Method

Simplex Method is one of the most powerful & popular methods for linear programming. The simplex method is an iterative procedure for getting the most feasible solution. In this method, we keep transforming the value of basic variables to get maximum value for the objective function.

Select the correct option

|                                  |                     |
|----------------------------------|---------------------|
| <input checked="" type="radio"/> | Algebraic (Simplex) |
| <input type="radio"/>            | Graphical           |



MTH601:Grand Quiz

Question # 24 of 30 ( Start time: 01:13:30 AM, 31 December 2020 )

Non-feasible solution----- associated with a given linear programming problem.

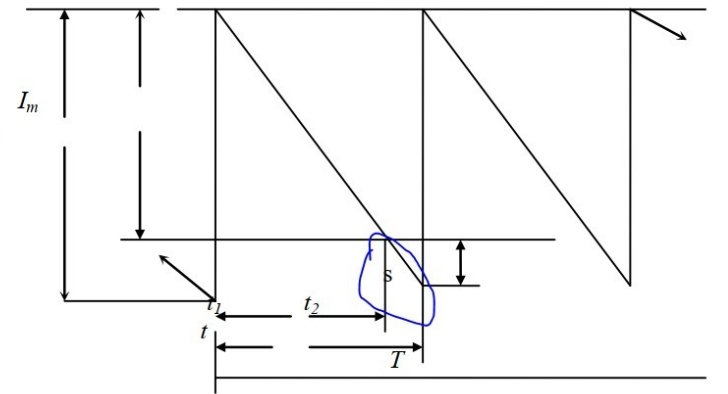
Select the correct option

- ☐ satisfy all the constraints
- ☐ does not satisfy all the constraints
- ☐ satisfy least one of the constraint
- ☒ does not satisfy least one of the constraint

Click t

In the graph of the Purchasing Model with shortages, the area under the horizontal axis represents -----.

|                                  |                       |
|----------------------------------|-----------------------|
| <input type="radio"/>            | carrying cost of item |
| <input type="radio"/>            | set up cost of item   |
| <input checked="" type="radio"/> | shortage cost of item |
| <input type="radio"/>            | purchase cost of item |



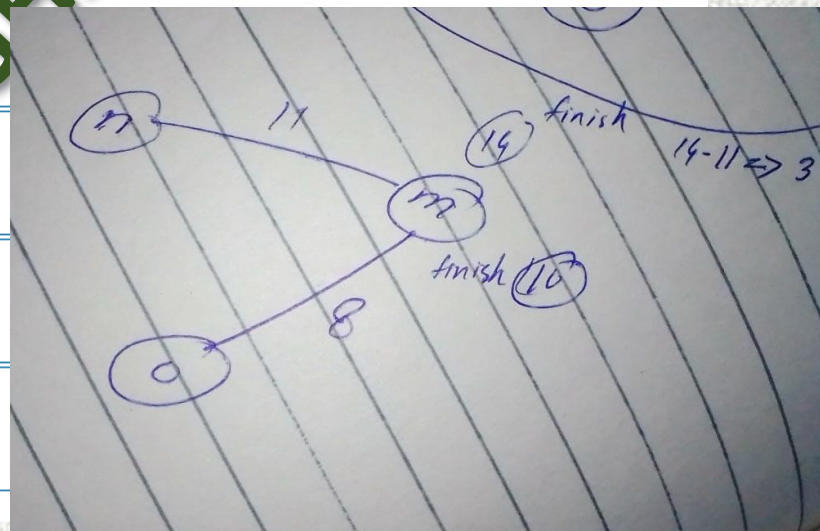
Question # 26 of 30 ( Start time: 01:14:10 AM, 31 December 2020 )

Total Mark

In a network flow diagram, two jobs (m,n) and (m,o) of '11' and '8' days duration leaves the node 'm' then which of the following will be Late start time for 'm', if it is provided that both (m,n) and (m,o) finish late on 14th and 10th day respectively?

Select the correct option

- |                                  |         |
|----------------------------------|---------|
| <input type="radio"/>            | 6th day |
| <input checked="" type="radio"/> | 3rd day |
| <input type="radio"/>            | 2nd day |
| <input type="radio"/>            | 1st day |





Question # 27 of 30 ( Start time: 01:14:26 AM, 31 December 2020 )

Total Marks: 1

Which of the following distribution gives the probability of completing the assigned project by considering the expected time of any activity as random variable?

Select the correct option

- |                                  |          |    |
|----------------------------------|----------|----|
| <input type="radio"/>            | Binomial | // |
| <input type="radio"/>            | Poisson  | // |
| <input type="radio"/>            | Beta     | // |
| <input checked="" type="radio"/> | Normal   | // |

MTH601:Grand Quiz

Question # 28 of 30 ( Start time: 01:14:45 AM, 31 December 2020 )

Delay in critical activities for a project management lead to \_\_\_\_\_.

Select the correct option

- |                                  |                                         |
|----------------------------------|-----------------------------------------|
| <input type="radio"/>            | project finish time will have to extend |
| <input type="radio"/>            | more resources have to employed         |
| <input type="radio"/>            | project cost will increase              |
| <input checked="" type="radio"/> | All above choices are equivalent        |

Question # 29 of 30 ( Start time: 01:14:58 AM, 31 December 2020 )

Total Marks:

In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 130 respectively then which of the following is true about the 2nd month's requirement?

Select the correct option

- |                                  |                                                           |    |
|----------------------------------|-----------------------------------------------------------|----|
| <input checked="" type="radio"/> | 2nd month demand can be included in 1st month             | // |
| <input type="radio"/>            | 2nd month demand will have to fulfill in 2nd month        | // |
| <input type="radio"/>            | 2nd month demand can be included in 3rd month             | // |
| <input type="radio"/>            | 2nd month demand can be included in any month of the year | // |



Question # 30 of 30 ( Start time: 01:15:16 AM, 31 December 2020 )

Total Marks:

In a network flow diagram, if two jobs 'a(l,n)' and 'b(m,n)' of '7' and '8' days durations respectively, start earlier simultaneously on 4th day, then the next activity containing 'n' as head event can't start until the entering activity ----- is completed.

Select the correct option

- |                                  |       |    |
|----------------------------------|-------|----|
| <input checked="" type="radio"/> | (m,n) | // |
| <input type="radio"/>            | (l,n) | // |
| <input type="radio"/>            | (m,l) | // |
| <input type="radio"/>            | (l,m) | // |

Question # 30 of 40 ( Start time: 04:53:26 PM, 11 July 2020 )

In a network flow diagram, a/an ————— is shown by dotted arrows and these consume no time or resources.

Select the correct option



activity



branch



dummy



nodes

Question # 29 of 40 ( Start time: 04:52:53 PM, 11 July 2020 )

Total M

A/An ----- is the collection of inter related activities to be performed in a particular sequence to completion.

05

Select the correct option

☐ node

☐ event

☒ project

☐ branch

Question # 28 of 40 ( Start time: 04:51:47 PM, 11 July 2020 )

For a LP say: 'Min:  $z = x+y$ ' under the constraints  $x,y \geq 0$ , the optimal solution would be-----.

Select the correct option

- |                                  |                        |
|----------------------------------|------------------------|
| <input type="radio"/>            | empty set              |
| <input type="radio"/>            | all xy-plane           |
| <input type="radio"/>            | all the first quadrant |
| <input checked="" type="radio"/> | point(0,0)             |

Question # 26 of 40 ( Start time: 04:49:41 PM, 11 July 2020 )

Total Marks: 1

While solving a problem by any of Linear Programming method, an objective function is to be -----.

Select the correct option

|                                  |              |                                                                                                                                                                                                                                                                                                 |  |
|----------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <input type="radio"/>            | maximized    | <p>Question # 3 of 10 ( Start time: 11:08:39 AM )      Total Marks: 1</p> <p>While solving a problem by any of Linear Programming method, an objective function is to be -----.</p> <p>Select correct option:</p> <p>maximized</p> <p>minimized</p> <p><b>Optimized</b></p> <p>standardized</p> |  |
| <input type="radio"/>            | minimized    |                                                                                                                                                                                                                                                                                                 |  |
| <input checked="" type="radio"/> | optimized    |                                                                                                                                                                                                                                                                                                 |  |
| <input type="radio"/>            | standardized |                                                                                                                                                                                                                                                                                                 |  |

Click to Save Answer & Move to Next Question

Question # 27 of 40 ( Start time: 04:50:15 PM, 11 July 2020 )

Total Marks

If the total inventory in one cycle of 't' units of time is  $(1/2)Qt$ , then which of the following is the average inventory at any time?

▶ Select the correct option

|                                  |          |                                                                                                                                                                                                                                                                                               |
|----------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input type="radio"/>            | $Q/t$    | Question # 10 of 10 ( Start time: 04:27:13 PM ) Total Marks: 1<br>If the total inventory in one cycle of 't' units of time is $(1/2)Qt$ , then which of the following is the average inventory at any time?<br>Select correct option:<br>$Q/t$<br><b><math>Q/2</math></b><br>$(Q/2)t$<br>$2Q$ |
| <input checked="" type="radio"/> | $Q/2$    |                                                                                                                                                                                                                                                                                               |
| <input type="radio"/>            | $(Q/2)t$ |                                                                                                                                                                                                                                                                                               |
| <input type="radio"/>            | $2Q$     |                                                                                                                                                                                                                                                                                               |



Which of the following is first step to model a linear programming problem?

Question # 7 of 10 ( Start time: 11:27:03 AM )      Total Marks: 1  
**Which of the following is first step to model a linear programming problem?**  
Select correct option:

Select the correct option

- |                                  |                                            |                                            |
|----------------------------------|--------------------------------------------|--------------------------------------------|
| <input checked="" type="radio"/> | Identifying the objective function         | <b>Identifying the objective function</b>  |
| <input type="radio"/>            | Identifying the non-negative constraints   | Identifying the non-negative constraints   |
| <input type="radio"/>            | Identifying the unknown decision variables | Identifying the unknown decision variables |
| <input type="radio"/>            | Identifying all the restrictions           | Identifying all the restrictions           |

Question # 24 of 40 ( Start time: 04:47:25 PM, 11 July 2020 )

Total Marks: 1

Which of the following method would remain impractical to solve a Linear Programming problem when there are more than two decision variable?

Select the correct option



Algebraic (Simplex)



Graphical

Question # 23 of 40 ( Start time: 04:46:02 PM, 11 July 2020 )

Total Marks: 1

In a network flow diagram, which of the following method through computations provides, i) start and completion times for each activity, ii) critical and non critical activities and iii) total and free slacks?

Select the correct option



Resource Scheduling

//



Resource Allocation

//



PERT

//



CPM

//

Question # 21 of 40 ( Start time: 04:42:54 PM, 11 July 2020 )

Total Mark

Solution region of the constraints:  $2x+3y>12$  or  $6x+9y=36$ , will be the half plane bisected by ' $2x+3y=12$ ', ----- all the points on ' $2x+3y=12$ '.

03:45

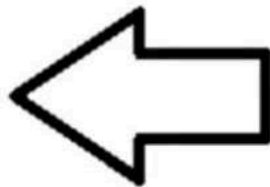
Select the correct option



excluding



including



Click to Save Answer & Move to Next Question

Question # 20 of 40 ( Start time: 04:41:23 PM, 11 July 2020 )

Total Marks:

For a development project, if its Standard Normal variable = 1.38, expected and the scheduled durations of the project are 100 and 110 days respectively, then Variance in the project length is -----.

Select the correct option

- |                       |        |   |
|-----------------------|--------|---|
| <input type="radio"/> | -7.246 |   |
| <input type="radio"/> | -52.50 |   |
| <input type="radio"/> | 52.50  | ↑ |
| <input type="radio"/> | 7.246  |   |

## Question # 19 of 40 ( Start time: 04:39:44 PM, 11 July 2020 )

In CPM each activity has one deterministic time while in PERT each activity has -----probabilistic time/times.

Select the correct option

- |                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | three    |
| <input type="radio"/>            | two      |
| <input checked="" type="radio"/> | also one |
| <input type="radio"/>            | no       |



Question # 17 of 40 ( Start time: 04:37:07 PM, 11 July 2020 )

Total Marks: 1

For any activity (i,j), if . a) Earliest start time of  $i$  = Latest finish time of  $i$ . b) Earliest start time of  $j$  = Latest finish time of  $j$ . c) difference of Earliest start times of events  $i$  and  $j$  = difference of Latest finish times of events  $i$  and  $j$  = time to complete the job, then the activity (i,j) is said to be -----.

Select the correct option

|                                  |                  |    |
|----------------------------------|------------------|----|
| <input type="radio"/>            | dummy            | // |
| <input checked="" type="radio"/> | critical         | // |
| <input type="radio"/>            | non-Critical     | // |
| <input type="radio"/>            | non of the above | // |

Question # 16 of 40 ( Start time: 04:35:48 PM, 11 July 2020 )

Total Marks: 1

While finding the critical path in a network flow diagram, if an activity (i,j) starts and finish late on 3rd and 9th day respectively, then it will be completed in -----days.

While finding the critical path in a network flow diagram, if an activity (i,j) starts and finish late on 3rd and 9th days respectively, then it will be completed in \_\_\_\_\_ days.

Three days

Twelve days

**Six days correct Answer**

Fifteen days

Select the correct option



Three days



Twelve days



Fifteen days



Six days

Question # 14 of 40 ( Start time: 04:32:43 PM, 11 July 2020 )

Total Marks: 1

In a network flow diagram, the precedence relationship among the activities are indicated through ————

02:39

Select the correct option

|                                  |         |  |
|----------------------------------|---------|--|
| <input type="radio"/>            | project |  |
| <input type="radio"/>            | branch  |  |
| <input checked="" type="radio"/> | dummy   |  |

Question # 13 of 40 ( Start time: 04:31:06 PM, 11 July 2020 )

Total Marks: 1

Which of the following distribution gives the probability of completing the assigned project by considering the expected time of any activity as random variable?

Select the correct option

|                       |          |   |
|-----------------------|----------|---|
| <input type="radio"/> | Binomial |   |
| <input type="radio"/> | Poisson  |   |
| <input type="radio"/> | Beta     | ↑ |
| <input type="radio"/> | Normal   |   |

Question # 12 of 40 ( Start time: 04:29:35 PM, 11 July 2020 )

Total Marks: 1

Which of the following relation is true among the probabilistic times in PERT?

Select the correct option

- ☐ Most Likely < Optimistic < Pessimistic
- ☒ Optimistic < Most Likely < Pessimistic
- ☐ Most Likely < Pessimistic < Optimistic
- ☐ Pessimistic < Most Likely < Optimistic

Question # 11 of 40 ( Start time: 04:28:03 PM, 11 July 2020 )

Total Marks.

If the point  $(6, t)$  lies in the feasible region associated with constraint  $2x + 3y \geq 12$ , then minimum value of 't' would be-----.

Select the correct option

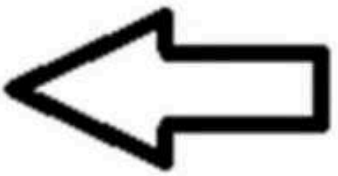
|                                  |   |  |
|----------------------------------|---|--|
| <input checked="" type="radio"/> | 0 |  |
| <input type="radio"/>            | 1 |  |
| <input type="radio"/>            | 3 |  |
| <input type="radio"/>            | 4 |  |



Question # 10 of 40 ( Start time: 04:26:36 PM, 11 July 2020 )

Non-feasible solution----- associated with a given linear programming problem.

Select the correct option

- |                       |                                              |                                                                                    |                    |
|-----------------------|----------------------------------------------|------------------------------------------------------------------------------------|--------------------|
| <input type="radio"/> | satisfy all the constraints                  |  | <b>not confirm</b> |
| <input type="radio"/> | does not satisfy all the constraints         |                                                                                    |                    |
| <input type="radio"/> | satisfy least one of the constraint          |                                                                                    |                    |
| <input type="radio"/> | does not satisfy least one of the constraint |                                                                                    |                    |

Question # 9 of 40 ( Start time: 03:46:47 PM, 11 July 2020 )

Total Marks:

Which of the following is the objective of Project Management by using PERT and CPM methods, for any project subject to resource constraints?

Select the correct option



To minimize the project time



To maximize the total project profit



To minimize the total project cost



To minimize the resource constraints

Question # 8 of 40 ( Start time: 03:46:18 PM, 11 July 2020 )

Which of the following is the major objective of the ABC analysis of inventory?

Select the correct option

- |                                         |                                              |
|-----------------------------------------|----------------------------------------------|
| <input checked="checked" type="radio"/> | To fulfill the demand of items               |
| <input type="radio"/>                   | To replace the required items in proper time |
| <input type="radio"/>                   | To control the inventories                   |

Question # 7 of 40 ( Start time: 03:44:45 PM, 11 July 2020 )

Which of the following would be the objective of the cost per unit of producing certain cameras?

Select the correct option



Maximization



Minimization

Question # 6 of 40 ( Start time: 03:43:30 PM, 11 July 2020 )

Total Marks: 1

While solving a Linear programming problem, we find -----number of basic feasible solution.

Select the correct option



even



odd



infinite



only finite

Question # 5 of 40 ( Start time: 03:42:10 PM, 11 July 2020 )

Total Marks: 1

Which of the following would be the objective of the daily loss of heat in a heating system?

Select the correct option

|                       |              |   |
|-----------------------|--------------|---|
| <input type="radio"/> | Maximization |   |
| <input type="radio"/> | Minimization | ← |
| <input type="radio"/> | Inflection   |   |
| <input type="radio"/> | Average      |   |



Question # 4 of 40 ( Start time: 03:41:06 PM, 11 July 2020 )

Which of the following is not a categories of the operations research (OR) techniques.

Select the correct option

|                                  |                                           |
|----------------------------------|-------------------------------------------|
| <input checked="" type="radio"/> | Linear mathematical programming technique |
| <input type="radio"/>            | Feasible solution                         |

Question # 3 of 40 ( Start time: 03:39:52 PM, 11 July 2020 )

Total Marks: 1

If a manufacturer company produces two types of product say 'A' and 'B' in 'x' and 'y' quantity respectively then which of the following would be the objective if the profit on one unit of 'A' is Rs.6 and on 'B' is Rs.11?

Select the correct option

- |                       |                    |    |
|-----------------------|--------------------|----|
| <input type="radio"/> | Min $z = 6x + 11y$ | // |
| <input type="radio"/> | Max $z = 66xy$     | // |
| <input type="radio"/> | Max $z = 6x + 11y$ | // |
| <input type="radio"/> | Max $z = 6x - 11y$ | // |

Question # 2 of 40 ( Start time: 03:38:25 PM, 11 July 2020 )

Total

Project Management, Critical Path method is based on ----- times.

ni smj lgi is ki

Select the correct option

|                       |               |
|-----------------------|---------------|
| <input type="radio"/> | deterministic |
| <input type="radio"/> | probabilistic |
| <input type="radio"/> | stochastic    |
| <input type="radio"/> | serial        |

Question # 1 of 40 ( Start time: 03:36:46 PM, 11 July 2020 )

Total Marks:

The Mean and Variance of probabilistic nature of activity times in PERT are expressed by -----distribution.

Select the correct option

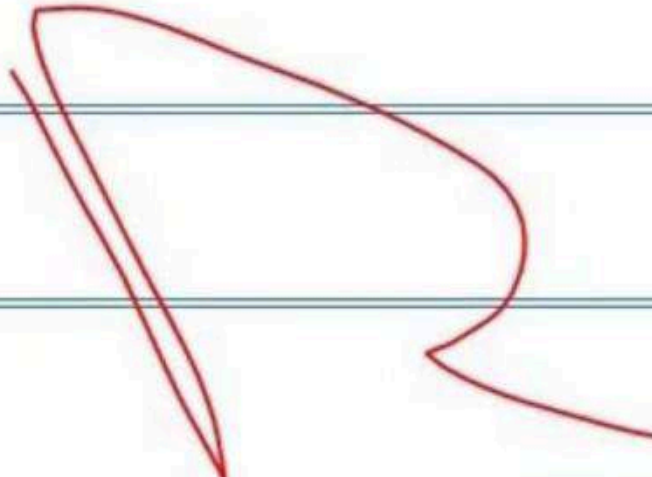
|                                  |           |    |
|----------------------------------|-----------|----|
| <input type="radio"/>            | Bernoulli | // |
| <input checked="" type="radio"/> | Beta      | // |
| <input type="radio"/>            | Chi       | // |
| <input type="radio"/>            | Binomial  | // |

Question # 8 of 40 ( Start time: 12:35:06 PM, 11 July 2020 )

Which of the following correspond to the practical limitations of available resources while modeling a real life situation into a linear program?

Select the correct option

|                                  |                    |
|----------------------------------|--------------------|
| <input type="radio"/>            | Objective function |
| <input checked="" type="radio"/> | Constraints        |
| <input type="radio"/>            | Decision variables |
| <input type="radio"/>            | Linear function    |



Question # 15 of 40 ( Start time: 12:38:16 PM, 11 July 2020 )

Duration of dummy activity is \_\_\_\_\_.

Select the correct option

- ☐ latest start time
- ☒ zero
- ☐ early start time
- ☐ 1

Question # 5 of 40 ( Start time: 12:33:49 PM, 11 July 2020 )

Excess of inventories are undesirable due to the following reasons;

Question # 4 of 40 ( Start time: 12:35:25 PM, 11 July 2020 )

In the Wilson's Model, an optimum order quantity for one period is found when item holding cost = \_\_\_\_\_.

Select the correct option

- ☐ shortage cost
- ☒ purchase cost
- ☐ set up cost
- ☐ All are equivalent

Question # 9 of 40 ( Start time: 12:35:34 PM, 11 July 2020 )

Which of the following would be the objective of the daily loss of heat in a heating system?

Activate Windows  
Go to Settings to activate Windows.



Average

Question # 13 of 40 ( Start time: 12:37:21 PM, 11 July 2020 )

In a development project, if an activity (i,j) of duration three days finish earlier on the tenth day then which of the following will be its early start day?

Select the correct option



Seventh day



Third day



First day



Fifth day

Question # 6 of 40 ( Start time: 12:34:16 PM, 11 July 2020 )

Activate Windows  
Go to Settings to activate Windows.

Question # 5 of 40 ( Start time: 12:33:49 PM, 11 July 2020 )

Excess of inventories are undesirable due to the following reasons;

Select the correct option

- ☐ It locks up the capital of the enterprise.
- ☐ It includes the expenses of stores, equipment, personnel, insurance etc..
- ☐ It may not lead to high volume of output instead it might hamper the production.
- ☒ All above reasons are equivalent

Question # 13 of 40 ( Start time: 12:37:21 PM, 11 July 2020 )

In development project, if activity (2) of duration three days finish earlier on the tenth day then which of the following will be its earliest start day?

Question # 9 of 40 ( Start time: 12:35:34 PM, 11 July 2020 )

Which of the following would be the objective of the daily loss of heat in a heating system?

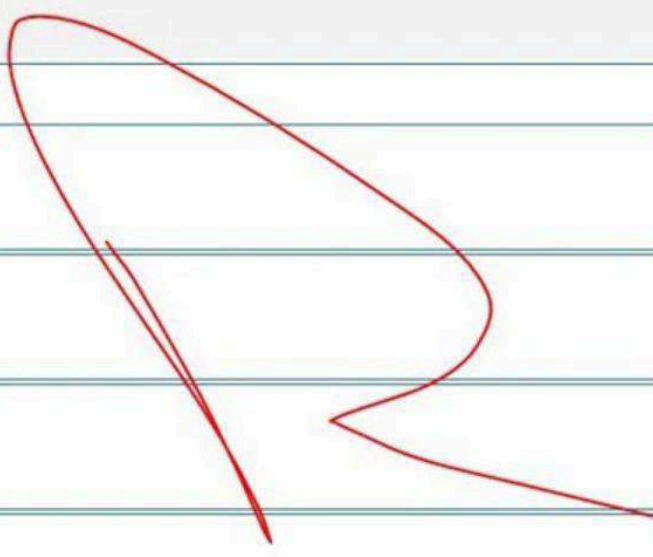
Select the correct option

- ☐ Maximization
- ☒ Minimization
- ☐ Inflection
- ☐ Average

Question # 10 of 40 ( Start time: 12:36:03 PM, 11 July 2020 )

Which of the following would be the objective of the revenue in a production of cell phones?

Select the correct option

- |                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | Inflection   |
| <input type="radio"/>            | Average      |
| <input checked="" type="radio"/> | Maximization |
| <input type="radio"/>            | Minimization |
- 

Question # 11 of 40 ( Start time: 12:36:27 PM, 11 July 2020 )

Question # 7 of 40 ( Start time: 12:34:42 PM, 11 July 2020 )

Activate Windows  
Go to Settings to activate Windows.

Question # 11 of 40 ( Start time: 12:36:27 PM, 11 July 2020 )

The amount of the inventory will always taken to be -----

Select the correct option

|                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | Negative |
| <input checked="" type="radio"/> | Positive |

Question # 2 of 40 ( Start time: 12:32:18 PM, 11 July 2020 )

Question # 7 of 40 ( Start time: 12:34:42 PM, 11 July 2020 )

Which of the following phases are performed to find the critical path in CPM?

Select the correct option

|                                  |                                                     |
|----------------------------------|-----------------------------------------------------|
| <input checked="" type="radio"/> | Forward and Backward(Early and Late finish program) |
| <input type="radio"/>            | Program Evaluation and Review                       |
| <input type="radio"/>            | Resource Scheduling and Allocation                  |
| <input type="radio"/>            | Precedence and succession                           |

Activate Windows  
Go to Settings to activate Windows.

Question # 1 of 30 ( Start time: 01:06:58 AM, 31 December 2020 )

Total Marks:

By putting the Replacement Rate ' $R = \infty$ ' and Shortage Cost ' $C_s = \infty$ ' in the Economic Order Quantity (EOQ) of Manufacturing Model with allowed Shortage, then the result is -----

Select the correct option

|                                  |                                               |    |
|----------------------------------|-----------------------------------------------|----|
| <input checked="" type="radio"/> | EOQ of Wilson's Model                         | // |
| <input type="radio"/>            | EOQ of Manufacturing Model without Shortage   | // |
| <input type="radio"/>            | EOQ of Purchasing Model with allowed Shortage | // |
| <input type="radio"/>            | Invariant                                     | // |



Question # 2 of 30 ( Start time: 01:07:22 AM, 31 December 2020 )

The solution set of the constraint:  $2x+3y>12$ , will lie in ----- bisected by the line:  $2x+3y=12$ .

Select the correct option

- |                                  |                                           |
|----------------------------------|-------------------------------------------|
| <input type="radio"/>            | origin side of half plane                 |
| <input checked="" type="radio"/> | opposite to the origin side in half plane |
| <input type="radio"/>            | both half planes                          |
| <input type="radio"/>            | first quadrant only                       |



MTH601:Grand Quiz

Quiz Start Time: 01:06 AM, 31 December 2020

Question # 3 of 30 ( Start time: 01:07:39 AM, 31 December 2020 )

Total Marks:

If the variance of the project length = 25, expected and scheduled times are 40 and 36 respectively, then the Standard Normal Variable of the given project is-----.

Select the correct option

- |                                  |       |    |
|----------------------------------|-------|----|
| <input type="radio"/>            | 0.80  | // |
| <input checked="" type="radio"/> | -0.80 | // |
| <input type="radio"/>            | 0.16  | // |
| <input type="radio"/>            | -0.16 | // |



Question # 4 of 30 ( Start time: 01:08:01 AM, 31 December 2020 )

Under which of the following condition, a mathematical Program would be non-linear?

Select the correct option



If least Objective function is non-linear



If both Objective function and constraints are non-linear

Which of the following would be the objective of the cost per unit of producing certain cameras?

Select the correct option

|                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | Maximization |
| <input checked="" type="radio"/> | Minimization |
| <input type="radio"/>            | Inflection   |
| <input type="radio"/>            | Average      |

In a development project, if an activity (i,j) of six days duration, starts late on 3rd day then which of the following will be its latest finish time?

Select the correct option

|                                  |          |
|----------------------------------|----------|
| <input checked="" type="radio"/> | 9th day  |
| <input type="radio"/>            | 2nd day  |
| <input type="radio"/>            | 3rd day  |
| <input type="radio"/>            | 18th day |



Solution of a Linear Programming problem is found by ----- methods.

Select the correct option

|                                  |                |
|----------------------------------|----------------|
| <input type="radio"/>            | Analytical     |
| <input type="radio"/>            | Probabilistic  |
| <input checked="" type="radio"/> | Iterative      |
| <input type="radio"/>            | Pure algebraic |

MTH601:Grand Quiz

Quiz Start Time: 01:06 AM, 31 December 2020

Question # 8 of 30 ( Start time: 01:08:57 AM, 31 December 2020 )

Total Marks:

In a development project, if an activity (q,r) of five days duration, starts late on 2nd day then which of the following will be its latest finish time?

Select the correct option

- |                                  |          |    |
|----------------------------------|----------|----|
| <input type="radio"/>            | 3rd day  | // |
| <input type="radio"/>            | 4th day  | // |
| <input checked="" type="radio"/> | 7th day  | // |
| <input type="radio"/>            | 10th day | // |





MTH601:Grand Quiz

Quiz Start Time: 01:06 AM, 31 December 2020

Question # 9 of 30 ( Start time: 01:09:13 AM, 31 December 2020 )

Total Marks: 1

Objective of inventory control is to balance between the advantage of having inventories and the ----- of carrying them to arrive at an optimal level of inventories to minimize the total inventory cost.

Select the correct option

- |                                  |                |    |
|----------------------------------|----------------|----|
| <input type="radio"/>            | demand         | // |
| <input type="radio"/>            | profit         | // |
| <input checked="" type="radio"/> | cost           | // |
| <input type="radio"/>            | transportation | // |



Question # 9 of 30 ( Start time: 01:09:13 AM, 31 December 2020 )

Total Marks: 1

Objective of inventory control is to balance between the advantage of having inventories and the ----- of carrying them to arrive at an optimal level of inventories to minimize the total inventory cost.

Select the correct option

|                                  |                |    |
|----------------------------------|----------------|----|
| <input type="radio"/>            | demand         | // |
| <input type="radio"/>            | profit         | // |
| <input checked="" type="radio"/> | cost           | // |
| <input type="radio"/>            | transportation | // |

Question # 10 of 30 ( Start time: 01:09:26 AM, 31 December 2020 )

Total Marks

If the annual demand of a product is 10000 items, and its set up and inventory costs are 200 and 100 respectively then its economic order quantity is -----provided that shortage is fulfilled INSTANTANEOUSLY.

Select the correct option

- |                                  |             |
|----------------------------------|-------------|
| <input checked="" type="radio"/> | 200 units   |
| <input type="radio"/>            | 100 units   |
| <input type="radio"/>            | 20000 units |
| <input type="radio"/>            | 40000 units |



Question # 11 of 30 ( Start time: 01:09:40 AM, 31 December 2020 )

The task which is executed by the usage of resources and time is called\_\_\_\_\_. Select correct option.

Select the correct option

|                       |         |
|-----------------------|---------|
| <input type="radio"/> | node    |
| <input type="radio"/> | event   |
| <input type="radio"/> | project |

Question # 11 of 30 ( Start time: 01:09:40 AM, 31 December 2020 )

The task which is executed by the usage of resources and time is called \_\_\_\_\_. Select correct option.

Select the correct option

|                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | node     |
| <input type="radio"/>            | event    |
| <input type="radio"/>            | project  |
| <input checked="" type="radio"/> | activity |



Question # 12 of 30 ( Start time: 01:09:53 AM, 31 December 2020 )

Which of the following is first step to model a linear programming problem?

Select the correct option

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| <input type="radio"/>            | Identifying the objective function         |
| <input type="radio"/>            | Identifying the non negative constraints   |
| <input checked="" type="radio"/> | Identifying the unknown decision variables |
| <input type="radio"/>            | Identifying all the restrictions           |



For a LP problem say,  $\text{Max } z = x + y$ , under the constraints  $x, y \geq 0$ , the feasible region would be-----.

Select the correct option

- |                                  |                        |
|----------------------------------|------------------------|
| <input type="radio"/>            | empty                  |
| <input type="radio"/>            | all xy-plane           |
| <input checked="" type="radio"/> | all the first quadrant |
| <input type="radio"/>            | point(0,0)             |

Question # 14 of 30 ( Start time: 01:10:26 AM, 31 December 2020 )

Total Marks:

While applying Simplex method to a LP of minimization type, we proceed stepwise from one -----solution to another in such a way that the objective function always increases its value.

Select the correct option

|                                  |                    |    |
|----------------------------------|--------------------|----|
| <input type="radio"/>            | optimal            | // |
| <input type="radio"/>            | non-basic feasible | // |
| <input checked="" type="radio"/> | basic feasible     | // |
| <input type="radio"/>            | degenerate         | // |

In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 120 respectively then which of the following is true about the 2nd month's requirement?

Select the correct option

- |                                  |                                                           |   |
|----------------------------------|-----------------------------------------------------------|---|
| <input type="radio"/>            | 2nd month demand can be included in 1st month             | / |
| <input checked="" type="radio"/> | 2nd month demand will have to fulfill in 2nd month        | / |
| <input type="radio"/>            | 2nd month demand can be included in 3rd month             | / |
| <input type="radio"/>            | 2nd month demand can be included in any month of the year | / |

A linear programming problem can be converted into a Transportation problem by restricting the coefficients ' $a_{ij}$ ' to have the values-----

Select the correct option

|                                  |                   |
|----------------------------------|-------------------|
| <input type="radio"/>            | Zero only         |
| <input type="radio"/>            | One only          |
| <input checked="" type="radio"/> | either '0' or '1' |
| <input type="radio"/>            | arbitrarily       |



By which of the following method, any complex linear programming problem can be handled?

Select the correct option

|                                  |                   |
|----------------------------------|-------------------|
| <input type="radio"/>            | Graphical method  |
| <input checked="" type="radio"/> | Simplex method    |
| <input type="radio"/>            | Dual method       |
| <input type="radio"/>            | Degenerate method |

Question # 19 of 30 ( Start time: 01:11:48 AM, 31 December 2020 )

Which of the following technique to solve the network flow diagrams, is event oriented?

Select the correct option



Programme Evaluation and Review Technique



Critical Path Method

RIZ MUGHAL



Question # 20 of 30 ( Start time: 01:12:18 AM, 31 December 2020 )

The parameters like demand and ordering quantity cost are quite ----- in the Probabilistic inventory models.

Select the correct option

☐ certain

☒ uncertain

RIZ MUGHAL

Question # 21 of 30 ( Start time: 01:12:34 AM, 31 December 2020 )

Total Marks

Which of the following correspond to the economic or physical quantities which are under the control of operations researcher while modeling a real life situation into a linear program?

Select the correct option



Objective function



Constraints



Decision variables



Linear function

Question # 22 of 30 ( Start time: 01:12:53 AM, 31 December 2020 )

The word 'Linear' in Linear programming problem describes ----- between two or more variables.

Select the correct option

- |                                  |                                   |
|----------------------------------|-----------------------------------|
| <input type="radio"/>            | that a Correlation exists         |
| <input type="radio"/>            | a recurrence relation             |
| <input checked="" type="radio"/> | a direct proportionate relation   |
| <input type="radio"/>            | an inverse proportionate relation |

RIZ MUGHAL

Question # 23 of 30 ( Start time: 01:13:07 AM, 31 December 2020 )

Which of the following method follows iterative procedure to solve a Linear Programming problem?

Select the correct option



Algebraic (Simplex)

Graphical

RIZ MUGHAL



Question # 24 of 30 ( Start time: 01:13:30 AM, 31 December 2020 )

Non-feasible solution----- associated with a given linear programming problem.

Select the correct option

- |                                  |                                              |
|----------------------------------|----------------------------------------------|
| <input type="radio"/>            | satisfy all the constraints                  |
| <input type="radio"/>            | does not satisfy all the constraints         |
| <input type="radio"/>            | satisfy least one of the constraint          |
| <input checked="" type="radio"/> | does not satisfy least one of the constraint |





Question # 25 of 30 ( Start time: 01:13:46 AM, 31 December 2020 )

In the graph of the Purchasing Model with shortages, the area under the horizontal axis represents -----.

Select the correct option



carrying cost of item



set up cost of item



shortage cost of item

RIZ MUGHAL

Question # 26 of 30 ( Start time: 01:14:10 AM, 31 December 2020 )

Total Mark

In a network flow diagram, two jobs (m,n) and (m,o) of '11' and '8' days duration leaves the node 'm' then which of the following will be Late start time for 'm', if it is provided that both (m,n) and (m,o) finish late on 14th and 10th day respectively?

Select the correct option

- |                                  |         |
|----------------------------------|---------|
| <input type="radio"/>            | 6th day |
| <input checked="" type="radio"/> | 3rd day |
| <input type="radio"/>            | 2nd day |
| <input type="radio"/>            | 1st day |

Question # 27 of 30 ( Start time: 01:14:26 AM, 31 December 2020 )

Total Marks: 1

Which of the following distribution gives the probability of completing the assigned project by considering the expected time of any activity as random variable?

Select the correct option

|                                  |          |  |
|----------------------------------|----------|--|
| <input type="radio"/>            | Binomial |  |
| <input type="radio"/>            | Poisson  |  |
| <input type="radio"/>            | Beta     |  |
| <input checked="" type="radio"/> | Normal   |  |

Delay in critical activities for a project management lead to \_\_\_\_\_.

Select the correct option

- |                                     |                                         |
|-------------------------------------|-----------------------------------------|
| <input type="radio"/>               | project finish time will have to extend |
| <input type="radio"/>               | more resources have to employed         |
| <input type="radio"/>               | project cost will increase              |
| <input checked="" type="checkbox"/> | All above choices are equivalent        |



Question # 30 of 30 ( Start time: 01:15:16 AM, 31 December 2020 )

Total Marks:

In a network flow diagram, if two jobs 'a(l,n)' and 'b(m,n)' of '7' and '8' days durations respectively, start earlier simultaneously on 4th day, then the next activity containing 'n' as head event can't start until the entering activity --

Select the correct option

|                                  |       |
|----------------------------------|-------|
| <input checked="" type="radio"/> | (m,n) |
| <input type="radio"/>            | (l,n) |
| <input type="radio"/>            | (m,l) |
| <input type="radio"/>            | (l,m) |

MTH601 Quiz solved with 100%  
Correct answers

MTH601 Quiz

Solved by  
Riz Mughal

⇒ 5 videos



Question # 18 of 30 ( Start time: 01:11:37 AM, 31 December 2020 )

Linear Programming is the backbone of Operations Research models due to-----

Select the correct option

- |                                  |                                                 |
|----------------------------------|-------------------------------------------------|
| <input type="radio"/>            | linearity of objective function and constraints |
| <input checked="" type="radio"/> | its fantastic computational efficiency          |
| <input type="radio"/>            | non-negativity of decision variables            |
| <input type="radio"/>            | unrestricted sign of decision variables         |

Question # 2 of 40 ( Start time: 12:32:18 PM, 11 July 2020 )

Convex set is the collection of all -----.

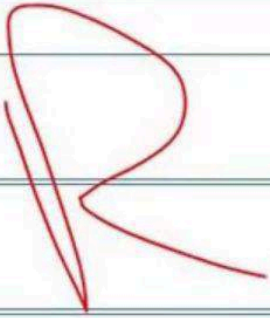
Select the correct option

- |                                  |                        |
|----------------------------------|------------------------|
| <input type="radio"/>            | optimal solutions      |
| <input checked="" type="radio"/> | feasible solutions     |
| <input type="radio"/>            | non-feasible solutions |
| <input type="radio"/>            | negative solutions     |

Question # 6 of 40 ( Start time: 12:34:16 PM, 11 July 2020 )

For a development project, if its Standard Normal variable = 1.38, expected and the scheduled durations of the project are 100 and 110 days respectively, then Variance in the project length is .....

Select the correct option

- |                                  |        |
|----------------------------------|--------|
| <input type="radio"/>            | -7.246 |
| <input type="radio"/>            | -52.50 |
| <input checked="" type="radio"/> | 52.50  |
| <input type="radio"/>            | 7.246  |
- 

Question # 10 of 40 ( Start time: 12:36:03 PM, 11 July 2020 )

Which of the following would be the objective of the revenue in a production of cell phones?

Activate Windows  
Go to Settings to activate Windows.



Question # 14 of 40 ( Start time: 12:37:50 PM, 11 July 2020 )

Total Marks

In a network flow diagram, if two jobs 'a(l,n)' and 'b(m,n)' of '7' and '8' days durations respectively, start earlier simultaneously on 4th day, then the next activity containing 'n' as head event can't start until the entering activity ----- is completed.

Select the correct option

- |                                  |       |  |  |
|----------------------------------|-------|--|--|
| <input checked="" type="radio"/> | (m,n) |  |  |
| <input type="radio"/>            | (l,n) |  |  |
| <input type="radio"/>            | (m,l) |  |  |
| <input type="radio"/>            | (l,m) |  |  |

Question # 16 of 40 ( Start time: 12:38:37 PM, 11 July 2020 )

Total Mark

For any activity (i,j), if . a) Earliest start time of 'i' = Latest finish time of 'i'. b) Earliest start time of 'j' = Latest finish time of 'j'. c) difference of Earliest start times of events 'i' and 'j' = difference of Latest finish times of events 'i' and 'j' time to complete the job, then the activity (i,j) is said to be -----.

Select the correct option

|                                  |                  |  |
|----------------------------------|------------------|--|
| <input type="radio"/>            | dummy            |  |
| <input checked="" type="radio"/> | critical         |  |
| <input type="radio"/>            | non-Critical     |  |
| <input type="radio"/>            | non of the above |  |

Question # 14 of 40 ( Start time: 12:37:50 PM, 11 July 2020 )

Total Mark



Question # 19 of 40 ( Start time: 12:39:58 PM, 11 July 2020 )

In the PERT, the expected time is classified as a----- variable.

Select the correct option

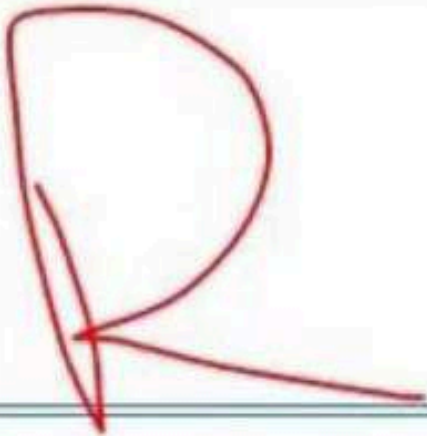
|                                  |                       |
|----------------------------------|-----------------------|
| <input type="radio"/>            | complex valued        |
| <input type="radio"/>            | fix valued (constant) |
| <input checked="" type="radio"/> | random                |
| <input type="radio"/>            | continuous            |

Question # 1 of 40 ( Start time: 12:31:44 PM, 11 July 2020 )

The parameters like demand and ordering quantity cost are quite ----- in the Probabilistic inventory models.

Select the correct option

|                                  |           |
|----------------------------------|-----------|
| <input type="radio"/>            | certain   |
| <input checked="" type="radio"/> | uncertain |



Question # 3 of 40 ( Start time: 12:32:55 PM, 11 July 2020 )

Which of the following property assumed that the coefficients of decision variables in a Linear Programming model are known with certainty?

Select the correct option

|                                  |                 |
|----------------------------------|-----------------|
| <input checked="" type="radio"/> | Deterministic   |
| <input type="radio"/>            | Additivity      |
| <input type="radio"/>            | Proportionality |
| <input type="radio"/>            | Divisibility    |

Question # 20 of 40 ( Start time: 12:40:26 PM, 11 July 2020 )

In PERT, the possible variation in activity times is measured from Standard Deviation which is -----of the difference between Pessimistic and Optimistic times.

Select the correct option

|                                  |            |
|----------------------------------|------------|
| <input checked="" type="radio"/> | one sixth  |
| <input type="radio"/>            | one fourth |
| <input type="radio"/>            | one third  |
| <input type="radio"/>            | one fifth  |

## Question #10 of 10 ( Start time: 11:37:39 PM, 24 December 2021 )

Total Marks: 1

If the cost function in terms of economic order quantity is  $C = Q^2 - 400Q$ , then the minimum cost of inventory per unit of time is ----- units.

Select the correct option

☐

200

☒

800

☐

400

☐

398

[Click to Save Answer & Move to Next Question](#)



in the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 130 respectively then which of the following is true about the 2nd month's requirement?

Select the correct option

- |                       |                                                           |
|-----------------------|-----------------------------------------------------------|
| <input type="radio"/> | 2nd month demand can be included in 1st month             |
| <input type="radio"/> | 2nd month demand can be included in 3rd month             |
| <input type="radio"/> | 2nd month demand will have to fulfill in 2nd month        |
| <input type="radio"/> | 2nd month demand can be included in any month of the year |

Click to Save Answer & Move to Next Question

MC200400721: MUFASAL HUSSAIN

Time Left 77  
sec(s)

MTH601 - Operations Research (Quiz No1)

Quiz Start Time: 11:27 PM, 24 December 2021

Question # 6 of 10 ( Start time: 11:32:35 PM, 24 December 2021 )

Total Marks: 1

In resource leveling when two or more jobs compete for same resources, first try to allocate an activity, which is of ----- duration and next to the activity which having next ----- duration.

Select the correct option

☐

maximum, minimum

☐

optimistic, most likely

☒

minimum, maximum

☐

most likely, pessimistic

Click to Save Answer & Move to Next Question

Question # 7 of 10 ( Start time: 11:33:34 PM, 24 December 2021 )

Total Marks: 1

Which of the following will be annual optimum cost of a product if the number of annual orders WITHOUT any shortage is 25 and the total cost of one cycle is 10,000?

Select the correct option

- ☒ Rs. 10025
- ☐ Rs. 9975
- ☐ Rs. 250000
- ☐ Rs. 400

Click to Save Answer & Move to Next Question

Question # 8 of 10 ( Start time: 11:34:55 PM, 24 December 2021 )

Total Marks: 1

In the Wilson's Model, an optimum order quantity for one period is found when item holding cost = -----.

Select the correct option

☐

shortage cost

☒

purchase cost

☐

set up cost

☐

All are equivalent

Click to Save Answer &amp; Move to Next Question

Question # 5 of 10 ( Start time: 11:31:37 PM, 24 December 2021 )

Total Marks: 1

In the context of Wilson's model at time  $t = 0$ , the quantity ' $Q$ ' of the product is----- in the inventory.

Select the correct option

|                                  |         |
|----------------------------------|---------|
| <input type="radio"/>            | Average |
| <input type="radio"/>            | Zero    |
| <input checked="" type="radio"/> | Maximum |
| <input type="radio"/>            | Minimum |

Click to Save Answer & Move to Next Question



Question # 4 of 10 ( Start time: 11:30:44 PM, 24 December 2021 )

Total Marks: 1

In the Purchasing Model with Shortages the Cost Function ' $C(S,Q)$ ' can also be expressed as function of -----.

Select the correct option

|                                  |         |
|----------------------------------|---------|
| <input type="radio"/>            | $(Q,D)$ |
| <input type="radio"/>            | $(S,t)$ |
| <input type="radio"/>            | $(t,D)$ |
| <input checked="" type="radio"/> | $(S,D)$ |

Click to Save Answer & Move to Next Question

Question # 3 of 10 ( Start time: 11:30:01 PM, 24 December 2021 )

Total Marks: 1

The parameters like demand and ordering quantity cost are quite ----- in the Probabilistic inventory models.

Select the correct option

certain

☐

uncertain

☒


Click to Save Answer & Move to Next Question

Under which of the following condition, the Economic Order Quantity (EOQ) of Manufacturing Model with allowed Shortage can be transformed into the same Model without Shortage?

Select the correct option

- |                                  |                                |
|----------------------------------|--------------------------------|
| <input checked="" type="radio"/> | When Shortage Cost is zero     |
| <input type="radio"/>            | When Setup Cost is zero        |
| <input type="radio"/>            | When Setup Cost is infinity    |
| <input type="radio"/>            | When Shortage Cost is infinity |



Click to Save Answer & Move to Next Question

Question # 2 of 10 ( Start time: 11:28:59 PM, 24 December 2021 )

Total Marks: 1

If the economic order quantity and the annual demand of a product are 2500 and 5000 units respectively, then WITHOUT any shortage the time between the consecutive orders is ----- months.

Select the correct option

|                                  |      |
|----------------------------------|------|
| <input type="radio"/>            | 1.50 |
| <input checked="" type="radio"/> | 0.50 |
| <input type="radio"/>            | 0.2  |
| <input type="radio"/>            | 2.50 |

Click to Save Answer & Move to Next Question

MTH601:Quiz

Quiz Start Time

Question # 1 of 10 ( Start time: 04:38:45 PM, 05 June 2021 )

Which of the following would be the objective of the revenue in a production of cell phones?

Select the correct option

- |                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | Average      |
| <input type="radio"/>            | Minimization |
| <input checked="" type="radio"/> | Maximization |
| <input type="radio"/>            | Inflation    |

Riz Mughal

R



Question # 2 of 10 ( Start time: 04:39:12 PM, 05 June 2021 )

Modeling of a Linear Programming problem from a real life world is a/an ----- .

Select the correct option



technology



science



engineering



art

R

Question # 3 of 10 ( Start time: 04:39:35 PM, 05 June 2021 )

To

For ABC analysis, in the Random Sample of some items, the ranks to the items are assigned with respect to their -----

Select the correct option

- |                                  |                            |
|----------------------------------|----------------------------|
| <input type="radio"/>            | largest annual unit cost   |
| <input type="radio"/>            | lowest annual unit cost    |
| <input checked="" type="radio"/> | largest annual usage value |
| <input type="radio"/>            | lowest annual usage value  |
- 

Question # 4 of 10 ( Start time: 04:39:56 PM, 05 June 2021 )

In the Purchasing Model with Shortages the Cost Function ' $C(S,Q)$ ' can also be expressed as function of \_\_\_\_\_

Select the correct option

- |                                  |       |
|----------------------------------|-------|
| <input type="radio"/>            | (L,D) |
| <input type="radio"/>            | (S,D) |
| <input checked="" type="radio"/> | (S,I) |
| <input type="radio"/>            | (Q,D) |
- 

MTH601:Quiz

Quiz Start Time: 04:38 PM, 05 June

Question # 5 of 10 ( Start time: 04:40:15 PM, 05 June 2021 )

Total M

If the economic order quantity and the annual demand of a product are 2500 and 5000 units respectively, then WITHOUT any shortage the time between the consecutive orders is ——— months.

Select the correct option

|                                  |      |
|----------------------------------|------|
| <input type="radio"/>            | 02   |
| <input checked="" type="radio"/> | 0.50 |
| <input type="radio"/>            | 2.50 |
| <input type="radio"/>            | 1.50 |

MTH601:Quiz

Quiz Start Time: 04:38 F

Question # 6 of 10 ( Start time: 04:40:37 PM, 05 June 2021 )

In an optimization problem, we seek for maximizing or minimizing a specific quantity, called the ———.

Select the correct option

- ☐ feasible solution
- ☐ constraints
- ☐ decision variables
- ☒ objective function

18

## Question # 7 of 10 ( Start time: 04:41:08 PM, 05 June 2021 )

Which of the following is true about the Manufacturing model without Shortage?

Select the correct option

- |                                  |                                |
|----------------------------------|--------------------------------|
| <input checked="" type="radio"/> | Demand rate < Replacement rate |
| <input type="radio"/>            | Demand rate > Replacement rate |
| <input type="radio"/>            | Demand rate = Replacement rate |
| <input type="radio"/>            | None of the above choices      |
- 



Question # 8 of 10 ( Start time: 04:41:32 PM, 05 June 2021 )

www.BANDICAM.com

Total Marks: 1

In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 120 respectively then which of the following is true about the 2nd month's requirement?

Select the correct option

- |                                  |                                                           |
|----------------------------------|-----------------------------------------------------------|
| <input checked="" type="radio"/> | 2nd month demand will have to fulfill in 2nd month        |
| <input type="radio"/>            | 2nd month demand can be included in 3rd month             |
| <input type="radio"/>            | 2nd month demand can be included in 1st month             |
| <input type="radio"/>            | 2nd month demand can be included in any month of the year |
- 

Question # 9 of 10 ( Start [www.BANDICAM.com](http://www.BANDICAM.com))

If we increase the inventory, then the shortage cost will \_\_\_\_\_.

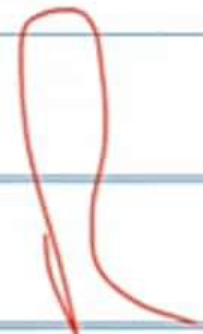
Select the correct option

|                                  |                 |
|----------------------------------|-----------------|
| <input type="radio"/>            | uncertain       |
| <input checked="" type="radio"/> | decrease        |
| <input type="radio"/>            | increase        |
| <input type="radio"/>            | remain constant |

Question # 1 of 10 ( Start time: 04:44:11 PM, 05 June 2021 )

In the ABC analysis, the items are classified into three categories with respect their———.

Select the correct option

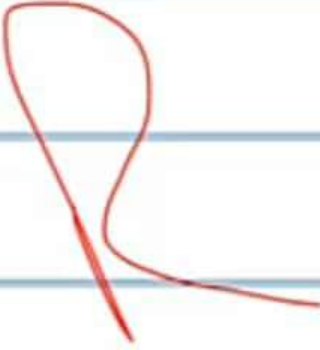
- |                                  |                 |
|----------------------------------|-----------------|
| <input checked="" type="radio"/> | turn over value |
| <input type="radio"/>            | cost value      |
| <input type="radio"/>            | demand value    |
| <input type="radio"/>            | mark up value   |
- 

Question # 2 of 10 ( Start time: 04:44:39 PM, 05 June 2021 )

In an optimization problem, we seek for maximizing or minimizing a specific quantity, called the ———.

Select the correct option

|                                  |                    |
|----------------------------------|--------------------|
| <input checked="" type="radio"/> | objective function |
| <input type="radio"/>            | feasible solution  |
| <input type="radio"/>            | constraints        |
| <input type="radio"/>            | decision variables |



MTH601:Quiz

Quiz Start Time: 04:38 PM, 05 June :  
04:38 PM, 05 June 2021

Question # 10 of 10 ( Start time: 04:42:14 PM, 05 June 2021 )

Total Mar

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " $C_2/q$ " which is ———proportional to the quantity ordered.

Select the correct option

☒ inversely☐ directly2<sup>nd</sup> account

MTH601:Quiz

Quiz Start Time: 04:44 PM,

Question # 3 of 10 ( Start time: 04:44:54 PM, 05 June 2021 )

For ABC analysis, in the Random Sample of some items, the ranks to the items are assigned with respect to their ———

Select the correct option

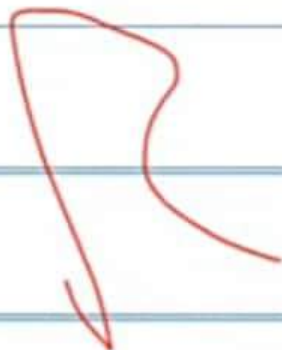
- |                                  |                            |
|----------------------------------|----------------------------|
| <input type="radio"/>            | lowest annual unit cost    |
| <input checked="" type="radio"/> | largest annual usage value |
| <input type="radio"/>            | largest annual unit cost   |
| <input type="radio"/>            | lowest annual usage value  |



Question # 4 of 10 ( Start time: 04:45:14 PM, 05 June 2021 )

Which of the following would be the objective of the revenue in a production of cell phones?

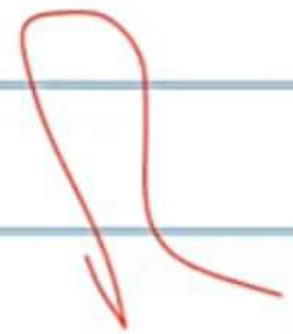
Select the correct option

- |                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | Average      |
| <input checked="" type="radio"/> | Maximization |
| <input type="radio"/>            | Inflection   |
| <input type="radio"/>            | Minimization |
- 

Question # 5 of 10 ( Start time: 04:45:33 PM, 05 June 2021 )

Which of the following is true about the Manufacturing model without Shortage?

Select the correct option

- |                                  |                                |
|----------------------------------|--------------------------------|
| <input type="radio"/>            | Demand rate = Replacement rate |
| <input type="radio"/>            | None of the above choices      |
| <input type="radio"/>            | Demand rate > Replacement rate |
| <input checked="" type="radio"/> | Demand rate < Replacement rate |
- 

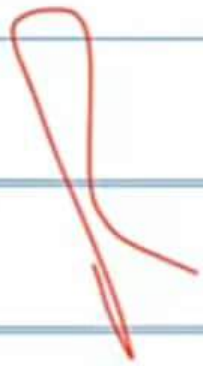
MTH601:Quiz

Quiz Start Time

Question # 6 of 10 ( Start time: 04:45:52 PM, 05 June 2021 )

Linear Programming is the backbone of Operations Research models due to——.

Select the correct option

- |                                  |                                                 |
|----------------------------------|-------------------------------------------------|
| <input checked="" type="radio"/> | its fantastic computational efficiency          |
| <input type="radio"/>            | non-negativity of decision variables            |
| <input type="radio"/>            | unrestricted signs of decision variables        |
| <input type="radio"/>            | linearity of objective function and constraints |
- 

MTH601:Quiz

Quiz Start Time: 04:44 PM, 05 Jun

Question # 7 of 10 ( Start time: 04:46:21 PM, 05 June 2021 )

Total A

Holding Cost is ————— proportional to the quantity to be kept in stock and the time to which the inventory is maintained.

Select the correct option

|                                  |           |
|----------------------------------|-----------|
| <input checked="" type="radio"/> | directly  |
| <input type="radio"/>            | inversely |



Question # 8 of 10 ( Start time: 04:46:37 PM, 05 June 2021 )

Total Marks:

Which of the following correspond to the practical limitations of available resources while modeling a real life situation into a linear program?

Select the correct option

- |                                  |                    |
|----------------------------------|--------------------|
| <input type="radio"/>            | Decision variables |
| <input checked="" type="radio"/> | Constraints        |
| <input type="radio"/>            | Linear function    |
| <input type="radio"/>            | Objective function |

Question # 9 of 10 ( Start time: 04:46:56 PM, 05 June 2021 )

For an industrialist, optimization means ———.

Select the correct option

- |                                  |                                                                |
|----------------------------------|----------------------------------------------------------------|
| <input type="radio"/>            | change the business strategy and looking forward new customers |
| <input type="radio"/>            | maximizing its profit only                                     |
| <input type="radio"/>            | minimizing its cost only                                       |
| <input checked="" type="radio"/> | both maximizing its profit and minimizing its cost             |
- 



Question # 10 of 10 ( Start time: 04:47:18 PM, 05 June 2021 )

Total Marks: -

Which of the following will be annual optimum cost of a product if the number of annual orders WITHOUT any shortage is 25 and the total cost of one cycle is 10,000?

Select the correct option

- |                                  |            |
|----------------------------------|------------|
| <input type="radio"/>            | Rs. 10025  |
| <input type="radio"/>            | Rs. 9975   |
| <input type="radio"/>            | Rs. 400    |
| <input checked="" type="radio"/> | Rs. 250000 |

3<sup>rd</sup> account

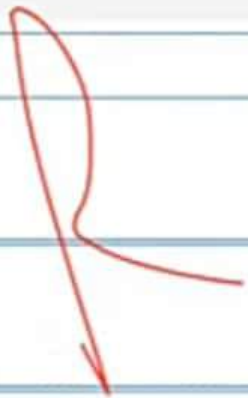
MTH601:Quiz

Quiz Start Time:

Question # 1 of 10 ( Start time: 04:50:37 PM, 05 June 2021 )

Which of the following is the major objective of the ABC analysis of inventory?

Select the correct option

- |                                  |                                               |
|----------------------------------|-----------------------------------------------|
| <input type="radio"/>            | To fulfill the demand of items                |
| <input checked="" type="radio"/> | To control the inventories                    |
| <input type="radio"/>            | To classify the items in different categories |
| <input type="radio"/>            | To replace the required items in proper time  |
- 

MTH601:Quiz

Quiz Start Time: 04:50 PM, 05 June :

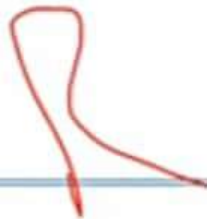
Question # 2 of 10 ( Start time: 04:51:00 PM, 05 June 2021 )

Total Mar

Holding Cost is ————— proportional to the quantity to be kept in stock and the time to which the inventory is maintained.

Select the correct option

|                                  |           |
|----------------------------------|-----------|
| <input checked="" type="radio"/> | inversely |
| <input type="radio"/>            | directly  |



Question # 3 of 10 ( Start time: 04:52:19 PM, 05 June 2021 )

Total

If the purchases are made in large quantities then the Economic Order Quantity model may fail because —————.

Select the correct option

- |                                  |                                |
|----------------------------------|--------------------------------|
| <input type="radio"/>            | the Holding Cost will decrease |
| <input type="radio"/>            | the Setup Cost will decrease   |
| <input type="radio"/>            | the Setup Cost will increase   |
| <input checked="" type="radio"/> | the Holding Cost will increase |
- 

Question # 4 of 10 ( Start time: 04:52:40 PM, 05 June 2021 )

Total Marks: -

The Economic Order Quantity in case of Manufacturing Model without Shortage can be transformed into Purchasing Model without Shortage by which of the following substitution?

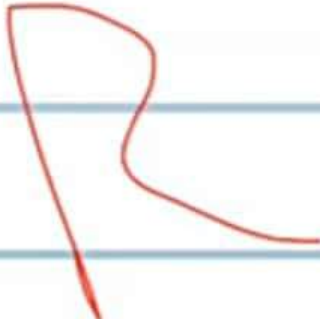
Select the correct option

- |                                  |                                               |
|----------------------------------|-----------------------------------------------|
| <input type="radio"/>            | When Replacement rate 'R' approaches zero     |
| <input type="radio"/>            | When Replacement rate 'R' approaches infinity |
| <input type="radio"/>            | When Demand rate approaches zero              |
| <input checked="" type="radio"/> | When Demand rate approaches infinity          |
- 

## Question # 5 of 10 ( Start time: 04:52:59 PM, 05 June 2021 )

In an optimization problem, we seek for maximizing or minimizing a specific quantity, called the \_\_\_\_\_.

Select the correct option

- |                                  |                    |
|----------------------------------|--------------------|
| <input type="radio"/>            | constraints        |
| <input type="radio"/>            | feasible solution  |
| <input checked="" type="radio"/> | objective function |
| <input type="radio"/>            | decision variables |
- 



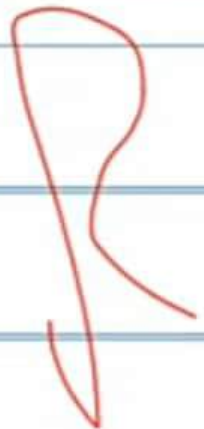
MTH601 Quiz

Quiz Start Time: 04:5

Question # 6 of 10 ( Start time: 04:53:18 PM, 05 June 2021 )

While solving a problem by any of Linear Programming method, an objective function is to be \_\_\_\_\_.

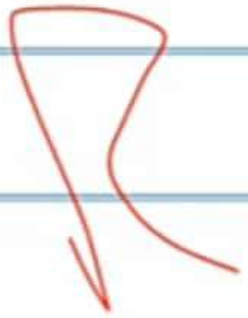
Select the correct option

- |                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | maximized    |
| <input type="radio"/>            | standardized |
| <input checked="" type="radio"/> | optimized    |
| <input type="radio"/>            | minimized    |
- 

Question # 7 of 10 ( Start time: 04:53:35 PM, 05 June 2021 )

Which of the following would be the objective of the cost per unit of producing certain cameras?

Select the correct option

- |                                  |              |
|----------------------------------|--------------|
| <input checked="" type="radio"/> | Minimization |
| <input type="radio"/>            | Maximization |
| <input type="radio"/>            | Infection    |
| <input type="radio"/>            | Average      |
- 

Question # 8 of 10 ( Start time: 04:53:52 PM, 05 June 2021 )

Total Marks: 1

By putting the Replacement Rate ' $R?$ infinity' and Shortage Cost ' $C4?$ infinity' in the Economic Order Quantity(EOQ) of Manufacturing Model with allowed Shortage, then the result is ———

Select the correct option

- |                                  |                                               |
|----------------------------------|-----------------------------------------------|
| <input type="radio"/>            | EOQ of Manufacturing Model without Shortage   |
| <input type="radio"/>            | EOQ of Purchasing Model with allowed Shortage |
| <input checked="" type="radio"/> | EOQ of Wilson's Model                         |
| <input type="radio"/>            | Invariant                                     |

Question # 9 of 10 ( Start time: 04:54:11 PM, 05 June 2021 )

In the Purchasing Model with Shortages the Cost Function ' $C(S,Q)$ ' can also be expressed as function of \_\_\_\_\_.

Select the correct option

- |                                  |           |
|----------------------------------|-----------|
| <input type="radio"/>            | ( $t,D$ ) |
| <input checked="" type="radio"/> | ( $S,t$ ) |
| <input type="radio"/>            | ( $Q,D$ ) |
| <input type="radio"/>            | ( $S,D$ ) |
- 

Question # 10 of 10 ( Start time: 04:54:29 PM, 05 June 2021 )

Total Marks

Which of the following correspond to the practical limitations of available resources while modeling a real life situation into a linear program?

Select the correct option

- |                                  |                    |
|----------------------------------|--------------------|
| <input checked="" type="radio"/> | Constraints        |
| <input type="radio"/>            | Objective function |
| <input type="radio"/>            | Linear function    |
| <input type="radio"/>            | Decision variables |
- 

# QUIZ NO 1 MTH 601 RESEARCH METHODS MCQ

## Lectures 10-16

1. Which of the following can NOT be considered to include in shortage cost?
  - a. Loss of goodwill of customers due to delay
  - b. Cost of expediting
  - c. **Cost of lost sales (Lecture no 3 Page no 32 Handouts)**
  - d. Overtime expenses
2. In the Wilison's, an optimum order quantity for one period is found when item holding costs = \_\_\_\_\_
  - a. set up costs
  - b. All are equivalent
  - c. **purchase cost (Lecture no 3 Page no 34 Handouts)**
  - d. shortage cost
3. If a firm produces its own goods instead of purchasing the same from an outside source, then the cost of resetting the equipment for production may referred to \_\_\_\_\_
  - a. Shortage Cost
  - b. **Set-up cost (From Google)**
  - c. Inventory holding cost
  - d. Item cost
4. If we increase the inventory, then the shortage cost will \_\_\_\_\_
  - a. remain constant
  - b. increase
  - c. **decrease**
  - d. uncertain
5. If an item is manufactured then the direct or indirect material, labor, and overhead expenses are referred to \_\_\_\_\_
  - a. Shortage Cost
  - b. Set-up cost
  - c. **Item cost (Lecture no 3 Page no 31 Handouts)**
  - d. Inventory holding cost
6. In the Purchasing Model with Shortages the Cost Function ' $C(S,Q)$ ' can also be expressed as function of \_\_\_\_\_
  - a.  $(t,D)$
  - b.  $(S,t)$
  - c.  $(Q,D)$
  - d.  $(S,D)$



7. Objective of inventory control is to balance between the advantage of having inventories and the \_\_\_\_\_ of carrying them to arrive at an optimal level of inventories to minimize the total inventory cost.

- a. profit
- b. demand
- c. cost**
- d. transportation

8. The amount of the inventory will always taken to be \_\_\_\_\_

- a. Positive**
- b. Negative

9. Which is the following reason is essential for an enterprise to have an inventory?

- a. It provides adequate service to the customers**
- b. It reduces the possibility of duplication of orders
- c. It helps in smooth and efficient running of the business
- d. All above reasons are equivalent

10. Which of the following is true about the Manufacturing model without Shortage?

- a. Demand rate = Replacement rate
- b. Demand rate > Replacement rate
- c. Demand rate < Replacement rate (Page no 40 Handouts)**
- d. None of the above

11. If we increase the inventory, then the shortage cost will decrease and the carrying cost will \_\_\_\_

☒ \_remain constant

☐ \_increase (Lecture no 3 Page no 45 Handouts)

☒ \_also decrease

☒ \_uncertain

12. If the economic order quantity and the annual demand of a product are 2500 and 5000 units respectively, then WITHOUT any shortage the number of orders are \_\_\_\_\_

The optimum number of orders per year is determined from  $N^* = D/Q^*$  pg 34

- a. 30
- b. 500
- c. 125
- d. 02 (5000/2500) = 2 Ans**

13. Holding cost is \_\_\_\_\_ proportional to the quantity to be kept in stock and the time to which the inventory is maintained.

a. directly

b. Inversely

**Item holding cost per period =  
(Holding cost)x(average inventory  
per period)x(time per period)**

$$= C_3 \times \frac{Q}{2} \times t$$

14. Which of the following times are directly related to the activity cost that is by reducing the activity duration, the direct cost of the corresponding activity is increased?

**CPM Times (Page no 09 Handouts)**

PERT Times

**Ref: If we employ more persons or skilled people or given overtime to the workers, the activity could be completed in a reduced duration known as crash duration. But this involves an increased cost in the form of additional resources. With CPM the amount needed to complete the various activities is assumed to be known with certainty. So, the direct costs for the activities increase and hence the cost of the project also increases.**

15. Which of the following technique to solve the net work flow diagrams, is activity oriented?

Programme Evaluation and Review Technique

**Critical Path Method (Page no 09 Handouts)**

**Ref:**

**PERT is an event-oriented technique. By 'event' we mean reaching a certain stage of completion of the project.**

**CPM, a network technique but it is concerned with obtaining the trade-off, between cost and completion date for large projects.**

16. Which of the following cost for an item is independent of the quantity ordered or produced?

a. Set-up cost

b. Item cost

c. **Shortage cost (Not sure)**

d. Inventory holding cost

17. Under which of the following condition, the Economic Order Quantity (EOQ) of Manufacturing Model with allowed Shortage can be transformed into the same Model without Shortage?

**When Shortage Cost is zero (page 40)**

When Setup Cost is zero

When Setup Cost is infinity

When Shortage Cost is infinity

18. In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 130 respectively then which of the following is true about the 2nd month's requirement?

**2nd month demand can be included in 1st month**

2nd month demand will have to fulfill in 2nd month

2nd month demand can be included in 3rd month

2nd month demand can be included in any month of the year

19. By putting the replacement rate 'R' infinity' in the Economic Order Quantity (EOQ) of Manufacturing Model with allowed Shortage, then the result is -----.

EOQ of Purchasing Model without Shortage

EOQ of Manufacturing Model without Shortage

**EOQ of Purchasing Model with allowed Shortage**

Invariant

20. The Economic Order Quantity in case of Manufacturing Model without Shortage can be transformed into Purchasing Model without Shortage by which of the following substitution?

When Replacement rate 'R' approaches zero

**When Replacement rate 'R' approaches infinity**

When Demand rate approaches zero

When Demand rate approaches infinity

21. If the total inventory in one cycle of 't' units of time is ' $(1/2)Qt$ ', then which of the following is the average inventory at any time?

$Q/t$

**$Q/2$**

$(Q/2)t$

$2Q$

22. In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 120 respectively then which of the following is true about the 2nd month's requirement?

**2nd month demand can be included in 1st month**

2nd month demand will have to fulfill in 2nd month

2nd month demand can be included in 3rd month

2nd month demand can be included in any month of the year

23. Wilson's model (Purchasing model with no shortages) is of the \_\_\_ type.

☐ \_Probabilistic

☒ \_Deterministic (Page 32 Handout)

**Ref: In the deterministic type of inventory control, the parameters like demand, ordering quantity cost, etc are already known or have been ascertained and there is no uncertainty.**

**In the Probabilistic /stochastic inventory control, the uncertain aspects are taken into account.**

24. Which of the following times are not directly related to the activity cost as there is no certainty of time in such activities so their duration cannot be reduced, hence the cost cannot be expressed correctly?

CPM Times

**PERT Times**

25. The parameters like demand and ordering quantity cost are quite ----- in the Probabilistic inventory models.

a. certain

**b. uncertain**

**Ref: Probabilistic techniques covers those problem in which all parameters are not known with certainty.**

26. Which of the following is the major assumed **difference** between the models of Purchasing and Manufacturing without shortage?

☒ **\_Replacement rate**

☐ \_Shortage

☐ \_Demand

☐ \_Item, Setup and Holding Costs

### **Model-1 Purchasing model with no shortages(Wilson's model)**

The following assumptions are made in deriving the formula for economic order quantity.

(1) Demand (D) is at a constant rate.

(2) Replacement of items is instantaneous (lead time is zero).

(3) The cost coefficients  $C_1$ ,  $C_2$ , and  $C_3$  are constant.

(4) There is no shortage cost or  $C_4 = 0$ .

### **Model-3(Manufacturing model with no shortages)**

In this model the following assumptions are made:

(1) Demand is at a constant rate (D).

(2) All cost coefficients ( $C_1$ ,  $C_2$ ,  $C_3$ ) are constants.

(3) There is no shortage cost, or  $C_4 = 0$ .

(4) The replacement rate is finite and greater than the demand rate. This is also called replenishment rate or manufacturing rate, denoted by  $R$ .

27. For ABC analysis, in the Random Sample of some items, the ranks to the items are assigned with respect to their -----.

lowest annual unit cost

largest annual unit cost

lowest annual usage value

**largest annual usage value (Page 47 Handout)**

28. The purpose of maintaining the inventory is to balance the inter-related

\_\_\_\_\_

a. items

**b. costs**

c. demands

d. times

Ref: Inventory Control: Primarily concerned with the inventory cost control

29. The graph of Wilson's Model (Purchasing Model with no shortage) is of the ---  
----- shape.

a. rectangular

b. isosceles triangular

**c. saw tooth**

d. equivalent triangular

Any process allocating objects at regular rate will result in a steady increase in heap consumption, followed by instantaneous drops

30. While estimating the time for a PERT activity, the estimator will give the three times estimates. When these three estimates are plotted on a graph, the probability distribution is closely associated with \_\_\_\_\_  
Distribution curve.
- a. **Beta (page 23 handout)**
  - b. Poisson
  - c. Binomial
  - d. Chi
31. Which is the following reason is essential for an enterprise to have an inventory?
- a. It helps in minimizing the loss due to deterioration, obsolescence, damage etc.
  - b. It helps in maintaining economy by absorbing some of the fluctuations when the demand of an item fluctuates or is seasonal.
  - c. Take advantage of price discounts by bulk purchasing.
  - d. **All above reasons are equivalent.**
32. Except the economic order quantity, the value of  $Q^*$  will result in \_\_\_\_\_
- a. zero cost
  - b. **higher cost**
  - c. average cost
  - d. lower cost
33. Excess of inventories are undesirable due to the following reasons:
- a. It may not lead to high volume of output instead it might hamper the production
  - b. It includes the expenses of stores, equipment, personnel, insurance etc.
  - c. It locks up the capital of the enterprise.
  - d. **All above reasons are equivalent.**
34. If someone asks you how much time you require to reach your VU campus from your house, you may say that it may take 1 to  $1^{1/2}$  hours. Then you meant which of the following variable?
- a. Discrete
  - b. Random
  - c. Complex
  - d. **Continuous [Time is a continuous variable]**



35. When we have to manage a project with available resources, we have the option of resource leveling. Here the resources availability is less than the maximum resources required for an activity, then it is feasible to divert the resources to \_\_\_\_\_

- a. **critical activities (page 29 handout)**
- b. head or tail activities
- c. dummy activities
- d. non-critical activities

Ref: The slack is usually the basis on which priority is accorded to various job. Hence, the jobs, which are most critical, will be scheduled first.

36. The parameters like demand and ordering quantity cost are quite ----- in the Deterministic inventory models.

- a. **Certain (Page 32 Handout)**
- b. uncertain

## Question # 2 of 10 ( Start time: 10:52:47 AM, 24 December 2021 )

Total Marks: 1

In resource leveling when two or more jobs compete for same resources, first try to allocate an activity, which is of ----- duration and next to the activity which having next ----- duration.

Select the correct option

☐

maximum, minimum

☐

most likely, pessimistic

☐

optimistic, most likely

☐

minimum , maximum

[Click to Save Answer & Move to Next Question](#)

Question # 7 of 10 ( Start time: 10:59:43 AM, 24 December 2021 )

Total Marks: 1

If the demand of an item is occurred in the period of stock out and it is satisfied at the beginning of the next cycle then such type of problem will discuss under which of the following?

Select the correct option

☐

Purchasing Model with no shortages

☒

Purchasing Model with shortages

Correct

[Click to Save Answer & Move to Next Question](#)

Question # 9 of 10 ( Start time: 11:01:19 AM, 24 December 2021 )

Total Marks: 1

If we decrease the inventory then the shortage cost and the -----per unit of time will increase.

Select the correct option

☐ carrying cost☐ set up cost

Correct

[Click to Save Answer & Move to Next Question](#)

Question # 10 of 10 ( Start time: 11:02:28 AM, 24 December 2021 )

Total Marks: 1

In the Dynamic Order Quantity problem, the Demands of all succeeding months can be included in the demand of preceding one provided that .....

Select the correct option

☐

Shortage and Item Costs are balanced

☐

Shortage and Setup Costs are balanced

☒

Setup and Holding Costs are balanced

☐

Item and Holding Costs are balanced

Correct

Click to Save Answer & Move to Next Question

Question # 10 of 10 ( Start time: 11:24:51 AM, 24 December 2021 )

Total Marks: 1

When we have to manage a project with available resources, we have the option of resource leveling. Here the resources availability is less than the maximum resources required for an activity, then it is feasible to delay the job having -----,

Select the correct option

☐

largest free float

☐

dummy branch

☒

largest total float

☐

largest independent float

Correct

Click to Save Answer &amp; Move to Next Question



Question # 1 of 10 ( **start time: 11:42:12 AM, 24 December 2021** )

Which of the following technique to solve the network flow diagrams, is event oriented?

Select the correct option



Programme Evaluation and Review Technique

**Correct**



Critical Path Method

For ABC analysis, In the Random Sample of some items, the ranks to the items are assigned with respect to their -----

Select the correct option

|                       |                            |
|-----------------------|----------------------------|
| <input type="radio"/> | lowest annual unit cost    |
| <input type="radio"/> | largest annual usage value |
| <input type="radio"/> | largest annual unit cost   |
| <input type="radio"/> | lowest annual usage value  |

Correct

Question # 7 of 10 ( start time: 11:49:08 AM, 24 December 2021 )

Tot

Which of the following category of items in ABC analysis has the maximum Holding Cost?

Select the correct option

|                                  |                                                |
|----------------------------------|------------------------------------------------|
| <input checked="" type="radio"/> | A category                                     |
| <input type="radio"/>            | C category                                     |
| <input type="radio"/>            | All categories" items have equal Holding Costs |
| <input type="radio"/>            | B category                                     |

Correct

In the graph of the Purchasing Model with shortages, the area under the horizontal axis represents -----.

Select the correct option

|                       |                       |
|-----------------------|-----------------------|
| <input type="radio"/> | carrying cost of item |
| <input type="radio"/> | shortage cost of item |
| <input type="radio"/> | set up cost of item   |
| <input type="radio"/> | purchase cost of item |

Correct



In which of the following model, the Carrying Cost per unit of time is reduced in the ratio  $(1-D/R):1$  ?

Select the correct option

|                       |                                           |
|-----------------------|-------------------------------------------|
| <input type="radio"/> | Purchasing Model with allowed Shortage    |
| <input type="radio"/> | Manufacturing Model without Shortage      |
| <input type="radio"/> | Purchasing Model without Shortage         |
| <input type="radio"/> | Manufacturing Model with allowed Shortage |

Correct



Type here to search





In the context of Wilson's model, at time  $t = 0$ , the quantity 'Q' of the product is----- in the inventory.

Select the correct option

|                       |         |
|-----------------------|---------|
| <input type="radio"/> | Average |
| <input type="radio"/> | Minimum |
| <input type="radio"/> | Maximum |
| <input type="radio"/> | Zero    |

Correct



Which of the following quantity will vary in case of Dynamic Order Quantity Model?

Select the correct option

|                       |               |
|-----------------------|---------------|
| <input type="radio"/> | Demand        |
| <input type="radio"/> | Item Cost     |
| <input type="radio"/> | Carrying Cost |
| <input type="radio"/> | Setup Cost    |

Correct

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " $C_2/q$ " which is -----proportional to the quantity ordered.

Select the correct option

- |                                  |           |
|----------------------------------|-----------|
| <input checked="" type="radio"/> | directly  |
| <input type="radio"/>            | inversely |

Correct

Click to S



Type here to search





Which of the following category of items in ABC analysis needs special attention by the management?

Select the correct option

- |                       |                                              |
|-----------------------|----------------------------------------------|
| <input type="radio"/> | C category                                   |
| <input type="radio"/> | B category                                   |
| <input type="radio"/> | A category                                   |
| <input type="radio"/> | All categories need equal and fair attention |

Correct



Type here to search



**Question # 7 of 10 ( Start time: 11:59:25 AM, 24 December 2021 )**

If the total inventory in one cycle of 't' units of time is ' $(1/2)Qt$ ', then which of the following is the average inventory at any time?

**Select the correct option**

|                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | $(Q/2)t$ |
| <input type="radio"/>            | $Q/t$    |
| <input checked="" type="radio"/> | $Q/2$    |
| <input type="radio"/>            | $2Q$     |

**Correct**

## Question # 8 of 10 ( Start time: 12:00:23 PM, 24 December 2021 )

In resource leveling when two or more jobs compete for same resources, first try to allocate an activity, which is of ----- duration and next to the activity which has ----- duration.

Select the correct option

- ☐ optimistic, most likely
- ☐ maximum, minimum
- ☐ minimum , maximum
- ☐ most likely, pessimistic

Correct



Question # 7 of 10 ( Start time: 11:07:45 AM, 24 December 2021 )

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " $C_2/q$ " which is -----proportional to the quantity ordered.

Select the correct option

|                       |           |
|-----------------------|-----------|
| <input type="radio"/> | directly  |
| <input type="radio"/> | inversely |

Correct



Which of the following category of items in ABC analysis has the maximum Holding Cost?

Select the correct option

- |                       |                                                |
|-----------------------|------------------------------------------------|
| <input type="radio"/> | B category                                     |
| <input type="radio"/> | A category                                     |
| <input type="radio"/> | C category                                     |
| <input type="radio"/> | All categories' items have equal Holding Costs |

Correct



Question # 10 of 10 ( Start time: 11:40:33 AM, 24 December 2021 )

In Wilson's model, Total cost = -----

Select the correct option

☐

Item cost + holding cost + shortage cost + set up cost

☐

Holding cost + shortage cost + set up cost

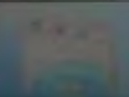
☐

Item cost + holding cost + shortage cost

☐

Item cost + set up cost + holding cost

Click to Save Answer & Move to Next Q





Quiz

<https://quiz.vu.edu.pk>

MC200204281: RAZWAN KHAN

Time Left 88 sec(s)

MTH601: Quiz

Quiz Start Time: 03:20 PM, 09 May 2021

Question # 1 of 10 ( Start time: 03:20:49 PM, 09 May 2021 )

Total Marks:

Dummy activity is represented by \_\_\_\_\_ .

Select the correct option

☐

doted line

☐

doted line with right to left arrow

☐

doted line with left to right arrow

☐

left to right arrow

Click to Save Answer &amp; Move to Next Question



 Quiz

MC200202537: RAFI UL HAQ

MTH601:Quiz

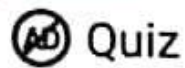
Question # 10 of 10 ( Start time: 04:21:08 PM, 09 May 2021 )

In Project Management, Critical Path method is based on ——— times.

Select the correct option

- |                       |               |
|-----------------------|---------------|
| <input type="radio"/> | deterministic |
| <input type="radio"/> | probabilistic |
| <input type="radio"/> | stochastic    |
| <input type="radio"/> | serial        |

[Click to Save Answer](#) [Click to Move to Next Question](#)



Quiz



MC200202537: RAFI UL HAQ

MTH601:Quiz

Question # 9 of 10 ( Start time: 04:20:17 PM, 09 May 2021 )

Dummy activity is represented by \_\_\_\_\_.

Select the correct option

- |                                  |                                     |
|----------------------------------|-------------------------------------|
| <input checked="" type="radio"/> | doted line                          |
| <input type="radio"/>            | doted line with right to left arrow |
| <input type="radio"/>            | doted line with left to right arrow |
| <input type="radio"/>            | left to right arrow                 |

Click to Save Answer &amp; Move to Next Question





 Quiz

MTH601:Quiz

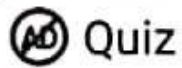
Question # 8 of 10 ( Start time: 04:19:17 PM, 09 May 2021 )

A/An \_\_\_\_\_ is the collection of inter related activities to be performed in a particular sequence to completion.

Select the correct option

|                                  |         |
|----------------------------------|---------|
| <input type="radio"/>            | node    |
| <input type="radio"/>            | event   |
| <input checked="" type="radio"/> | project |
| <input type="radio"/>            | branch  |

[Click to view Answer & Move to Next Question](#)



Quiz



MC200202537: RAFI UL HAQ

MTH601:Quiz

Question # 7 of 10 ( Start time: 04:18:25 PM, 09 May 2021 )

Which of the following is an operations research (OR) process?

Select the correct option

|                       |                    |
|-----------------------|--------------------|
| <input type="radio"/> | Linear programming |
| <input type="radio"/> | Networking         |
| <input type="radio"/> | Observation        |
| <input type="radio"/> | Simplex method     |

[Click to show Answer & Move to Next Question](#)

MC200202537: RAFI UL HAQ

MTH601 Quiz

Quiz Start Time

Question # 6 of 10 ( Start time: 04:17:54 PM, 09 May 2021 )

The Operational Research Society of U.K. defines OR as "Operational Research is the application of the methods of \_\_\_\_\_ to complex problems arising in the direction and management of large systems of men, machines, materials and money in industry, business, government and defense."

Select the correct option

|                       |             |
|-----------------------|-------------|
| <input type="radio"/> | arts        |
| <input type="radio"/> | statistics  |
| <input type="radio"/> | mathematics |
| <input type="radio"/> | science     |

[Go to Previous Answer](#) [Mark as Incorrect Question](#)

MC200202537: RAFI UL HAQ

MTH601:Quiz

Question # 5 of 10 ( Start time: 04:17:04 PM, 09 May 2021 )

In a network flow diagram, events are just the points in ———be represented by circles.

Select the correct option

|                       |            |
|-----------------------|------------|
| <input type="radio"/> | xy-plane   |
| <input type="radio"/> | space only |
| <input type="radio"/> | time only  |
| <input type="radio"/> | space-time |

Click to Save Answer &amp; Move to Next Question





Quiz



MC200202537: RAFI UL HAQ

MTH601:Quiz

Question # 4 of 10 ( Start time: 04:16:16 PM, 09 May 2021 )

The length of a path in a network is the total \_\_\_\_\_ it takes to travel along the path.

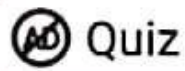
Select the correct option

- |                       |        |
|-----------------------|--------|
| <input type="radio"/> | length |
| <input type="radio"/> | days   |
| <input type="radio"/> | time   |
| <input type="radio"/> | months |

Click to view Answer &amp; Move to Next Question







Quiz



MC200202537: RAFI UL HAQ

MTH601:Quiz

Question # 4 of 10 ( Start time: 04:16:16 PM, 09 May 2021 )

The length of a path in a network is the total \_\_\_\_\_ it takes to travel along the path.

Select the correct option

- |                       |        |
|-----------------------|--------|
| <input type="radio"/> | length |
| <input type="radio"/> | days   |
| <input type="radio"/> | time   |
| <input type="radio"/> | months |
- no idea

Click to show Answer &amp; Move to Next Question



## Question 3 of 10 ( Start time: 04:15:18 PM, 09 May 2021 )

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, for the first node/event, we start with time ———.

Select the correct option

- ☐  $t = \text{infinity}$
- ☐  $t = 0$
- ☐  $t = a$  (arbitrary)
- ☐ with strict positive value

Click on the correct answer to receive 10 points



MC200202537: RAFI UL HAQ

MTH601:Quiz

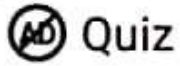
Question # 2 of 10 ( Start time: 04:14:04 PM, 00 May 2021 )

In 1948, Blackett defined OR as "OR is a \_\_\_\_\_ of providing executives with any \_\_\_\_\_ and objective basis for decisions"

Select the correct option

|                                  |                               |
|----------------------------------|-------------------------------|
| <input type="radio"/>            | computational method, control |
| <input checked="" type="radio"/> | scientific method, analytical |
| <input type="radio"/>            | graphical method, home        |
| <input type="radio"/>            | none of these                 |

[Click to Save Answer & Move to Next Question](#)



Quiz



MC200202537: RAFI UL HAQ

MTH601:Quiz

Question # 1 of 10 ( Start time: 04:12:50 PM, 09 May 2021 )

Which of the following is an operations research (OR) process?

Select the correct option

- |                                  |                    |
|----------------------------------|--------------------|
| <input type="radio"/>            | Linear programming |
| <input type="radio"/>            | Solution           |
| <input checked="" type="radio"/> | Networking         |
| <input type="radio"/>            | Simplex method     |





MC200203588: MUHAMMAD SOHAIL

MTH601:Quiz

Time Left 90 sec(s)

Quiz Start Time: 04:06 PM, 09 May 2021

Question # 10 of 10 ( Start time: 04:12:47 PM, 09 May 2021 )

Total Marks: 1

In a project, a network diagram shows the precedence relations of inter related activities along with their corresponding activity \_\_\_\_.

Select the correct option

- ☒ times
- ☐ cost
- ☐ profit
- ☐ quantity

Click to Save Answer &amp; Move to Next Question





MC200203588: MUHAMMAD SOHAIL

MTH601 Quiz

Time Left 90 sec(s)

Quiz Start Time: 04:06 PM, 09 May 2021

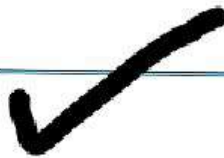
Question # 9 of 10 (Start time: 04:12:17 PM, 09 May 2021)

Total Marks:

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, the computations are proceeded from ----- to the final event.

Select the correct option

- |                       |               |
|-----------------------|---------------|
| <input type="radio"/> | bottom to top |
| <input type="radio"/> | right to left |
| <input type="radio"/> | left to right |
| <input type="radio"/> | top to bottom |

[Click to Save Answer & Move to Next Question](#)



MC200203588: MUHAMMAD SOHAIL

Time Left 90 sec(s)

MTH601 Quiz

Quiz Start Time: 04:06 PM, 09 May 2021

Question # 8 of 10 ( Start time: 04:11:42 PM, 09 May 2021 )

Total Marks: 1

CPM is abbreviation of \_\_\_\_\_ .

Select the correct option

- |                       |                             |
|-----------------------|-----------------------------|
| <input type="radio"/> | Constant Process Method     |
| <input type="radio"/> | Critical Path Method        |
| <input type="radio"/> | Continuous Path Membership  |
| <input type="radio"/> | Critical Performance Method |

Click to Save Answer &amp; Move to Next Question



MC200203588: MUHAMMAD SOHAIL

Time Left 90  
SEC(s)

MTH601:Quiz

Quiz Start Time: 04:06 PM, 09 May 2021

Question # 7 of 10 (Start time: 04:10:42 PM, 09 May 2021)

Total Marks: 1

If both jobs 'a(l,n)' and 'b(m,n)' of '7' and '8' days durations respectively, start earlier simultaneously on 4th day, then 'n' can start earlier on \_\_\_\_\_day.

Select the correct option

|                       |      |
|-----------------------|------|
| <input type="radio"/> | 8th  |
| <input type="radio"/> | 11th |
| <input type="radio"/> | 15th |
| <input type="radio"/> | 12th |

[Click to Solve AnyWhere Move to Next Question](#)



MC200203588: MUHAMMAD SOHAIL

Time Left 89 sec(s)

MTH601: Quiz

Quiz Start Time: 04:06 PM, 09 May 2021

Question # 6 of 10 ( Start time: 04:10:07 PM, 09 May 2021 )

Total Marks:

The Operational Research Society of U.K. defines OR as  
"Operational Research is the application of the methods of \_\_\_\_\_ to complex problems arising in the direction and management of large systems of men, machines, materials and money in industry, business, government and defense."

Select the correct option

- ☐ arts
- ☐ statistics
- ☐ mathematics
- ☐ science



Click to Save Answer &amp; Move to Next Question



MC200203588: MUHAMMAD SOHAIL

MTH601 Quiz

Time Left 89 sec(s)

Quiz Start Time: 04:06 PM, 09 May 2021

Question # 5 of 10 (Start time: 04:08:55 PM, 09 May 2021)

Total Marks: 1

If for path 1-3-4 time is  $5+3=8$  days, for path 1-2-3-4 time is  $6+7+3=16$  days and for path 1-2-4 time is  $6+6=12$  days, then which of the followings is true.

Select the correct option

- |                       |         |
|-----------------------|---------|
| <input type="radio"/> | 16 days |
| <input type="radio"/> | 8 days  |
| <input type="radio"/> | 12 days |
| <input type="radio"/> | 36 days |

[Click to Save Answer & Move to Next Question](#)





MC200203588: MUHAMMAD SOHAIL

MTH601:Quiz

Time Left 90  
sec(s)

Quiz Start Time: 04:06 PM, 09 May 2021

Question # 4 of 10 ( Start time: 04:08:18 PM, 09 May 2021 )

Total Marks: 1

In a network flow diagram, a/an \_\_\_\_\_ is shown by dotted arrows and these consume no time or resources.

Select the correct option

- |                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | activity |
| <input type="radio"/>            | branch   |
| <input checked="" type="radio"/> | dummy    |
| <input type="radio"/>            | nodes    |

[Click to Solve Answer & Move to Next Question](#)



MC200203588: MUHAMMAD SOHAIL

MTH601:Quiz

Time Left 89 sec(s)

Quiz Start Time: 04:06 PM, 09 May 2021

Question # 3 of 10 ( Start time: 04:07:35 PM, 09 May 2021 )

Total Marks: 1

In a project, a network diagram shows the \_\_\_\_\_ relations of the inter related activities along with their corresponding activity times.

Select the correct option

- ☐ deterministic and probabilistic
- ☐ precedence or succession
- ☐ union and intersection
- ☐ dummy and artificial

[Click to Save Answer & Move to Next question](#)



MC200203588: MUHAMMAD SOHAIL

MTH601:Quiz

Time Left 89 sec(s)

Quiz Start Time: 04:06 PM, 09 May 2021

Question # 2 of 10 ( Start time: 04:06:40 PM, 09 May 2021 )

Total Marks: 1

About which of the following it is true that, "they only signify with the passage of time, the beginning and ending of some activities under no consumption of resources" ?

Select the correct option

|                       |         |
|-----------------------|---------|
| <input type="radio"/> | project |
| <input type="radio"/> | nodes   |
| <input type="radio"/> | dummy   |
| <input type="radio"/> | branch  |

Click to Save Answer &amp; Move to Next Question

Question # 1 of 10 ( Start time: 04:06:05 PM, 09 May 2021 )

Total Marks:

Which of the following is not a problem of probabilistic technique.

Select the correct option

- |                       |                    |
|-----------------------|--------------------|
| <input type="radio"/> | Linear programming |
| <input type="radio"/> | Decision analysis  |
| <input type="radio"/> | Game theory        |
| <input type="radio"/> | Forecasting        |

[Click to Save Answer & Move to Next Question](#)



# Quiz

<https://quiz.vu.edu.pk>



MC200203376 MAAZ

Time Left

88

sec(s)

MTH601:Quiz

Quiz Start Time: 03:48 PM, 09 May 2021

Question # 10 of 10 ( Start time: 03:56:23 PM, 09 May 2021 )

Total Marks:

In a network flow diagram, the precedence relationship among the activities are indicated through \_\_\_\_\_.

Select the correct option

- ☐ project
- ☐ branch
- ☐ dummy
- ☒ nodes

Click on the Answered All Questions





## Quiz

<https://quiz.vu.edu.pk>

MC200203376 MAAZ

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 03:48 PM, 09 May 2021

Question # 9 of 10 ( Start time: 03:55:21 PM, 09 May 2021 )

Total Marks: 1

Which of the following is not a problem of probabilistic technique.

Select the correct option

- |                       |                    |
|-----------------------|--------------------|
| <input type="radio"/> | Linear programming |
| <input type="radio"/> | Decision analysis  |
| <input type="radio"/> | Game theory        |
| <input type="radio"/> | Forecasting        |

Click on the correct answer to mark the question



## Quiz

<https://quiz.vu.edu.pk>

MC200203376 MAAZ

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 03:48 PM, 09 May 2021

Question # 8 of 10 ( Start time: 03:54:29 PM, 09 May 2021 )

Total Marks: 1

CPM is abbreviation of \_\_\_\_\_ .

Select the correct option

- |                                  |                             |
|----------------------------------|-----------------------------|
| <input type="radio"/>            | Constant Process Method     |
| <input checked="" type="radio"/> | Critical Path Method        |
| <input type="radio"/>            | Continuous Path Membership  |
| <input type="radio"/>            | Critical Performance Method |

Previous Question Next Question



# Quiz

<https://quiz.vu.edu.pk>



MC200203376: MAAZ

MTH601:Quiz

Time Left 88 sec(s)

Quiz Start Time: 03:48 PM, 09 May 2021

Question # 7 of 10 ( Start time: 03:53:43 PM, 09 May 2021 )

Total Marks: 1

Operations research (OR) is the application of quantitative methods to \_\_\_\_\_ making.

Select the correct option

- |                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | formula      |
| <input checked="" type="radio"/> | decision     |
| <input type="radio"/>            | observation  |
| <input type="radio"/>            | all of these |

Go Back / Save Answer / Show / Hide Question



# Quiz

<https://quiz.vu.edu.pk>



MC200203376 MAAZ

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 03:48 PM, 09 May 2021

Question # 6 of 10 ( Start time: 03:53:10 PM, 09 May 2021 )

Total Marks: 1

Dummy activity is represented by \_\_\_\_\_ .

Select the correct option

- ☐ dotted line
- ☐ dotted line with right to left arrow
- ☐ dotted line with left to right arrow
- ☐ left to right arrow

Click on the correct answer to mark it as correct.



## Quiz

<https://quiz.vu.edu.pk>

MC200203376 MAAZ

Time Left 86 sec(s)

MTH601:Quiz

Quiz Start Time: 03:48 PM, 09 May 2021

Question # 5 of 10 ( Start time: 03:51:56 PM, 09 May 2021 )

Total Marks:

In a network flow diagram, two jobs 'a(i,k)' and 'b(j,k)' of durations '4' and '5' days respectively, enter an event 'k' then which of the following will be earliest start day of 'k' provided that (i,k) and (j,k) have started earlier at 3rd and 4th day respectively?

Select the correct option

- |                       |     |
|-----------------------|-----|
| <input type="radio"/> | 7th |
| <input type="radio"/> | 8th |
| <input type="radio"/> | 9th |
| <input type="radio"/> | 6th |

Question Bank / Answer &amp; Marking / Mark Questions





## Quiz

<https://quiz.vu.edu.pk>

MC200203376 MAAZ

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 03:48 PM, 09 May 2021

Question # 4 of 10 ( Start time: 03:51:00 PM, 09 May 2021 )

Total Marks: 1

In a network flow diagram, if an activity (i,j) starts earlier after two days and finish earlier on fifth day, then which of the following will be its completion time?

Select the correct option

- |                                  |            |
|----------------------------------|------------|
| <input checked="" type="radio"/> | Three days |
| <input type="radio"/>            | Five days  |
| <input type="radio"/>            | Two days   |
| <input type="radio"/>            | Ten days   |

Click on the Answer &amp; Move to Next Question



## Quiz

<https://quiz.vu.edu.pk>

MC200203376: MAAZ

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 03:48 PM, 09 May 2021

Question # 3 of 10 ( Start time: 03:50:25 PM, 09 May 2021 )

Total Marks: 1

Duration of the project is described by the \_\_\_\_\_ to complete the project.

Select the correct option

- |                                  |                   |
|----------------------------------|-------------------|
| <input type="radio"/>            | shortest time     |
| <input checked="" type="radio"/> | longest time      |
| <input type="radio"/>            | latest start time |
| <input type="radio"/>            | quick time        |

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Quiz

<https://quiz.vu.edu.pk>

MC200203376 MAAZ

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 03:48 PM, 09 May 2021

Question # 2 of 10 ( Start time: 03:49:36 PM, 09 May 2021 )

Total Marks: 1

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, the computations are proceeded from ————— to the final event.

Select the correct option

|                                  |               |
|----------------------------------|---------------|
| <input type="radio"/>            | bottom to top |
| <input type="radio"/>            | right to left |
| <input checked="" type="radio"/> | left to right |
| <input type="radio"/>            | top to bottom |

Click on the correct answer to mark the question



## Quiz

<https://quiz.vu.edu.pk>

MC200203376 MAAZ

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 03:48 PM, 09 May 2021

Question # 1 of 10 ( Start time: 03:48:31 PM, 09 May 2021 )

Total Marks: 1

If for path 1-3-4 time is  $5+3=8$  days, for path 1-2-3-4 time is  $6+7+3=16$  days and for path 1-2-4 time is  $6+6=12$  days, then which of the followings is true.

Select the correct option

|                       |         |
|-----------------------|---------|
| <input type="radio"/> | 16 days |
| <input type="radio"/> | 8 days  |
| <input type="radio"/> | 12 days |
| <input type="radio"/> | 36 days |

Click on the correct answer to mark the question



## Quiz

<https://quiz.vu.edu.pk>

MC200204281: RAZWAN KHAN

Time Left: 86 sec(s)

MTH601: Quiz

Quiz Start Time: 03:20 PM, 09 May 2021

Question # 10 of 10 ( Start time: 03:28:35 PM, 09 May 2021 )

Total Marks: \*

In 1958, Saaty defined OR as "The art of giving \_\_\_\_\_ answer to problems to which, otherwise, \_\_\_\_\_ answers are given."

Select the correct option

- |                       |                  |
|-----------------------|------------------|
| <input type="radio"/> | bad, worse       |
| <input type="radio"/> | worse, bad       |
| <input type="radio"/> | good, excellent  |
| <input type="radio"/> | excellent, worse |

Click to Save Answer &amp; Move to Next Question





Quiz

<https://quiz.vu.edu.pk>

MC200204281: RAZWAN KHAN

Time Left 88 sec(s)

MTH601 Quiz

Quiz Start Time: 03:20 PM, 09 May 2021

Question # 9 of 10 ( Start time: 03:28:00 PM, 09 May 2021 )

Total Marks:

Which of the following is not a categories of the operations research (OR) techniques.

Select the correct option

- |                                  |                                           |
|----------------------------------|-------------------------------------------|
| <input type="radio"/>            | Linear mathematical programming technique |
| <input checked="" type="radio"/> | Feasible solution                         |
| <input type="radio"/>            | Inventory techniques                      |
| <input type="radio"/>            | Network techniques                        |

Click to Save Answer &amp; Move to Next Question



## Quiz

<https://quiz.vu.edu.pk>

MC200204281: RAZWAN KHAN

Time Left 88  
sec(s)

MTH601:Quiz

Quiz Start Time: 03:20 PM, 09 May 202

Question # 8 of 10 ( Start time: 03:27:20 PM, 09 May 2021 )

Total Marks:

In 1948, Morse defined OR as "The term OR, has here-to fore been used to connote various attempts to study operations of war by \_\_\_\_\_. From a more general point of view, OR can be considered to be an attempt to study those operations of modern society which involved organizations of men or men and machines."

Select the correct option

- |                                  |                       |
|----------------------------------|-----------------------|
| <input type="radio"/>            | computational methods |
| <input type="radio"/>            | graphical methods     |
| <input checked="" type="radio"/> | scientific methods    |
| <input type="radio"/>            | none of these         |

Click to Save Answer &amp; Move to Next Question



Quiz

<https://quiz.vu.edu.pk>

MC200204281: RAZWAN KHAN

Time Left 88 sec(s)

MTH601 Quiz

Quiz Start Time: 03:20 PM, 09 May 2021

Question # 7 of 10 ( Start time: 03:26:11 PM, 09 May 2021 )

Total Marks:

If for path 1-3-4 time is  $5+3=8$  days, for path 1-2-3-4 time is  $6+7+3=16$  days and for path 1-2-4 time is  $6+6=12$  days, then which of the followings is true.

Select the correct option

☐

1-3-4 is critical path

☐

1-2-4 is critical path

☒

1-2-3-4 is critical path

☐

1-4 is critical path

Click to Save Answer &amp; Move to Next Question



## Quiz

<https://quiz.vu.edu.pk>

MC200204281: RAZWAN KHAN

Time Left 87 sec(s)

MTH601: Quiz

Quiz Start Time: 03:20 PM, 09 May 2021

Question # 6 of 10 ( Start time: 03:25:32 PM, 09 May 2021 )

Total Marks:

In a project, a network diagram shows the ----- relations of the inter related activities along with their corresponding activity times.

Select the correct option

☐

deterministic and probabilistic

☐

precedence or succession

☐

union and intersection

☐

dummy and artificial

Click to Save Answer &amp; Move to Next Question



## Quiz

<https://quiz.vu.edu.pk>

MC200204281: RAZWAN KHAN

Time Left 87 sec(s)

MTH601-Quiz

Quiz Start Time: 03:20 PM, 09 May 2021

Question # 5 of 10 ( Start time: 03:24:51 PM, 09 May 2021 )

Total Marks:

PERT is an acronym for \_\_\_\_\_ .

Select the correct option

☐

Program Evaluation and Review Technique

☐

Performance Evaluation and Review Technique Observation

☐

Program Execution and Revenue Technique

☐

Program Evaluation and Review Teacher

Click on Back Answer &amp; Move to Next Question





## Quiz

<https://quiz.vu.edu.pk>



MC200204281: RAZWAN KHAN

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 03:20 PM, 09 May 2021

Question # 4 of 10 ( Start time: 03:24:07 PM, 09 May 2021 )

Total Marks: -

In Project Management, Critical Path method is based on ---  
----- times.

Select the correct option

- ☒ deterministic
- ☐ probabilistic
- ☐ stochastic
- ☐ serial

Click to Save Answer & Move to Next Question



## Quiz

<https://quiz.vu.edu.pk>

MC200204281: RAZWAN KHAN

Time Left 74 sec(s)

MTH601:Quiz

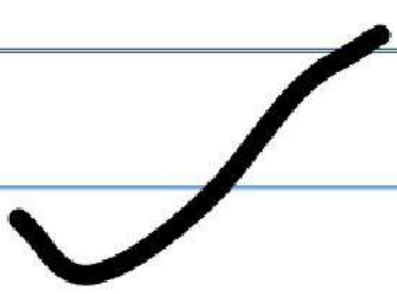
Quiz Start Time: 03:20 PM, 09 May 2021

Question # 3 of 10 ( Start time: 03:23:07 PM, 09 May 2021 )

Total Marks: 1

Operations research is based on \_\_\_\_\_.

Select the correct option

- |                       |              |
|-----------------------|--------------|
| <input type="radio"/> | engineering  |
| <input type="radio"/> | economics    |
| <input type="radio"/> | computing    |
| <input type="radio"/> | all of these |
- 

Click to save Answer &amp; Move to Next question



Quiz

<https://quiz.vu.edu.pk>

MC200204281: RAZWAN KHAN

Time Left: 73 sec(s)

MTH601 Quiz

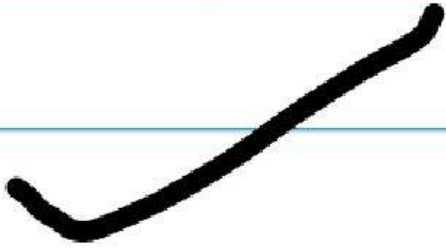
Quiz Start Time: 03:20 PM, 09 May 2021

Question # 2 of 10 ( Start time: 03:21:41 PM, 09 May 2021 )

Total Marks:

In a network flow diagram, which of the following method through computations provides, i) start and completion times for each activity, ii) critical and non critical activities and iii) total and free slacks?

Select the correct option

- |                       |                     |
|-----------------------|---------------------|
| <input type="radio"/> | Resource Scheduling |
| <input type="radio"/> | Resource Allocation |
| <input type="radio"/> | PERT                |
| <input type="radio"/> | CPM                 |
- 

[Click to Save Answer & Move to Next Question](#)

---

MTH601 MID TERM SOLVED MCQS  
(VISIT VURANK FOR MORE)

---

1. The first step in a problem solving exercises in OR is the \_\_\_\_\_that exist in the system.  
Construction of the model of problem  
**Identification o f the problem**  
Solution of the problem  
None of these
- 2 .A requirement for the definition of the problem in OR is that \_\_\_\_\_must be clearly defined which helps to focus attention on what the problem is  
Method  
**Objective**  
Solution  
None of these
- 3 . In a network flow diagram, time follows from\_\_\_\_\_  
**Left to right**  
Right to left  
Any direction
4. The dummy activities consume  
**No time,no resources**  
No time but some resources  
Some resources in minimum time  
All resources in a given time
5. In a national highway system interestion of two or more roads has the same analogy as the \_\_\_\_\_in the network flow diagram.  
Branch  
Critical activity  
Flow  
**Node**

6. Best possible time estimate that a given activity would take under normal conditions which often exist, is called

**Most likely time estimate**

Pessimistic time estimate

Smallest time estimate

Activity time estimate

7. In \_\_\_\_\_, activity time estimates are distributed according to beta distribution

**PERT**

CPM

Project Network

Float

8. If we have 2,4 and 12 hours as the optimistic( $t_0$ ) most likely ( $t_m$ ) and pessimistic( $t_p$ ) then the expected time for the activity would be

7

6

**5**

4

9. If  $t_0=2$   $t_m=5$  and  $t_p=14$  then  $SD=$ \_\_\_\_\_

**2**

4

6

8

10. IF  $t_0=6$ ,  $t_m=12$  and  $t_p=18$  then  $V_t=$ \_\_\_\_\_

12

2

**4**

144

11. The graphical method for solving a linear programming problem is applicable when

\_\_\_\_\_

**Two variables are involved**

Three variables are involved

Any number of variables are involved

None of these

12. Which of the following is equivalent relation to the constraint  $x=2$  of a linear programming problem?

$X \geq 2$

$x \leq 2$

$x < 2$  and  $x \geq 2$

**$x < 2$  and  $X > 2$**



13. While using simplex method for a LPP if there are 4 standard equations in 8 variables then number of non basic variables will be \_\_\_\_\_

32

12

4

2

15. In the purchasing model without any shortage(Wilsonsmodel) which of the following quatitiy does not vary?

Supply

**Demand**

Inventory

Transportation cost

16. How many decision variables are involved in the inventory model with shortages?

One

Two

Three

Four

17. Which of the following rule is used in order to minimize the decision variables in any inventory model?

Derivative test

Simplex method

PERT

Graphical method

19. ABC analysis depends on the

Quality of materials

Cost of materials

**Annual consumption value of materials**

Quantity of materials used

---

(THE END)

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**MTH601 Solved Midter MCQS File**

1. \_\_\_\_\_ employs a different modeling and solution logic than linear programming
  - (a). Transportation Model
  - (b). Inventory Control Model
  - (c). Dynamic Programming**
  - (d). None of the above
2. To identify and maintain the proper precedence relationship between activities those are not connected by event, we introduce
  - (a). Parallel Activity
  - (b). Dummy Activity**
  - (c). Sequential Activity
  - (d). None of the above
3. EST and EFT of activities are calculated in
  - (a). Forward pass**
  - (b). Backward pass
4. Critical path is obtained by connecting the jobs having
  - (a). Activities having same EST and LST
  - (b). Activities having same EFT and LFT
  - (c). Activities having zero slack
  - (d). All of the above**
5. The Variance  $V_t$  of expected time is calculated as
  - (a).  $V_t = \left(\frac{t_m - t_o}{6}\right)^2$
  - (b).  $V_t = \left(\frac{t_o - t_p}{6}\right)^2$**
  - (c).  $V_t = \left(\frac{t_p - t_m}{6}\right)^2$
  - (d). None of the above
6. In LP problems Additivity means that
  - (a). The effect of two different programs of production is the same as that of a joint program**
  - (b). The doubling (or tripling) the product will exactly double (or triple) the profit and the required resource
  - (c). Both (a)& (b)
  - (d). None of the above
7. Two of the first steps of OR process encompass the actual use of OR techniques. These steps are
  - (a). Model Construction and Model Solution**
  - (b). Observation and Implementation
  - (c). Definition of the problem and Model Solution
  - (d). Model Solution and Implementation of results
8. Let FS = Free Slack, TS = Total Slack, INDS = Independent Slack, then which relation is true
  - (a).  $TS \leq FS$
  - (b).  $INDS \leq FS$
  - (c).  $FS \leq TS$
  - (d). Both (b) & (c)**
9. Best possible time estimate that a given activity would take under normal conditions which often exist, is called
  - (a). Most Likely time estimate**
  - (b). Pessimistic time estimate
  - (c). Smallest time estimate

(d). None of the above

10. Standard Deviation S.D is

**(a). One sixth of the difference between pessimistic time estimates and optimistic time estimates**

(b). One sixth of the difference between pessimistic time estimates and most likely time estimates

(c). One sixth of the difference between optimistic time estimates and most likely time estimates

(d). One sixth of the difference between most likely time estimates and optimistic time estimates

Which one is best describe Sectoral planning

? **Inventory Planning in agriculture**

? Improving the layout of a workshop in a company

? Simulation Modeling of the Economy of the country

? None of these.

\_\_\_\_\_ is the most appropriate to situations where we maintain a relative stable employment levels and utilize the resource at a more constant rate

? **(a). Resource Leveling Program**

? (b). Resource Allocation Program

? (c). Both a & b

? (d). None of these.

If the slack time is zero, it means that the project will be

? Delayed

? **Completed on schedule**

The amount of an activity can be delayed without affecting the early start time of any other job, is called

? **Free Slack**

? Independent Slack

? Total Slack

? None of these.

**Question No: 1 ( Marks: 1 ) - Please choose one**

---

EST and EFT of activities are calculated in

► **Forward pass**

► Backward pass

- ▶ Path does not effected

**Question No: 2 ( Marks: 1 ) - Please choose one**

---

\_\_\_\_\_ may be less than most likely time estimate

- ▶ **Pessimistic time estimate**
- ▶ Smallest time estimate
- ▶ Optimistic Time estimate

**Question No: 3 ( Marks: 1 ) - Please choose one**

---

The dummy activities consume

- ▶ **No time, no resources**
- ▶ No time but some resources
- ▶ Some resources in minimum time
- ▶ None of these

**Question No: 4 ( Marks: 1 ) - Please choose one**

---

If an activity consumes no time and no resources then this activity is called \_\_\_\_\_.

- ▶ **dummy activity**
- ▶ sequential activity
- ▶ critical activity
- ▶ cyclic activity

**Question No: 5 ( Marks: 1 ) - Please choose one**

---

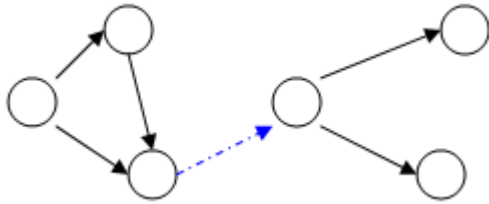
Cost period = ----- × ( No of ordered items)

- ▶ Holding cost
- ▶ Set up cost
- ▶ Stock out cost
- ▶ **Item cost**

**Question No: 6 ( Marks: 1 ) - Please choose one**

---

The following network is an example of

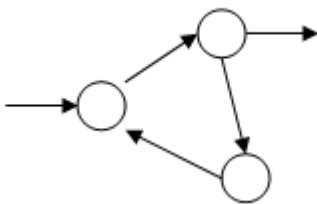


- ▶ Redundancy
- ▶ Cycling
- ▶ Looping
- ▶ **Merging**

**Question No: 7 ( Marks: 1 ) - Please choose one**

---

The following network is an example of



- ▶ **Redundancy**
- ▶ Dangling
- ▶ Cycling
- ▶ Dummy

**Question No: 8 ( Marks: 1 ) - Please choose one**

---

Which one is best describe Micro Economic Planning?

- ▶ Distribution of fertilizer
- ▶ **Improving the layout of a workshop in a company**
- ▶ Investment planning of the country
- ▶ PERT

**Question No: 9 ( Marks: 1 ) - Please choose one**

---

If  $t_0 = 6$ ,  $t_m = 12$  and  $t_p = 18$ , then  $V_t =$  \_\_\_\_\_



- ▶ 12
- ▶ 2
- ▶ **4**
- ▶ 144

**Question No: 10 ( Marks: 1 ) - Please choose one**

---

Which inventory model also known as a saw tooth model?

- ▶ **Purchasing Model with no shortages**
- ▶ Purchasing Model with shortages
- ▶ Manufacturing Model with no shortages
- ▶ Manufacturing Model with shortages

**Question No: 11 ( Marks: 1 ) - Please choose one**

---

For backward pass computations

- ▶ Earliest start time  $\geq$  Latest start time
- ▶ **Earliest start time  $\leq$  Latest start time**
- ▶ Earliest start time  $+$  Latest start time = 0
- ▶ Earliest start time  $=$  Latest start time

**Question No: 12 ( Marks: 1 ) - Please choose one**

---

For an activity if optimistic time, most likely time estimate and pessimistic time estimate are 3, 6 and 15 respectively then expected time is

- ▶ 4
- ▶ 3
- ▶ **7**
- ▶ 20

**Question No: 13 ( Marks: 1 ) - Please choose one**

---

In a quadratic programming problem unlike linear programming problem

- ▶ Only objective function is quadratic
- ▶ Both objective function and constraints are quadratic
- ▶ Only constraints are quadratic
- ▶ **At least one of objective function or constraint must be quadratic**

**Question No: 14 ( Marks: 1 ) - Please choose one**

---

Solution region of the constraint  $x \geq 0$  is

- ▶ Half plane to the right of straight line  $x = 0$
- ▶ **Half plane to the right of y-axis**
- ▶ Half plane to the region where abscissas are non-negative
- ▶ All are equivalent

**Question No: 15 ( Marks: 1 ) - Please choose one**

---

A dummy activity is a simulated activity of sorts, one that is of \_\_\_\_\_duration and is created for the sole purpose of demonstrating a specific relationship and path of action on the arrow diagramming method.

- ▶ **Zero**
- ▶ Minimum
- ▶ Maximum
- ▶ Average

**Question No: 16 ( Marks: 1 ) - Please choose one**

---

Activity definition refers to the process of parsing a project into a number of individual tasks which must be completed \_\_\_\_\_ the deliverables can be considered completed. Activity definitions rely on a number of specific input processes.

- ▶ **before**
- ▶ both before and after
- ▶ after

**Question No: 17 ( Marks: 1 ) - Please choose one**

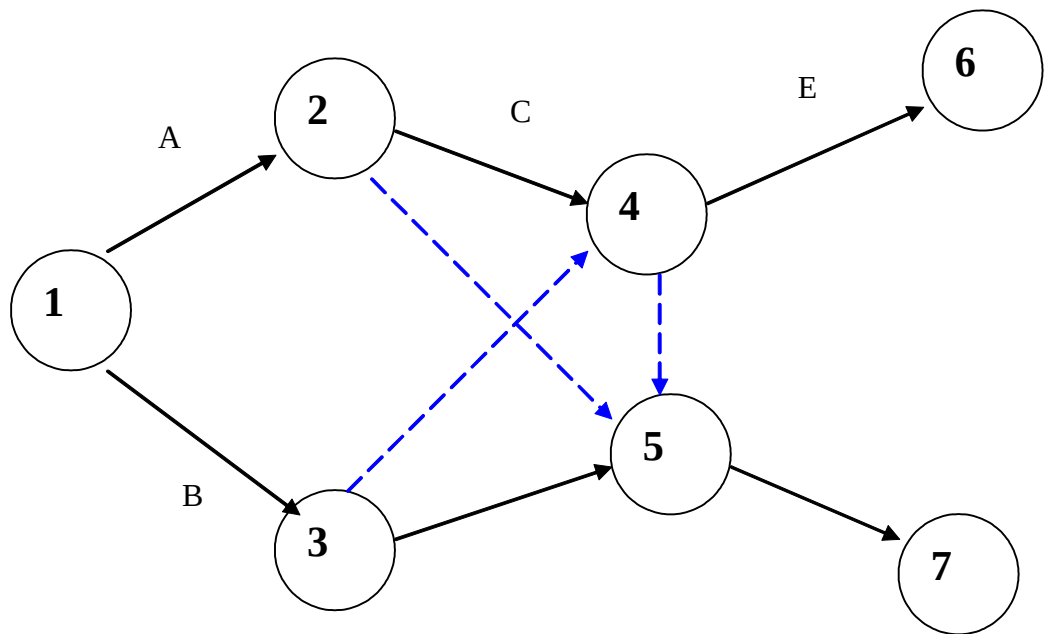
---

A forward pass is used to determine and calculate the \_\_\_\_\_ dates, through utilization of a previously specified start date.

- ▶ **early start and early finish**
- ▶ late start and early finish
- ▶ early start and late finish
- ▶ late start and late finish

**Question No: 18    ( Marks: 1 ) - Please choose one**

Is this Network legal?



- ▶ Yes.
- ▶ **No.**

**Question No: 19    ( Marks: 1 ) - Please choose one**

Total cost per period = Item cost + Order cost + Holding cost + \_\_\_\_\_.

- ▶ **Shortage cost**
- ▶ Optimum Shortage (S\*)
- ▶ Economic Order Quantity. (Q\*)
- ▶ Maximum Inventory. (I<sub>max.</sub>)

**Question No: 20    ( Marks: 1 ) - Please choose one**

$K = Z \times (-----)$

Where K is called service factor.



$\sqrt{\pi/2}$

---



$\sqrt{2/\pi}$



$\sqrt{2\pi/3}$



$\sqrt{3\pi/2}$

**MTH601 Midterm Exam Solved Paper 2005 – Operations Research**  
**SEMESTER SPRING**

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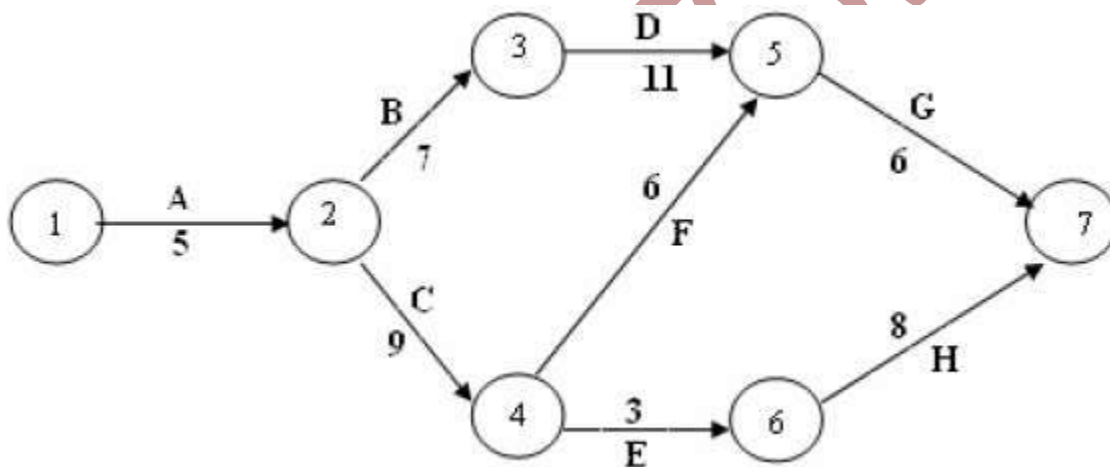
[quiz.vuzs.net](http://quiz.vuzs.net)

Total Marks = M= 50

Duration: 90mins

**MTH601 - Operations Research - Q. No. 1 M -10**

**For the following project diagram find out Earliest start time (EST), Earliest finish time (EFT), Latest start time (LST), Latest finish time (LFT) for each activity. Also find the Critical Path.**



**NOTE:** There is no need of making diagram corresponds to each step, you have to write down only the required terms correspond to each activity and the justification for the critical path.

**Solution:**

| Activity | EST | EFT | LST | LFT |
|----------|-----|-----|-----|-----|
| A(1 – 2) | 0   | 5   | 0   | 5   |
| B(2 – 3) | 5   | 12  | 5   | 12  |
| C(2 – 4) | 5   | 14  | 8   | 17  |
| D(3 – 5) | 12  | 23  | 12  | 23  |
| E(4 – 6) | 14  | 17  | 18  | 21  |
| F(4 – 5) | 14  | 20  | 17  | 23  |
| G(5 – 7) | 23  | 29  | 23  | 29  |
| H(6 – 7) | 17  | 25  | 21  | 29  |

To find the **Critical Path** we'll find the Slack (Float) of each activity.

| Activity | Slack |
|----------|-------|
| A(1 – 2) | 0     |
| B(2 – 3) | 0     |
| C(2 – 4) | 3     |
| D(3 – 5) | 0     |
| E(4 – 6) | 4     |
| F(4 – 5) | 3     |
| G(5 – 6) | 0     |
| H(6 – 7) | 4     |

**MTH601 - Operations Research - Q. No. 2 M -10**

Suppose the following estimates of activity times of a project,

| Activity | Optimistic Time | Most Likely Time | Pessimistic Time |
|----------|-----------------|------------------|------------------|
| A        | 1               | 2                | 3                |
| B        | 3               | 6                | 8                |
| C        | 5               | 8                | 20               |
| D        | 10              | 13               | 28               |
| E        | 4               | 7                | 12               |
| F        | 2               | 4                | 5                |
| G        | 8               | 12               | 16               |
| H        | 2               | 2                | 2                |

If the critical path of the project is A, D, G, H, then calculate the Expected Completion Time (Length of the path) and corresponding Variance of the project.



**Solution:**

| Critical Activities | Optimistic Time ( $t_o$ ) | Most Likely Time ( $t_m$ ) | Pessimistic Time ( $t_p$ ) | Expected Duration ( $t_e$ ) | Variance |
|---------------------|---------------------------|----------------------------|----------------------------|-----------------------------|----------|
| A                   | 1                         | 2                          | 3                          | 2                           | 0.11     |
| D                   | 10                        | 13                         | 28                         | 15                          | 9        |
| G                   | 8                         | 12                         | 16                         | 12                          | 1.78     |
| H                   | 2                         | 2                          | 2                          | 2                           | 0        |

$$\text{Expected Completion time} = 2 + 15 + 12 + 2 \\ = 31$$

$$\text{Variance (of the project)} = 0.11 + 9 + 1.78 + 0 \\ = 10.89$$

<http://www.vuzs.net/study-portals/bscs-study-portal.html>

**MTH601 - Operations Research - Q. No. 3 M -10**

**Demand for the product is 600 units per year, and the items are withdrawn at a constant rate. The setup cost for placing an order to replenish inventory is \$25. The unit cost of each item is \$3, and the inventory holding cost is \$0.05 per item per week.**

(a). Assuming shortages are not allowed, determine how often to make a production run and what size it should be.

(b). If shortages are allowed but cost \$2 per item per month, determine how often to order and what size the order should be.

**Solution:**

D=600 unit/year

Ordering Cost ( $C_2$ ) = \$25/order

Item Cost ( $C_1$ ) = \$3/item

Holding Cost ( $C_3$ ) = 0.05/item/week  
= 0.05\*52/item/year  
= 2.6/item/year

(a). Shortages are not allowed:

$$Q^* = ?$$

$$t^* = ?$$

$$Q^* = \sqrt{2C_2D / C_3}$$

Putting all the values in the above formula we have

$$Q^* = \sqrt{2 * 25 * 600 / 2.6}$$

$$Q^* = 107.42$$

$$t^* = Q^*/D$$

$$= 107.41/600$$

$$= 0.1796$$

(b). Shortages are allowed:

Shortage Cost ( $C_4$ ) = \$2 /item/month  
= 24 item/year

$$Q^* = \sqrt{2C_2D / C_3} * \sqrt{(C_3 + C_4) / C_4}$$

$$= 107.41 * \sqrt{(2.6 + 24) / 24}$$

$$= 223.606$$

$$t^* = Q^*/D$$

$$= 0.3726$$

#### MTH601 - Operations Research - Q. No. 4 M -10

Rabbani and Sons of Sialkot manufacture sports goods like hockey sticks, cricket bats etc .Each hockey stick yields a profit of Rs.20 and each cricket bat Rs 25.A hockey stick requires four hours of processing at machine A and two hours at machine B. A bat requires six hours of processing at machine A and five hours at machine B and one hour at machine C. Machine A has a maximum of 135 hours of availability capacity, machine B has 75 hours and machine C has 20 hours.

Read the above problem carefully and formulate it as an LP model to maximize the company's profit.

**Solution:**

Decision variables

$x_1$  = Number of hockey stick

$x_2$  = number of cricket bats

Objective function

Since each hockey stick yields a profit of Rs.20 so  $x_1$  number of hockey stick yields a profit of Rs.20 $x_1$

Since each cricket bat yields a profit of Rs 25  $x_2$  number of cricket bats yields a profit of Rs.25 $x_2$

As we want to maximize the company profit so objective function is

Maximize  $Z = 20x_1 + 25x_2$

**Constraints**

| Product | Machine A | Machine B | Machine C |
|---------|-----------|-----------|-----------|
| Hockey  | 4         | 2         | 0         |
| Bat     | 6         | 5         | 1         |

Hourly Availability of machine A = 135

Hourly Availability of machine B = 75

Hourly Availability of machine C = 20

$$4x_1 + 6x_2 \leq 135$$

$$2x_1 + 5x_2 \leq 75$$

$$0x_1 + x_2 \leq 20$$

$$x_1, x_2 \geq 0$$

Now LP problem is

Maximize  $Z = 20x_1 + 25x_2$

Subject to

$$4x_1 + 6x_2 \leq 135$$

$$2x_1 + 5x_2 \leq 75$$

$$0x_1 + x_2 \leq 20$$

$$x_1, x_2 \geq 0$$

**MTH601 - Operations Research - Q. No. 5 M -10**

Select the best choice (Only one) from the given four choices against each question.

**1. \_\_\_\_\_ employs a different modeling and solution logic than linear programming**

- (a). Transportation Model
- (b). Inventory Control Model
- (c). Dynamic Programming**
- (d). None of the above

**2. To identify and maintain the proper precedence relationship between activities those are not connected by**

event, we introduce

- (a). Parallel Activity
- (b). Dummy Activity**
- (c). Sequential Activity
- (d). None of the above

**3. EST and EFT of activities are calculated in**

- (a). Forward pass**
- (b). Backward pass

**4. Critical path is obtained by connecting the jobs having**

- (a). Activities having same EST and LST
- (b). Activities having same EFT and LFT
- (c). Activities having zero slack
- (d). All of the above**

**5. The Variance  $V_t$  of expected time is calculated as**

- $V_t = (t_m - t_0 / 6)^2$
- **$V_t = (t_0 - t_p / 6)^2$**
- $V_t = (t_p - t_m / 6)^2$
- None of the Above

**6. In LP problems Additivity means that**

**(a). The effect of two different programs of production is the same as that of a joint program**

(b). The doubling (or tripling) the product will exactly double (or triple) the profit and the required resource

(c). Both (a)& (b)

(d). None of the above

**7. Two of the first steps of OR process encompass the actual use of OR techniques. These steps are**

**(a). Model Construction and Model Solution**

(b). Observation and Implementation

(c). Definition of the problem and Model Solution

(d). Model Solution and Implementation of results

**8. Let FS = Free Slack, TS = Total Slack, INDS = Independent Slack, then which relation is true**

- (a).  $TS \leq FS$
- (b).  $INDS \leq FS$
- (c).  $FS \leq TS$
- (d). Both (b) & (c)**

**9. Best possible time estimate that a given activity would take under normal conditions which often exist, is called**

- (a). Most Likely time estimate**
- (b). Pessimistic time estimate
- (c). Smallest time estimate
- (d). None of the above

**10. Standard Deviation S.D is**

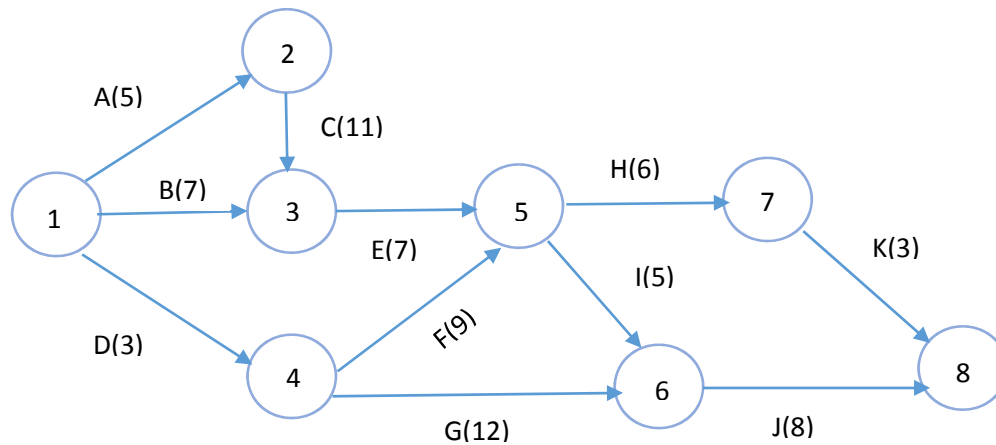
- (a). One sixth of the difference between pessimistic time estimates and optimistic time estimates**
- (b). One sixth of the difference between pessimistic time estimates and most likely time estimates
- (c). One sixth of the difference between optimistic time estimates and most likely time estimates
- (d). One sixth of the difference between most likely time estimates and optimistic time estimates



## Solution of Practice Questions Lecture#7-9

### Question 1:

Find the critical path and the expected duration of the following project.



### Solution

There are 7 possible paths to get from the starting point at node 1 and travel through the network to end at node 8.

#### Possible Paths

$$1-2-3-5-7-8=A-C-E-H-K=5+11+7+6+3=32$$

$$1-2-3-5-6-8=A-C-E-I-J=5+11+7+5+8=36$$

$$1-3-5-7-8=B-E-H-K=7+7+6+3=23$$

$$1-3-5-6-8=B-E-I-J=7+7+5+8=27$$

$$1-4-5-6-8=D-F-I-J=3+9+5+8=25$$

$$1-4-5-7-8=D-F-H-K=3+9+6+3=21$$

$$1-4-6-8=D-G-J=3+12+8=23$$

Hence the critical path is the longest distance with the shortest possible time i.e.

$$=1-2-3-5-6-8$$

$$=A-C-E-I-J$$

$$=5+11+7+5+8$$

$$=36$$

**Question 2:**

A project has the following time schedule.

| Activity | Time(hours) |
|----------|-------------|
| 1-2      | 2           |
| 1-3      | 2           |
| 1-4      | 1           |
| 2-5      | 4           |
| 3-6      | 8           |
| 3-7      | 5           |
| 4-6      | 3           |
| 5-8      | 1           |
| 6-9      | 5           |
| 7-8      | 4           |
| 8-9      | 3           |

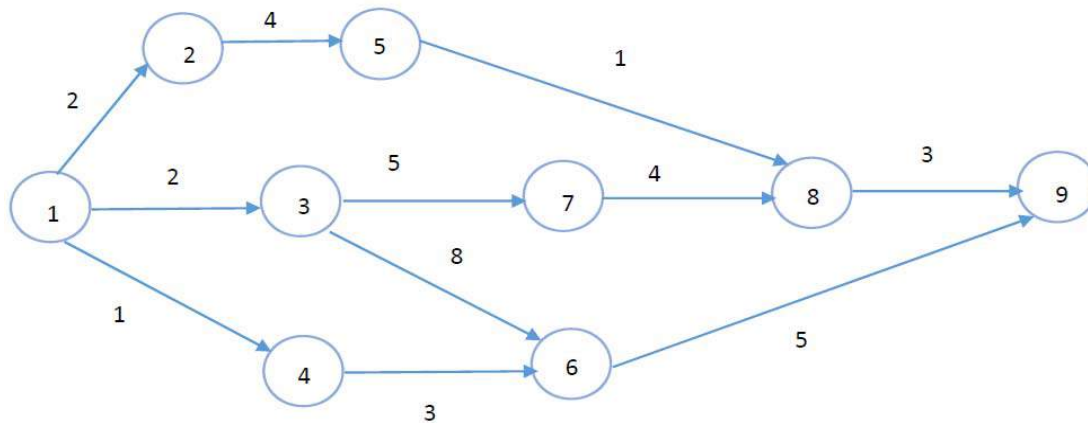
a) Draw the network

b) Calculate the following time estimates.

- Earliest Start Time (EST) of each activity.
- Earliest Finish Time (EFT) of each activity.

**Solution**

a) Network



b)

| Event | EST |
|-------|-----|
| 1     | 0   |
| 2     | 2   |
| 3     | 2   |
| 4     | 1   |
| 5     | 6   |
| 6     | 10  |
| 7     | 7   |
| 8     | 11  |
| 9     | 15  |

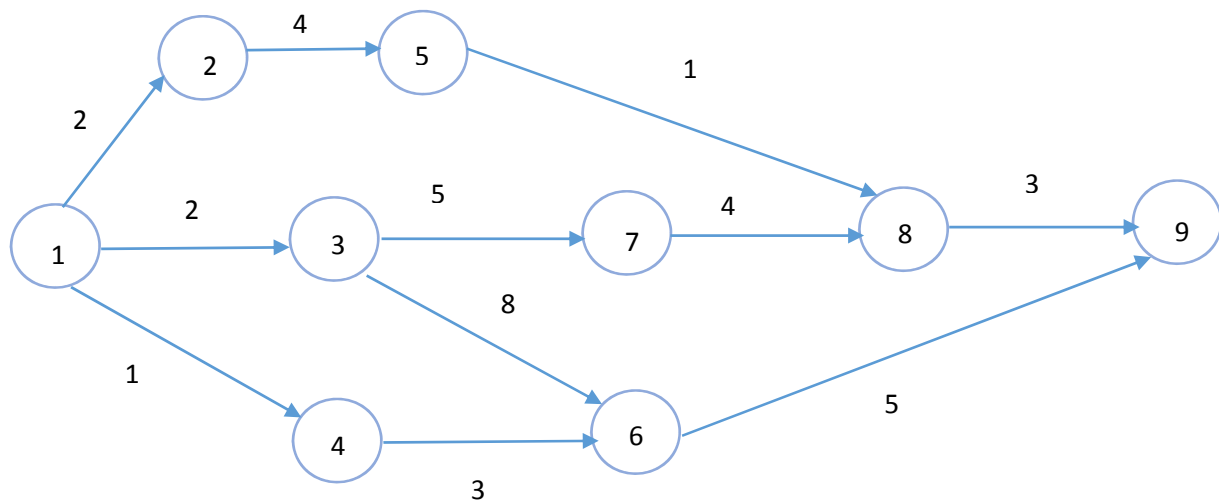
EST of any activity is the EST of its tail event

| Activity | Time in Hours | EST | EFT |
|----------|---------------|-----|-----|
| 1-2      | 2             | 0   | 2   |
| 1-3      | 2             | 0   | 2   |
| 1-4      | 1             | 0   | 1   |
| 2-5      | 4             | 2   | 6   |
| 3-6      | 8             | 2   | 10  |
| 3-7      | 5             | 2   | 7   |
| 4-6      | 3             | 1   | 4   |
| 5-8      | 1             | 6   | 7   |
| 6-9      | 5             | 10  | 15  |
| 7-8      | 4             | 7   | 11  |
| 8-9      | 3             | 11  | 14  |

$EFT = EST + \text{Duration of the activity}$

### Question 3:

Calculate the LST and LFT for each activity for the network given below.



Find the critical path and the expected duration of the project.

### Solution

| Event | LFT |
|-------|-----|
| 1     | 0   |
| 2     | 7   |
| 3     | 2   |
| 4     | 7   |
| 5     | 11  |
| 6     | 10  |
| 7     | 8   |
| 8     | 12  |
| 9     | 15  |

LFT of any activity is the LFT of its head event

| Activity | Time in Hours | LST | LFT |
|----------|---------------|-----|-----|
| 1-2      | 2             | 5   | 7   |
| 1-3      | 2             | 0   | 2   |
| 1-4      | 1             | 6   | 7   |
| 2-5      | 4             | 7   | 11  |
| 3-6      | 8             | 2   | 10  |
| 3-7      | 5             | 3   | 8   |
| 4-6      | 3             | 7   | 10  |
| 5-8      | 1             | 11  | 12  |
| 6-9      | 5             | 10  | 15  |
| 7-8      | 4             | 8   | 12  |
| 8-9      | 3             | 12  | 15  |

$LST = LFT - \text{Duration of the activity}$

**Possible Paths:**

- $1-2-5-8-9 = 2+4+1+3 = 10$
- $1-3-7-8-9 = 2+5+4+3 = 14$
- $1-3-6-9 = 2+8+5 = 15$
- $1-4-6-9 = 1+3+5 = 9$

Hence 1-3-6-9 is the critical path with duration 15.

**Question 4:**

A small marketing project consists of the jobs as shown in the table below.

| Activity | Predecessor | Duration |
|----------|-------------|----------|
| X        | -           | 4        |
| Y        | -           | 5        |
| Z        | X           | 6        |

a) Draw the Network diagram.

b) Calculate the following time estimates.

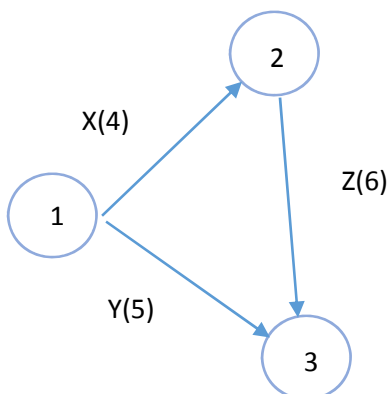
- Earliest Start Time of each activity
- Earliest Finish Time of each activity
- Latest Start Time of each activity
- Latest Finish Time of each activity

c) Critical Path and its duration.

d) Find Total Float, Independent Float and free Float for each activity.

**Solution**

a) Network:



b) Time Estimates

| Activity | Duration | EST | EFT | LST | LFT |
|----------|----------|-----|-----|-----|-----|
| 1-2      | 4        | 0   | 4   | 0   | 4   |
| 1-3      | 5        | 0   | 5   | 5   | 10  |
| 2-3      | 6        | 4   | 10  | 4   | 10  |

c) Possible Paths:

Time taken to complete the activities in two paths are:

- 1-2-3=4+6=10
- 1-3=5

Hence 1-2-3 is the critical path and the duration is 10 days.

d)

| Activity | Duration | Total Float | Free Float | Independent Float |
|----------|----------|-------------|------------|-------------------|
| 1-2      | 4        | 0           | 0          | 0                 |
| 1-3      | 5        | 5           | 5          | 5                 |
| 2-3      | 6        | 0           | 0          | 0                 |

**Question 5:**

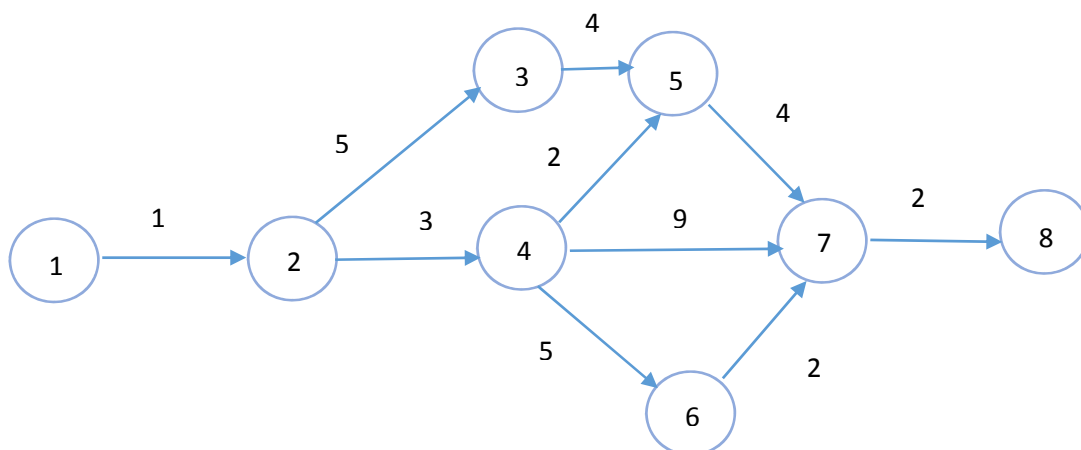
A project with 10 activities is given below

| Activity | 1-2 | 2-3 | 2-4 | 3-5 | 4-5 | 4-6 | 4-7 | 5-7 | 6-7 | 7-8 |
|----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Duration | 1   | 5   | 3   | 4   | 2   | 5   | 9   | 4   | 2   | 2   |

a) Construct the Network

b) Find the Critical Path.

**Solution** a) Network





b) **Possible Paths:**

- 1-2-3-5-7-8=1+5+4+4+2=16
- 1-2-4-5-7-8=1+3+2+4+2=12
- 1-2-4-7-8=1+3+9+2=15
- 1-2-4-6-7-8=1+3+5+2+2=13

Hence the critical path is 1-2-3-5-7-8=16

**Question 6**

How many types of floats are there? Also mention their formulas.

**Solution**

1. Independent float

$$I = j_E - i_L - D$$

2. Free float

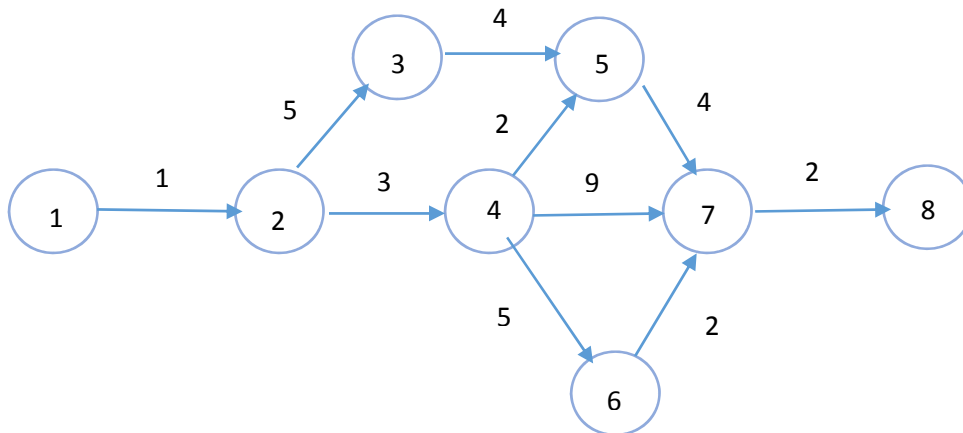
$$F = j_E - i_E - D$$

3. Total float

$$I = j_L - i_E - D.$$

**Question 7:**

A project is represented by the network shown below



Find total float and free float.

### Solution

| Activity | Total Float | Free Float |
|----------|-------------|------------|
| 1-2      | 0           | 0          |
| 2-3      | 0           | 0          |
| 2-4      | 1           | 0          |
| 3-5      | 0           | 0          |
| 4-5      | 4           | 4          |
| 4-6      | 3           | 0          |
| 4-7      | 1           | 1          |
| 5-7      | 0           | 0          |
| 6-7      | 3           | 3          |
| 7-8      | 0           | 0          |

### Question 8:

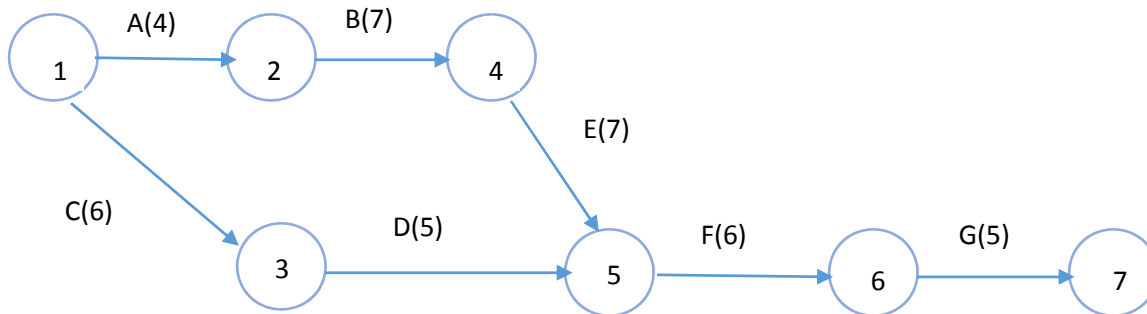
A project consist of the seven activities for which the relevant data are given below:

| Activity | Preceding activity | Duration |
|----------|--------------------|----------|
| A        | -                  | 4        |
| B        | A                  | 7        |
| C        | -                  | 6        |
| D        | C                  | 5        |
| E        | B                  | 7        |
| F        | D,E                | 6        |
| G        | F                  | 5        |

- (a) Draw an arrow diagram representing the project and find the project completion time.  
(b) Calculate the total float for each activity.

### Solution

a)



b)

| Activity | Total Float |
|----------|-------------|
| 1-2      | 0           |
| 1-3      | 7           |
| 2-4      | 0           |
| 3-5      | 7           |
| 4-5      | 0           |
| 5-6      | 0           |
| 6-7      | 0           |

***Critical path:***

Path of zero total float through the project is called the critical path. Total float is zero on the activities A, B, E, F and G. Therefore

=A-B-E-F-G

=1-2-4-5-6-7

=4+7+7+6+5

=29

### **Solution No 1:-**

In the time subinterval AC, the area of triangle AOC above the x- axis represent the availability of items

In the time subinterval BC, the area of triangle CEB below the x- axis represent the shortage of items

### **Solution No 2:-**

$$D = 100$$

$$C_3 = 0.02$$

$$C_2 = 100$$

$$C_4 = 5$$

$$EOQ = ?$$

$$EOQ = \sqrt{\frac{2C_2D}{C_3}} * \sqrt{\frac{C_3 + C_4}{C_4}}$$

$$EOQ = \sqrt{\frac{2 * 100 * 100}{0.02}} * \sqrt{\frac{0.02 + 5}{5}}$$

$$EOQ = \sqrt{\frac{20000 * 5.02}{0.02 * 5}}$$

$$EOQ = \sqrt{\frac{20000 * 5.02}{0.02 * 5}}$$

$$EOQ = \sqrt{1004000}$$

$$EOQ = 1001.998$$

### **Solution No 3:-**

$$1) \quad EOQ = \sqrt{\frac{2C_2D}{C_3}} * \sqrt{\frac{C_3 + C_4}{C_4}}$$

$$EOQ = \sqrt{\frac{2 * 400 * 15000}{1.2}} * \sqrt{\frac{1.2 + 10}{10}}$$

$$EOQ = \sqrt{\frac{800 * 15000 * 11.2}{1.2 * 10}}$$

$$EOQ = \sqrt{11200000}$$

$$EOQ = 3346.640$$

$$Q^* = 3346.640$$

$$2) \quad t = \frac{Q^*}{D}$$

$$t = \frac{3346.640}{15000} = 0.223109 \text{Years}$$

$$t = 2.67 \text{Months}$$

$$3) \quad N = \frac{12}{t}$$

$$N = \frac{12}{2.67} = 4.5$$

$$4) \quad S^* = \sqrt{\frac{2C_2 D}{C_4}} * \sqrt{\frac{C_3}{C_3 + C_4}}$$

$$S^* = \sqrt{\frac{2 * 400 * 15000}{10}} * \sqrt{\frac{1.2}{1.2 + 10}}$$

$$S^* = \sqrt{80 * 15000} * \sqrt{\frac{1.2}{11.2}}$$

$$S^* = \sqrt{128571.4286}$$

$$S^* = 358.568$$

$$5) \quad I = Q^* - S^*$$

$$I = 3346.640 - 358.568$$

$$I = 2988.072$$

$$6) \quad \text{Total Cost per period:}$$

$$= C_1 Q^* + C_2 + \frac{C_3 (Q^* - S^*)^2}{2D} + \frac{C_4 S^{*2}}{2D}$$

$$= (2 * 3346.64) + 400 + \frac{1.2(2988.071)^2}{30,000} + \frac{10(358.568)^2}{30,000}$$

$$= 6693.28 + 400 + 357.1429 + 42.857 = 7493.0099$$

**Solution No 4:-**

$$C(S, Q) = C_1 D + \frac{C_2 D}{Q} + \frac{C_3 (Q-S)^2}{2Q} + \frac{C_4 S^2}{2Q}$$

$$\frac{\partial C}{\partial Q} = 0 - \frac{C_2 D}{Q^2} + \frac{C_3}{2} \left( \frac{2Q(Q-S) - (Q^2 - 2QS + S^2)}{Q^2} \right) - \frac{C_4 S^2}{2Q^2}$$

$$= -\frac{C_2 D}{Q^2} + \frac{C_3}{2} \left( \frac{2Q^2 - 2QS - Q^2 + 2QS - S^2}{Q^2} \right) - \frac{C_4 S^2}{2Q^2}$$

$$= -\frac{C_2 D}{Q^2} + \frac{C_3}{2} \left( \frac{Q^2 - S^2}{Q^2} \right) - \frac{C_4 S^2}{2Q^2}$$

$$\frac{\partial C}{\partial S} = 0 + 0 - C_3 \left( \frac{(Q-S)}{Q} \right) + \frac{C_4 S}{Q}$$

$$= -C_3 \left( \frac{Q-S}{Q} \right) + \frac{C_4 S}{Q}$$



**Solution No 5:**

$$\frac{\partial C}{\partial Q} = -\frac{C_2 D}{Q^2} + \frac{C_3}{2} \left( \frac{Q^2 - S^2}{Q^2} \right) - \frac{C_4 S^2}{2Q^2}$$

For optimal value putting  $\frac{\partial C}{\partial Q}$  and  $\frac{\partial C}{\partial S}$  equals to zero.

$$-2C_2 D + C_3(Q^2 - S^2) - C_4 S^2 = 0$$

$$C_3 Q^2 = 2C_2 D + C_3 S^2 + C_4 S^2$$

$$C_3 Q^2 = 2C_2 D + S^2(C_3 + C_4)$$

$$Q^2 = \frac{2C_2 D}{C_3} + S^2 \left( \frac{C_3 + C_4}{C_3} \right) \quad (A)$$

$$\frac{\partial C}{\partial S} = 0 + 0 - C_3 \left( \frac{(Q - S)}{Q} \right) + \frac{C_4 S}{Q}$$

$$\frac{\partial C}{\partial S} = -C_3 \left( \frac{Q - S}{Q} \right) + \frac{C_4 S}{Q}$$

$$C_3 Q = C_3 S + C_4 S$$

$$S = \frac{C_3 Q}{C_3 + C_4} \quad (B)$$

Putting the value of  $S$  in (A)

$$Q^2 = \frac{2C_2 D}{C_3} + \left( \frac{C_3 Q}{C_3 + C_4} \right)^2 \left( \frac{C_3 + C_4}{C_3} \right)$$

$$= \frac{2C_2 D}{C_3} + \left( \frac{C_3 Q^2}{C_3 + C_4} \right)$$

$$Q^2 \left( 1 - \frac{C_3}{C_3 + C_4} \right) = \frac{2C_2 D}{C_3}$$

$$Q = \sqrt{\frac{2C_2 D}{C_3}} \sqrt{\frac{C_3 + C_4}{C_4}}$$

Putting the value of  $Q$  in (B).

$$S = \frac{C_3}{C_3 + C_4} \sqrt{\frac{2C_2 D}{C_3}} \sqrt{\frac{C_3 + C_4}{C_4}}$$

$$S = \sqrt{\frac{2C_2 D}{C_4}} \sqrt{\frac{C_3}{C_3 + C_4}}$$

**Question No.01:**

Find the minimum value of “x” for the constraint  $-4x + 2y \leq 11$  if the point (x, 4) satisfy the above constraint.

**Question No.02:**

Find the maximum value of “y” for the constraint  $4x - 2y \leq 11$  if the point (2, -y) satisfy the above constraint.

**Question No.03:**

Find the corner points of the feasible region for the following constraints:

a)  $4x + 2y \leq 12$

$x, y \geq 0$

b)  $x + 2y \leq 16$

$x + y \leq 9$

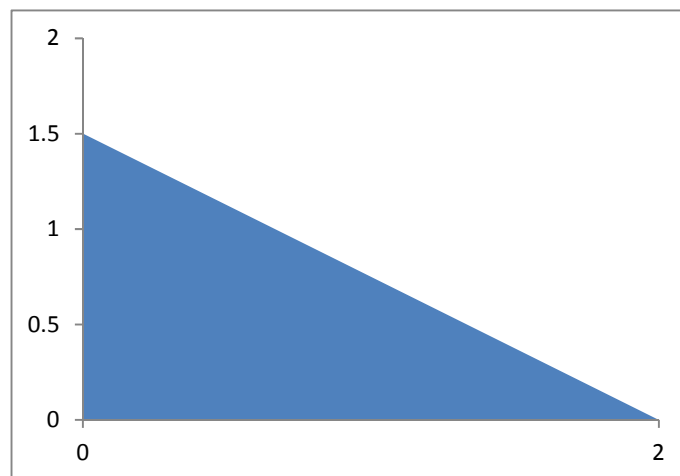
$x, y \geq 0$

c)  $3x + 5y \leq 15$

$x, y \geq 0$

**Question No.04:**

Find the constraint expression for the below shaded region:



**Question No.05:**

Find the optimal values of “x” and “y” for the below given constraints:

a)  $4x + 9y \leq 36$

$$x, y \geq 0$$

b)  $4x + 9y \geq 36$

$$x, y \geq 0$$

**Question No 06:**

Maximize  $Z = 4x + 3y$

Subject to the following constraints

$$x + 4y \leq 20$$

$$2x + 3y \leq 30$$

$$x, y \geq 0$$

**Question No.07:**

Minimize  $Z = 4x + 3y$

Subject to the following constraints

$$x + y \geq 6$$

$$3x + 8y \geq 24$$

$$x, y \geq 0$$

**Question No.08:**

Find the corner points of feasible region for the given constraints:

$$2x + 5y \leq 10$$

$$4x + 5y \geq 20$$

$$x, y \geq 0$$

**Solution No.01:**

The given constraint is

$$-4x + 2y \leq 11$$

*As the point  $(x, 4)$  satisfy the given constraint*

$$-4(x) + 2(4) \leq 11$$

$$-4x + 8 \leq 11$$

$$-4x \leq 11 - 8$$

$$-4x \leq 3$$

$$x \geq -\frac{3}{4}$$

Hence

$$X_{\min} = -3/4$$

**Solution No.02:**

The given constraint is

$$4x - 2y \leq 11$$

*As the point  $(2, -y)$  satisfy the given constraint*

$$4(2) - 2(-y) \leq 11$$

$$8 + 2y \leq 11$$

$$2y \leq 11 - 8$$

$$2y \leq 3$$

$$y \leq \frac{3}{2}$$

Hence

$$y_{\max} = 3/2$$

### **Solution No.03:**

#### **Part-a**

The associated equation of the given constraint is:

$$4x + 2y = 12$$

As the feasible region of the given constraints comes only in the 1<sup>st</sup> quadrant because x and y are positive only in 1<sup>st</sup> quadrant. (  $x, y \geq 0$  )

On Y-Axis,  $x = 0$ ;

$$\implies 4(0) + 2y = 12$$

$$2y = 12$$

$$y = 6$$

**1<sup>st</sup> corner point is A(0,6)**

At intersection of line with X-Axis,  $y = 0$ ;

$$\implies 4x + 2(0) = 12$$

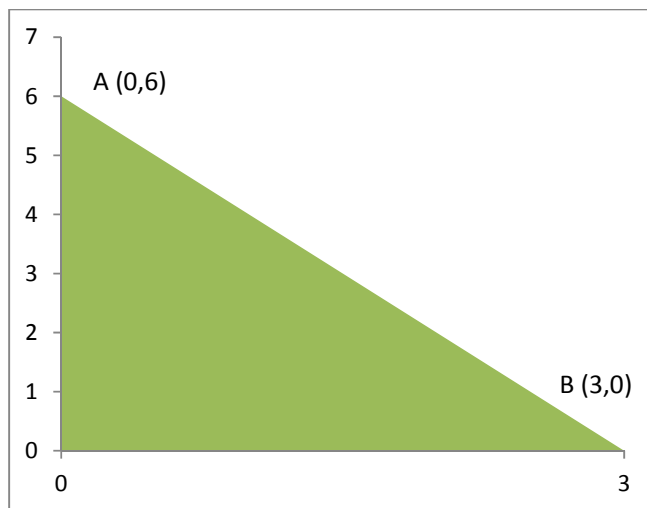
$$4x = 12$$

$$x = 3$$

**2<sup>nd</sup> corner point is B(3,0)**

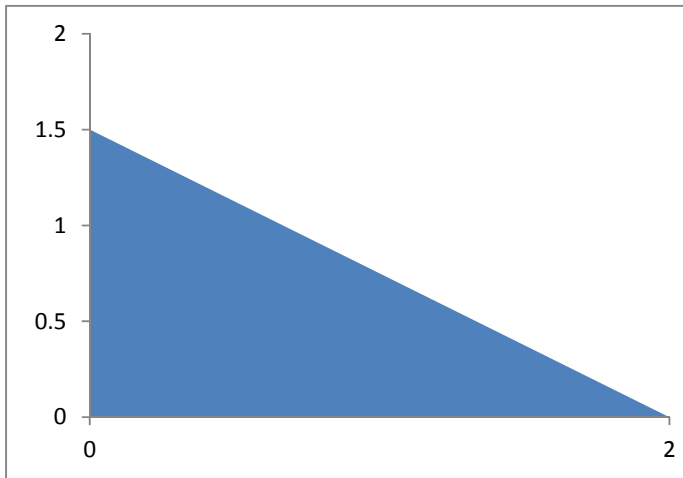
3<sup>rd</sup> corner point will be origin C(0,0)

**The feasible region with corner points is shown below**



Do **Part-b&c** by yourself (Similar to part-a)

**Solution No.04:**



From the graph, it is clear that

$$\text{X-intercept} = a = 2$$

$$\text{Y-intercept} = b = 1.5$$

So the associated equation of constraint can be written as:

$$\frac{x}{a} + \frac{y}{b} = 1$$

$$\frac{x}{2} + \frac{y}{1.5} = 1$$

$$\frac{x}{2} + \frac{1.0 y}{1.5} = 1$$

$$\frac{x}{2} + \frac{2 y}{3} = 1$$

$$3x + 4y = 6$$

**So the set of constraints that will represent the above shaded region will be:**

$$3x + 4y \leq 6$$

$$x, y \geq 0$$



### **Solution No.05:**

#### **Part-a**

The optimal value of “x” will be on X-Axis where  $y = 0$ ;

$$\implies 4x + 9(0) \leq 36$$

$$4x \leq 36$$

$$x \leq 9$$

$$x_{\max} = 9$$

The optimal value of “y” will be on Y-Axis where  $x = 0$ ;

$$\implies 4(0) + 9y \leq 36$$

$$9y \leq 36$$

$$y \leq 4$$

$$y_{\max} = 4$$

As “x” and “y” cannot be negative as given by 2<sup>nd</sup> constraint equation i.e  $x, y \geq 0$

So optimal values of “x” and “y” are:

$$x_{\max} = 9$$

$$y_{\max} = 4$$

$$x_{\min} = 0$$

$$y_{\min} = 0$$

#### **Part-b**

The optimal value of “x” will be on X-Axis where  $y = 0$ ;

$$\implies 4x + 9(0) \geq 36$$

$$4x \geq 36$$

$$x \geq 9$$

$$x_{\min} = 9$$

The optimal value of “y” will be on Y-Axis where  $x = 0$ ;

$$\implies 4(0) + 9y \geq 36$$

$$9y \geq 36$$

$$y \geq 4$$

$$y_{\min} = 4$$

So optimal values of “x” and “y” are:

$$x_{\max} = \infty$$

$$y_{\max} = \infty$$

$$x_{\min} = 9$$

$$y_{\min} = 4$$

### **Solution No.06:**

First, it is needed to find the corner points of the feasible region for the given constraint equations:

The associated equations of the given constraints are:

$$x + 4y = 20 \dots\dots\dots(1)$$

$$2x + 3y = 30 \dots\dots\dots(2)$$

**For equation (1)**

Putting  $x=0$ ; we get

$$(0) + 4y = 20 \implies y = 5$$

**A (0, 5)**

Putting  $y=0$ ; we get

$$x + 4(0) = 20 \implies x = 20$$

**B(20, 0)**

So the two points of equation (1) are **A (0, 5) and B(20, 0)**

**For equation (2)**

Putting  $x=0$ ; we get

$$2(0) + 3y = 30 \implies y = 10$$

**C (0, 10)**

Putting  $y=0$ ; we get

$$2x + 3(0) = 30 \implies x = 15$$

**D(15, 0)**

So the two points of equation (2) are **A (0, 10) and B(15, 0)**

The corner points of the feasible region are A(0, 5), D(15, 0), O(0, 0) and E(a, b).

Point “E(a, b)” is the intersection of given two lines.

Multiplying equation (1) by “2” and subtracting from equation (2)

$$\begin{array}{r} 2x + 3y = 30 \\ \pm 2x \pm 8y = \pm 40 \\ \hline -5y = -10 \implies y = 2 \end{array}$$

Putting value of “y” in equation (1), we get

$$x + 4(2) = 20$$

$$x = 20 - 8$$

$$x = 12$$

So point E is E(12, 2)

**The value of function Z at point “A(0, 5)”**

$$Z = 3(0) + 2(5) \implies Z = 10$$

**The value of function Z at point “D(15, 0)”**

$$Z = 3(15) + 2(0) \implies Z_{\max} = 45$$

**The value of function Z at point “E(12, 2)”**

$$Z = 3(12) + 2(2) \implies Z = 40$$

**So the objective function has maximum value at point D(15, 0).**

### **Question No.07:**

First, it is needed to find the corner points of the feasible region for the given constraint equations:

The associated equations of the given constraints are:

$$x + y = 6 \dots\dots\dots(1)$$

$$3x + 8y = 24 \dots\dots\dots(2)$$

**For equation (1)**

Putting  $x=0$ ; we get

$$(0) + y = 6 \implies y = 6$$

**A (0, 6)**

Putting  $y=0$ ; we get

$$x + (0) = 6 \implies x = 6$$

**B(6, 0)**

So the two points of equation (1) are **A (0, 6) and B(6, 0)**

**For equation (2)**

Putting  $x=0$ ; we get

$$3(0) + 8y = 24 \implies y = 3$$

**C (0, 3)**

Putting  $y=0$ ; we get

$$3x + 8(0) = 24 \implies x = 8$$

**D(8, 0)**

So the two points of equation (2) are **A (0, 3) and B(8, 0)**

The corner points of the feasible region are B(6, 0), C(0, 3), O(0, 0) and E(a, b).

Point “E(a, b)” is the intersection of given two lines.

Multiplying equation (1) by “3” and subtracting from equation (2)

$$\begin{array}{r} 3x + 8y = 24 \\ \pm 3x \pm 3y = \pm 18 \\ \hline \end{array}$$

$$5y = 6 \quad \implies y = 6/5$$

Putting value of “y” in equation (1), we get

$$x + 6/5 = 6$$

$$x = 6 - 6/5$$

$$x = 24/5$$

**So point E is E(24/5, 6/5)**

**The value of function Z at point “B(6, 0)”**

$$Z = 4(6) + 3(0) \quad \implies Z = 24$$

**The value of function Z at point “C(0, 3)”**

$$Z = 4(0) + 3(3) \quad \implies Z_{\min} = 9$$

**The value of function Z at point “E(24/5, 6/5)”**

$$Z = 4 * \frac{24}{5} + 3 * \frac{6}{5}$$

$$Z = \frac{96}{5} + \frac{18}{5}$$

$$Z = \frac{96+18}{5}$$

$$Z = \frac{114}{5}$$

**So the objective function has minimum value at point C(0, 3).**

**Question No.08:**

Try yourself



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# Rana Abubakar Khan

If you want to learn computer programming then contact with me

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MTH601 MID TERM PAPER SHARED BY STUDENT

ON DECEMBER 22, 2018 AT 8:35PM

Mth601

2 or 3 mcqs from past and questions are

- 1.write predecessors.2 and 5 marks
- 2.equations to solve karni ten.2 marks
- 3.linear programming ka ta. 5 marks
- 4.write independent float. 3 marks

MTH601 MID TERM PAPER SHARED BY STUDENT

ON DECEMBER 27, 2018 AT 1:43PM

nth601

most of mcq from past papers

- 1.network dia hua tha EFT find krna tha 2 marks
- 2.economic order quantity AND  $N^*$  find krna ti manufacturing model without shortage ki 3 marks
3. $Q^*$  aur  $t^*$  find krna tha with shortage 3 marks



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4.network diagram ka critical path find krna tha r completion time 5 marks

5.ABC analysis find krna tha 5 marks

MTH601 MID TERM PAPER SHARED BY STUDENT

ON JUNE 6, 2017 AT 5:42PM

Assalam o Alaikum!

Mcqs handouts ky 10 lectures masy zeyda aye thy aik 2 sirf linear programming ka aya tha.

Draw network diagramme thi?

kuch diagraphme di thi oski activity ka relationship btana tha?

Optimum Manufacturing of no shortage find krna tha?

Time period of purchasing aur No of orders find krna tha?

aur aik sawal linear programming ka tha bs mean and average find krna tha kuch asa he sawal tha?

MTH601 MID TERM PAPER SHARED BY STUDENT

MY MTH601 TODAY PAPER 24 DEC, 2016

1- Game of theory problems are converted by which of the following OR technique-----

2- A ----- is a network where every pair of nodes is connected [option Closed network, open network , connected network ]

3- A project is defined as the combination of [interrelated activities ] all of which must be executed in a certine order for its completion

4- EST and EFT of activities are calculated in [forward pass]

5- x

MTH601 MID TERM PAPER SHARED BY STUDENT

ON DECEMBER 24, 2016 AT 5:22PM

Subjective

Key issue wit inventory model? 2 marks





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2- EOQ ka question ta 2 marks

3- Activity Diagram sy table bana ta 3 marks

4 Dummy activities ka question ta MCQS ma b ur Subjective ma b 3 marks

Network diagram bana ta table sy 5 marks

REMEMBER ME IN YOUR PRAYERS IF still need help contact me

MTH601 MID TERM PAPER SHARED BY STUDENT

ON JUNE 20, 2015 AT 9:44PM

paper was not so difficult. mcqs were mostly easy and subjctv was

1 application of inventory

2. what happend if we increase the inventorygive reasons

1) shortage cost 2) holdng cost 3) setup cost

3. ek q tha standard form m likhna tha

4. ek q tha LP m krna tha scenior tha

bqi yad ni. no q from abc r big M method mostly from LP thy

MTH601 MID TERM PAPER SHARED BY STUDENT

ON JANUARY 20, 2015 AT 8:32PM

my todays paper

key issues of inventory methods **2marks**

a solution table was given for  $z = x + y$ , we had to tell whether the solution is optimal or not **2 marks**

$2x \leq$  (less than and equal to) 3

convert this inequality into standard form by introducing slack variable **3 marks**

in no shortage model;

$Q^* = 4470$ ;  $D = 1500/\text{unit/month}$ ;  $R = 3000/\text{month}$



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find  $t$ ,  $N$ , and Total time of manufacturing... **3 marks**

find the minimum value of  $z = 5x - 2y$

conditions:  $2x + 3y \geq 1$ ;  $x, y \geq 0$ ; **5marks**

when to use deterministic model and stochastic models.

a graph was given;

1. we had to find size of order in each cycle.

2. rate of replenishment. **5marks**

MTH601 MID TERM PAPER SHARED BY STUDENT

ON MAY 25, 2013 AT 10:26PM

total 26 questions

20 mcqs

2 marks question(2)

3 marks (2)

5marks (2)

majority portion of questions were from 15 to 21 lectures don't miss linear programming i had total 4 questions from this ,5 ,5 ,3,2 marks k.. ALLAH BLESS YOU ALL... PRAY FOR ALL AND KEEP PREPARING..

MCQS were from 1 to 15 lectures estimated time, MAD, etc.



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MTH601 MID TERM PAPER SHARED BY STUDENT

ON MAY 30, 2013 AT 11:09AM

mcqs are mostly from pert cpm concept base n few are shown graphs about critical path n and eoq equations related mcqs s.d, mad ,forumla over all paper was too easy

subjective questions are :

1.  $D = \text{Rs } 70,000/\text{order}$ ,  $c=0.12$ , where  $p$  is unit price,  $s=10$

| Order size    | Unit price |
|---------------|------------|
| Less than 100 | Rs. 100    |
| 100 or more   | Rs. 50     |

find EOQ for Rs. 50..... 2 marks

2. find the corner points of feasible region bounded by following constraints;

$$4x+3y=12$$

$$x,y \geq 0 \dots\dots\dots 2 \text{ marks}$$

3. if in manufacturing model with no shortages ,  $Q^*=5000$  units,  $D= 1800$  units/month,  $R= 4000$  per month , find

(i). time between orders

(ii). the no of orders per year.

(iii). the time of manufacturing ..... 3 marks

4. a person want to decide the constituents of a diet which will fullfill his daily requirments of protein, fats and carbohyderate at the minimum cost. the choice is to be made from four different type of food . the yield per unit of these foods are given in table .

| Food type | Yield per unit |      |                | Cost per unit |
|-----------|----------------|------|----------------|---------------|
|           | Protein        | fats | carbohyderates |               |
| 1         | 3              | 2    | 6              | 30            |



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|                        |    |    |    |    |
|------------------------|----|----|----|----|
| 2                      | 4  | 2  | 4  | 25 |
| 3                      | 8  | 7  | 7  | 70 |
| 4                      | 6  | 5  | 4  | 50 |
| Minimum<br>requirement | 80 | 20 | 70 |    |

write the model constraints ? ..... 3 marks

5. the demand rate for a particular item 1000 unit/month. the ordering cost is Rs. 100 per order and holding cost is 0.80 Rs per item/month . if no shortages are allowed and replacement is instantaneous , determine:

(i). the economic order quantity.

(II). the time between orders. .... 5 marks

6. Minimize  $Z=2x+ y$ ..... 5 marks

subject to the constraints;

$$x+y \geq 3$$

$$7x+ 5 y \leq 35$$

$$x,y \geq 0$$